



Review

Mixed-integer programming in motion planning

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ABSTRACT

This paper presents a review of past and present results and approaches in the area of motion planning using MIP (Mixed-integer Programming). Although in the early 2000s MIP was still seen with reluctance as method for solving motion planning-related problems, nowadays, due to increases in computational power and theoretical advances, its extensive modeling capabilities and versatility are coming to the fore and enjoy increased application and appreciation. This class of control problems involves, essentially, either a selection from a limited number of alternatives or a constrained optimization problem over a non-convex domain. In both situations, MIP has proven to be an efficient modeling technique as it will be shown in the present review paper. Furthermore, an emphasis is laid on the existing alternatives for implementation and on various experimental validations documented in the literature.

1. Introduction

This section introduces some general real-world problems/situations actually reducing to optimization problems which contain both integer and real variables and can be assimilated to open-loop or closed-loop control problems. Also, for a more general perspective it presents the class of problems which are tackled through MIP formulations.

1.1. Motivation and background

During more than 60 years of existence, the field of integer programming was extensively studied in the mathematics community due to its promising modeling capability and flexibility. In recent years (mainly the last two decades), mostly owing to the growing computational capabilities, the integer programming was brought to the attention of the control and robotic communities. There exists a broad variety of decision making problems that can be dealt through a MIP framework/approach.

MIP (Mixed-integer Programming) is a mathematical optimization problem in which some or all the variables are integers.

As its name indicates, MIP (Mixed-integer Programming) represents a mathematical optimization problem in which the objective is a linear,

quadratic function or sometimes a more general criterion to be minimized or maximized, the constraints are linear (or non-linear) equalities (or inequalities) and there exist some (non-empty) subsets of integer and real variables playing the role of arguments (Jünger et al., 2009; Williams, 2013).

MIP is used to model several design problems and decision processes.

In a larger perspective, MIP is used to model several design problems and decision processes. Consider a typical logistics problem: an airport, which serves on average 50 flights per hour. The airport has only four runways. The task assignment problem that appears is to assign flights to runways, such that the runways are efficiently and uniformly used, while respecting some regulations (e.g., time separation between two consecutive landings/takeoffs or a minimum distance between two runways for simultaneous takeoffs). Another classical situation is described by the well-known *traveling salesman problem* and its variations, where the salesman wants to visit a number of customers in a minimal time or to cover a minimal distance. This has applications in several domains (e.g, overhauling gas turbine engines or X-ray crystallography Matai, Singh, & Mittal (2010)). The above problems can be solved either intuitively, based on experience or by a trial and error method, but for critical situations an accurate mathematical formulation is necessary in view of certification. There are of course many other use

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综述

运动规划中的混合整数规划

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任务分配
碰撞避免

ABSTRACT

本文回顾了使用混合整数规划 (MIP) 进行运动规划领域过去和现在的成果与方法。尽管在21世纪初, MIP仍被视为一种不太受欢迎的解决运动规划相关问题的方法, 但由于计算能力的提升和理论进展, 如今其广泛的建模能力和通用性日益凸显, 并获得了更多的应用和认可。这类控制问题本质上涉及从有限数量的选项中选择, 或是在非凸域上进行约束优化问题。在这两种情况下, MIP都已被证明是一种高效的建模技术, 正如本文的综述所展示的那样。此外, 本文还重点介绍了现有的实现替代方案以及文献中记录的各种实验验证。

1. 引言

本节介绍了一些实际可简化为优化问题的通用现实问题/情况, 这些问题包含整数和实变量, 并可归约为开环或闭环控制问题。此外, 为了更广泛的视角, 还介绍了通过MIP公式解决的问题类别。

1.1. 动机和背景

在六十多年的发展历程中, 整数规划领域由于其有前景的建模能力和灵活性, 在数学界得到了广泛研究。近年来 (主要是过去二十年), 主要得益于计算能力的增长, 整数规划引起了控制和机器人学界的关注。存在大量可以通过MIP框架/方法处理的决策问题。

MIP (混合整数规划) 是一种数学优化问题, 其中部分或全部变量为整数。

从其名称可以看出, MIP (混合整数规划) 代表一种数学优化问题, 其目标是一个线性、

二次函数或有时是一个更通用的标准, 需要被最小化或最大化, 约束条件是线性 (或非线性) 等式 (或不等式), 并且存在一些 (非空) 整数和实变量子集充当参数 (Jünger 等人, 2009; Williams, 2013)。

MIP用于建模多种设计问题和决策过程。

从更宏观的角度来看, MIP 用于建模多种设计问题和决策过程。考虑一个典型的物流问题: 一个机场, 平均每小时服务 50 架航班。该机场只有四条跑道。出现的问题是分配航班到跑道上, 使得跑道得到高效和均匀的使用, 同时遵守一些规定 (例如, 连续两次着陆/起飞之间的时间间隔或同时起飞的两条跑道之间的最小距离)。另一个经典的场景是由著名的旅行商问题及其变体所描述, 其中旅行商希望在最短的时间内访问若干客户或覆盖最短的距离。这适用于多个领域 (例如, 涡轮发动机翻新或 X 射线晶体学 Matai, Singh, & Mittal (2010))。上述问题可以通过直觉、经验或试错法解决, 但对于关键情况, 出于认证目的, 需要精确的数学表述。当然, 还有许多其他用途

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cases which may employ MIP. For example, the optimal power flow in the energy transmission networks [Bahense, Oliveira, Pereira, and Granville \(2001\)](#) or the transportation problems in a cluttered environment. Consider a boat moving within a fjord region. In order to safely arrive to its destination, the boat should follow a given path and avoid collision with the fjords. Thus, the feasible region is non-convex and should be efficiently described.

In the following, a brief classification of the types of problems, which can be modeled through MIP, is provided. A first class of problems is designated by those that involve integer quantities (i.e. discrete/quantified inputs or outputs), e.g. the *knapsack problem* ([Williams, 2013](#)). For this type of problem, MIP does not seem the obvious, natural, first choice, but, usually, it represents a better solution than a classical approach (e.g., use of the classical linear programming and approximate the provided solution to the nearest integer value).

Another MIP-modelisable class of problems is the one involving logical conditions, extensively treated in: [Bemporad and Morari \(1999\)](#), [Williams \(2013\)](#), [Smith and Taskin \(2008\)](#). For example, in [Bemporad and Morari \(1999\)](#), using the notations from [Williams \(2013\)](#), boolean algebra tools are aggregated, which allows to transform logical conditions on continuous variables into *mixed-integer inequalities* (linear inequalities involving continuous and binary variables).

As well, MIP is a popular modeling tool for sequencing and/or allocation problems (also, named combinatorial problems) ([Smith & Taskin, 2008](#)), including here the typical *task assignment* problem and its variations (e.g travelling salesman problem [Dantzig, Fulkerson, & Johnson \(1954\)](#)). This class of problems can be easily extended to networks (and graph theory) problems: *resource allocation on a PERT (Project Evaluation and Review Techniques) network* ([Williams, 2013](#)).

Lastly, but most importantly for this paper’s purpose, MIP turned out to be a captivating method to model non-linearity ([Bemporad & Morari, 1999](#); [Vielma, 2015](#)) and/or non-convexity ([Prodan, Stoican, Oлару, & Niculescu, 2015](#); [Richards & How, 2005](#)). A plethora of control engineering problem are naturally and intrinsically characterized by non-linearity and/or non-convexity. For this reason and due to the increasing interest in optimization-based control ([Mayne, Rawlings, Rao, & Scokaert, 2000](#)), MIP has become an essential technique, which allows to include logical decisions and non-convex constraints in the optimization problem. Therefore, MIP’s presence in control can be perceived in: piecewise-affine system identification ([Bemporad, Roll, & Ljung, 2001](#); [Roll, Bemporad, & Ljung, 2004](#)), assignment problems ([Alighanbari, Kuwata, & How, 2003](#)), persisting exciting control ([Marafioti, Stoican, Bitmead, & Hovd, 2012](#)), control of hybrid systems ([Bemporad & Morari, 1999](#)), fault detection ([Stoican, Oлару, Seron, & De Doná, 2012](#)) or motion planning ([Prodan et al., 2015](#); [Richards & How, 2002](#)). As the title suggests, in the present review, the main objective is to identify and summarize the state of the art of MIP-based motion planning. Hence, in what follows, we place less emphasis on the other control areas employing MIP, even if throughout the paper we occasionally refer the interested readers to the references covering the other MIP-based control topics.

1.2. Contributions

This paper offers a detailed literature review of breakthrough research results and open issues in the field of multi-agent motion planning in a mixed-integer framework. This work can be employed to the benefit of both control and optimization research communities allowing to swiftly identify previous, timely and relevant research topics in the field and, at the same time, decreasing the time for literature review, although we acknowledge that the present review effort is not exhaustive but merely covers the experience of the authors in the last 10 years in these topics.

To the best of the authors’ knowledge, there are no other exhaustive surveys in this topic although valuable attempts with different objectives are to be found in the literature. For instance, the tutorial session

[Richards and How \(2005\)](#) offers a brief overview on how MIP can be employed for (feedback) control. As well, the paper of [Smith and Taskin \(2008\)](#) represents a concise introduction to the MIP modeling, providing the basic concepts regarding MIP formulations (the principles and some handful recipes) and, at the same time, discussing the techniques widely-used in the resolution of MIP problems. Moreover, the work “50 Years of integer programming 1958–2008: From the early years to the state-of-the-art” ([Jünger et al., 2009](#)) presents a historical perspective of the field of integer programming and discusses the theoretical, algorithmic and computational aspects of MIP throughout more than five decades of existence. Additionally, but with significant influence, there exist surveys, as, e.g., [Vielma \(2015\)](#), reviewing the advanced MIP formulations techniques, aiming to provide the guidelines for obtaining stronger and/or smaller formulations for a certain class of decision making problems. These works either limit themselves looking to the programming side, either to decision making in general or to a narrow control design topic. We henceforth decide to focus our review on control problems based on MIP and in particular those emerging from the active research field of motion planning which has a plethora of applications in automotive, robotics or multi-agent systems to mention just a few.

1.3. Outline

The remainder of the paper is organized as follows. We begin in [Section 2](#) with a brief delineation of the evolution of MIP mathematical descriptions, providing the necessary prerequisites used in these formulations. [Section 3](#) presents and details the standard MIP-based problems in motion planning, introducing the generic control strategy employed in such problems. Next, [Section 4](#) makes the transition to multi-agent systems and to the formation control problems involving MIP. [Section 5](#) gives a concise overview of the control architectures exploited in MIP-based navigation problems. Further, we proceed in [Section 6](#) with a brief presentation of both software and hardware implementations of MIP solutions in the literature. [Section 7](#) succinctly presents the motion planning alternatives to MIP. We close our review with the conclusions and with some challenges in MIP-based motion planning and some suggestions for future research.

Notation

Throughout this paper we use the following standard notations. The logical operators: $\wedge$ (AND), $\vee$ (OR) and $\neg$ (negation). The Minkowski sum of two sets: $A \oplus B = \{x : x = a + b, a \in A, b \in B\}$. For $x \in \mathbb{R}^d$ we denote $\|x\|_Q^2 = x^\top Qx$. Given a compact set $S \subset \mathbb{R}^n$, $C_X(S)$ denotes the complement of S over X , while $cl(S)$ is the closure of the subset S . For any polyhedron $P \subset \mathbb{R}^d$, $\mathcal{V}(P)$ is the (finite) set of its vertices. Any polytope (i.e. a bounded polyhedron) has a dual representation in terms of intersection of half-spaces or convex hull of extreme points: $P = \{x : s_i^\top x \leq r_i, \forall i\} = \{x : x = \sum \alpha_j v_j, \sum \alpha_j = 1, \alpha_j \geq 0, \forall j\}$. For a discrete set $\mathcal{I}$, $\#\mathcal{I}$ represents its cardinality.

2. MIP Formulations

There are various MIP formulations which go back to the early ’80 (or even earlier), each emphasizing the modeling capabilities of MIP. All these formulations share a common characteristic: the encoding of discrete decisions using binary and/or integer variables. These decisions appear in different problems, each using a certain formulation. This section provides a brief description of the most used MIP techniques and, concurrently, introduces some basic theoretical notions and tools.

Although generalized disjunctive programming (GDP) is not explicitly used in motion planning, we succinctly present it for the sake of the generality of modeling disjunctions through MIP.

可能使用 MIP 的情况。例如，能源输电网络中的最优潮流（Bahense, Oliveira, Pereira 和 Granville, 2001 年）或杂乱环境中的运输问题。考虑一艘船在峡湾区域内移动。为了安全到达目的地，船应遵循给定路径并避免与峡湾碰撞。因此，可行区域是非凸的，需要有效地描述。

以下，提供了一种可通过MIP建模的问题类型的简要分类。第一类问题是涉及整数量数的那些问题（即离散/量化输入或输出），例如背包问题（Williams, 2013）。对于这类问题，MIP似乎不是显而易见、自然的第一选择，但通常它比经典方法（例如，使用经典线性规划并将近似解逼近到最近的整数值）能提供更好的解决方案。

另一种可通过MIP建模的问题类型是涉及逻辑条件的问题，在Bemporad和Morari (1999)、Williams (2013)、Smith和Taskin (2008)中得到了广泛讨论。例如，在Bemporad和Morari (1999)中，使用Williams (2013)的符号，布尔代数工具被整合，这使得可以将关于连续变量的逻辑条件转换为混合整数不等式（涉及连续变量和二元变量的线性不等式）。

此外，MIP 是一种流行的建模工具，用于解决排序和/或分配问题（也称为组合问题）（Smith & Taskin, 2008），包括这里的典型任务分配问题及其变体（例如旅行商问题 Dantzig, Fulkerson, & Johnson (1954)）。这类问题可以很容易地扩展到网络（和图论）问题：PERT（项目评估和审查技术）网络上的资源分配（Williams, 2013）。

最后，但最重要的是，对于本文的目的而言，MIP 被证明是一种引人入胜的建模非线性（Bemporad & Morari, 1999; Vielma, 2015）和/或非凸性（Prodan, Stoican, Oлару, & Niculescu, 2015; Richards & How, 2005）的方法。大量控制工程问题自然且本质上具有非线性性和/或非凸性特征。由于这个原因以及基于优化的控制日益增长的兴趣（Mayne, Rawlings, Rao, & Scokaert, 2000），MIP 已成为一项基本技术，它允许在优化问题中包含逻辑决策和非凸约束。因此，MIP 在控制中的存在体现在：分段仿射系统辨识（Bemporad, Roll, & Ljung, 2001；Roll, Bemporad, & Ljung, 2004）、分配问题（Alighanbari, Kuwata, & How, 2003）、持续激励控制（Marafioti, Stoican, Bitmead, & Hovd, 2012）、混合系统控制（Bemporad & Morari, 1999）、故障检测（Stoican, Oлару, Seron, & De Doná, 2012）或运动规划（Prodan et al., 2015；Richards & How, 2002）。如标题所示，在本文的综述中，主要目标是识别和总结基于 MIP 的运动规划的现状。因此，在以下内容中，我们较少强调采用 MIP 的其他控制领域，即使在整个论文中，我们偶尔会向感兴趣的读者提及涵盖其他基于 MIP 的控制主题的参考文献。

1.2. 贡献

本文对混合整数框架下多智能体运动规划领域的突破性研究成果和开放性问题进行了详细的文献综述。这项工作可用于控制与优化研究社区，帮助快速识别该领域先前、及时且相关的研究主题，同时减少文献综述时间，尽管我们承认目前的综述工作并不全面，仅涵盖了作者在过去10年中对这些主题的经验。

据作者所知，尽管在文献中可以找到具有不同目标的宝贵尝试，但在这个主题上还没有其他详尽的综述。例如，教程环节

Richards和How (2005) 提供了关于MIP如何用于（反馈）控制的简要概述。此外，Smith和Taskin (2008) 的论文代表了对MIP建模的简洁介绍，提供了关于MIP公式的基本概念（原则和一些实用的方法），同时讨论了广泛用于解决MIP问题的技术。此外，著作《50年的整数规划 1958–2008：从早期到最前沿》（Jünger等人，2009）展示了整数规划领域的历史视角，并讨论了MIP在半个多世纪存在期间的理论、算法和计算方面。此外，但具有显著影响，存在综述，例如Vielma (2015)，回顾了高级MIP建模技术，旨在为特定类别的决策问题提供获得更强和/或更小公式的指南。这些工作要么局限于编程方面，要么局限于一般决策或狭窄的控制设计主题。因此，我们决定将我们的综述集中在基于MIP的控制问题，特别是那些来自运动规划这一活跃研究领域的那些问题，该领域在汽车、机器人或多智能体系统中有着众多应用，仅举几例。

1.3. 概述

本文的其余部分安排如下。我们在第2节简要概述了MIP数学描述的演变，提供了这些公式中使用的必要前提。第3节介绍了运动规划中的标准MIP问题，并详细介绍了此类问题中采用的通用控制策略。接下来，第4节过渡到多智能体系统和涉及MIP的编队控制问题。第5节简要概述了MIP导航问题中使用的控制架构。此外，我们在第6节简要介绍了文献中MIP解决方案的软件和硬件实现。第7节简要介绍了MIP的运动规划替代方案。我们以结论结束本综述，并讨论MIP运动规划中的一些挑战以及未来研究的建议。

符号表示法

在本文中，我们使用以下标准符号表示。逻辑运算符： $\wedge$ (与)， $\vee$ (或)和 $\neg$ (否定)。两个集合的Minkowski和： $A \oplus B = \{x : x = a + b, a \in A, b \in B\}$ 。对于 $x \in \mathbb{R}^d$ ，我们表示 $\|x\|_Q^2 = x^\top Qx$ 。给定一个紧致集合 $S \subset \mathbb{R}^n$ ， $C_X(S)$ 表示 S 在 X 上的补集，而 $cl(S)$ 是子集 S 的闭包。对于任何多面体 $P \subset \mathbb{R}^d$ ， $\mathcal{V}(P)$ 是其（有限）顶点集。任何多面体（即有界多面体）都可以用半空间的交集或极端点的凸包来表示其对偶表示： $P = \{x : s_i^\top x \leq r_i, \forall i\} = \{x : x = \sum \alpha_j v_j, \sum \alpha_j = 1, \alpha_j \geq 0, \forall j\}$ 。对于离散集 $\mathcal{I}$ ， $\#\mathcal{I}$ 表示其基数。

2. MIP配方

存在多种MIP配方，可追溯至20世纪80年代初（甚至更早），每种配方都强调MIP的建模能力。所有这些配方都具有一个共同特征：使用二进制和/或整数变量对离散决策进行编码。这些决策出现在不同的问题中，每个问题使用特定的配方。本节简要介绍了最常用的MIP技术，同时介绍了一些基本的理论概念和工具。

尽管广义析取规划（GDP）在运动规划中并未明确使用，但为了通过MIP进行析取建模的普遍性，我们简要介绍了它。

2.1. Generalized disjunctive programming

Generalized Disjunctive Programming (GDP) has appeared for the first time in [Raman and Grossmann \(1994\)](#) in the research effort to use both quantitative and qualitative information in order to optimally solve chemical engineering problems. To this end, the qualitative information is represented using disjunctions and logic propositions. Compared to MIP, the GDP approach has a relatively more compact formulation, because the logical conditions are not transformed using boolean algebra and inequalities but rather in their natural (logic) form. In other words, GDP represents a combination of algebraic and logical equation as shown by the typical GDP from [\(1\)](#):

$$\min_x f(x) + \sum_k c_k \quad (1a)$$

$$\text{s.t. } r(x) \leq 0, x \in \mathbb{R}^n, \quad c_k \in \mathbb{R}, \quad (1b)$$

$$\bigvee_{j \in J_k} \begin{bmatrix} Y_{jk} \\ g_{jk}(x) \leq 0 \\ c_k = \gamma_{jk} \end{bmatrix}, \quad k \in K, \quad (1c)$$

$$\Omega(Y) = \text{true}, \quad Y_{jk} \in \{\text{true}, \text{false}\}, \quad (1d)$$

where $r(x)$ is a generic constraint, which does not depend on logical decisions; c_k cost variables, γ_{jk} fixed charges and $k \in K$ is the number of disjunctions. The logical function $\Omega(Y)$ corresponds to logical decisions in terms of boolean algebra and it is expressed in conjunctive normal form (CNF- “product of sums”). The idea of GDP can be summarized as follows: If $Y_{jk} = \text{true}$, then the constraints $g_{jk}(x) \leq 0$ and $c_k = \gamma_{jk}$ are imposed. Otherwise, they are ignored.

It is worth mentioning that the disjunction $\bigvee_{j \in J_k}$ represents, in fact, an exclusive relationship, in a disjunction k only a boolean variable Y_{jk} can be *true*. Some formulations state this requirement explicitly (as a separate constraint), others include it implicitly in the boolean function $\Omega(Y)$.

The problem [\(1\)](#) can be written as MIP, employing the binary variables $y_{jk} \in \{0, 1\}$ instead of the boolean ones Y_{jk} and replacing the constraint [\(1c\)](#) with:

$$g_{jk}(x) \leq M_{jk}(1 - y_{jk}) \quad (2a)$$

$$\sum_{j \in J_k} y_{jk} = 1, \quad k \in K \quad (2b)$$

where M_{jk} are “big-M” parameters, sufficiently large constants.

Remark 1. The “big-M” formulation¹ ([Williams, 2013](#)) consists in choosing a very large positive number M which plays the role of a relaxation constant ([Richards & How, 2005](#); [Vielma & Nemhauser, 2011](#)). In this formulation, the binary variables play the role of a “switch”, activating/deactivating the corresponding constraints. This is a powerful technique able to encode logical conditions but one should proceed with caution in selection of the value of M . A too large value may hinder the resolution of the MIP. For a detailed analysis regarding the “big-M” technique, we refer the interested reader to, e.g. [Vielma and Nemhauser \(2011\)](#), [Hooker \(2011\)](#). ♦

Remark 2. It is worth discussing the value of the coefficient M . For illustration, we consider a common case of modeling the logical implication: “ $x > 0 \rightarrow \alpha = 1$ ” for a variable x which evolves implicitly within a bounded domain² of variation $\mathbb{X} = \{x | |x| \leq \bar{x}\}$. The “big-M” formulation simply reverts the logical implication in terms of an inequality $x - M\alpha \leq 0$. Intuitively, as long as M is chosen to satisfy $M \geq \bar{x}$, the

implication follows naturally. Nonetheless, by considering a too large value of the coefficient M we increase the size of the searching domain. Thus, the idea is to take M as small as possible (but still large enough to fulfill its relaxation role). In our simple example a proper choice is: $M =$

$\max_{x \in \mathbb{X}} x$, i.e., $M = \bar{x}$. ♦

Moreover, the cost is reformulated by rewriting each cost variable c_k as the product $\gamma_{jk}y_{jk}$. As well, the condition [\(1d\)](#) is written in an algebraic form $Ay \leq a$.

The apparent advantage of the GDP over MIP is diminished because all the existing GDP-dedicated solvers are based on a MIP reformulation. It is enough in general to assume that a direct MIP modeling of the problem at hand could lead to a better and more compact model. In spite of eluding GDP in such a way, there is another possibility to benefit from GDP. The problem can be modeled as a GDP and the resulting MIP reformulation can be parsed using one of the several tools, as detailed in ([Williams, 2013](#)).

2.2. MIP - a geometric viewpoint

Complex control synthesis or design leads to a constrained optimization problems and whenever a non-convex feasible domain needs to be handled, the ability of MIP to mathematically formalize disjunction constraints can be exploited. An active research topic was the MIP efficient description of these non-convex regions. Preliminary results make use of hyperplane arrangements to characterize these regions. Hereinafter, we briefly recall these results which involve besides hyperplane arrangements some set-theoretic notions. It is beyond the scope of this review to provide an exhaustive presentation of these techniques and refer the interested readers to complementary materials referenced in this manuscript (e.g., [Vielma and Nemhauser \(2011\)](#), [Vielma \(2015\)](#) or [Prodan et al. \(2015\)](#)).

Most sets involved in MIP formulations are polyhedral ones due to their ability to provide a geometrical representation of the linear constraints. In what follows, we use the notion of polytope which is a bounded polyhedral set and has a dual representation in terms of intersection of half-spaces or convex hull of extreme points: $P = \{x : s_i^\top x \leq r_i, \forall i\} = \{x : x = \sum \alpha_i v_i, \sum \alpha_i = 1, \alpha_i \geq 0\}$.

Consider a finite collection of hyperplanes from $\mathbb{R}^d$

$$\mathcal{H}_i = \{x \in \mathbb{R}^d : h_i x = k_i\}, \quad i \in \mathcal{I} \quad (3)$$

where $\mathcal{I} \triangleq \{1 \dots N\}$, and $(h_i, k_i) \in \mathbb{R}^{1 \times d} \times \mathbb{R}$.

Each of these hyperplanes divides the space in two disjoint³ regions:

$$\mathcal{R}_i^+ = \{x \in \mathbb{R}^d : h_i x \leq k_i\}, \quad (4a)$$

$$\mathcal{R}_i^- = \{x \in \mathbb{R}^d : -h_i x \leq -k_i\}. \quad (4b)$$

A polytope P is a bounded intersection of these half-spaces⁴:

$$P = \bigcap_{i \in \mathcal{I}} \mathcal{R}_i^-, \quad (5)$$

Its complement (up to its relative interior, see also footnote 1) $\mathcal{C}(P) \triangleq \mathcal{C}(\mathbb{R}^d \setminus P)$ over $\mathbb{R}^d$ represents the union of all regions covering the entire space except P . Recalling the fact that the union and intersection operators interchange under complement operation, we write:

³ The relative interiors of these regions are disjoint, but their closures have as a common boundary the hyperplane $\mathcal{H}_i$

⁴ The sign “-” was chosen for the simplicity of notation, any other feasible combination of signs from [\(4a\)-\(4b\)](#) could be chosen in order to describe the polytope P .

2.1. 广义析取规划

广义析取规划 (GDP) 首次出现在Raman和Grossmann (1994年)的研究工作中, 该研究旨在利用定量和定性信息来优化解决化学工程问题。为此, 定性信息被表示为析取和逻辑命题。与MIP相比, GDP方法具有相对更紧凑的公式, 因为逻辑条件不是通过布尔代数和不等式进行转换, 而是以它们自然 (逻辑) 的形式表示。换句话说, GDP表示了代数方程和逻辑方程的组合, 如 (1) 中典型的GDP所示:

$$\min_x f(x) + \sum_k c_k \quad (1a)$$

$$\text{s.t. } r(x) \leq 0, x \in \mathbb{R}^n, \quad c_k \in \mathbb{R}, \quad (1b)$$

$$\bigvee_{j \in J_k} \begin{bmatrix} Y_{jk} \\ g_{jk}(x) \leq 0 \\ c_k = \gamma_{jk} \end{bmatrix}, \quad k \in K, \quad (1c)$$

$$\Omega(Y) = \text{true}, \quad Y_{jk} \in \{\text{true}, \text{false}\}, \quad (1d)$$

其中 $r(x)$ 是一个通用约束, 它不依赖于逻辑决策; c_k 成本变量, γ_{jk} 固定费用和 $k \in K$ 是析取的数量。逻辑函数 $\Omega(Y)$ 对应于布尔代数中的逻辑决策, 并以合取范式 (CNF- “和的积”) 表示。GDP 的思想可以总结如下: 如果 $Y_{jk} = \text{true}$, 则施加约束 $g_{jk}(x) \leq 0$ 和 $c_k = \gamma_{jk}$ 。否则, 它们将被忽略。

值得一提的是, 析取 $\bigvee_{j \in J_k}$ 实际上表示一种互斥关系, 在析取 k 中, 只有一个布尔变量 Y_{jk} 可以为真。一些公式明确陈述这一要求 (作为一个独立的约束), 而另一些则将其隐含地包含在布尔函数 $\Omega(Y)$ 中。

问题 (1) 可以写成 MIP, 使用二元变量 $y_{jk} \in \{0, 1\}$ 代替布尔变量 Y_{jk} , 并将约束 (1c) 替换为:

$$g_{jk}(x) \leq M_{jk}(1 - y_{jk}) \quad (2a)$$

$$\sum_{j \in J_k} y_{jk} = 1, \quad k \in K \quad (2b)$$

M_{jk} 在哪里, “大M” 参数是指足够大的常数。

注释1. “大M” 公式¹ (威廉姆斯, 2013)包括选择一个非常大的正数 M , 它充当松弛常数的角色 (理查兹和霍, 2005; 维尔马和内姆哈USER, 2011)。在这种公式中, 二元变量充当 “开关”, 激活/禁用相应的约束。这是一种强大的技术, 能够编码逻辑条件, 但在选择 M 的值时应谨慎行事。 M 值过大可能会阻碍MIP的求解。关于 “大M” 技术的详细分析, 我们建议感兴趣的读者参考, 例如维尔马和内姆哈USER (2011)、霍克 (2011)。♦

备注2. 值得讨论系数 M 的价值。为说明起见, 我们考虑一种常见的逻辑蕴涵建模情况: “ $x > 0 \rightarrow \alpha = 1$ ” 对于一个在变化域² 内隐式演化的变量 $x \in \mathbb{X} = \{x | |x| \leq \bar{x}\}$ 。 “大 M ” 公式将逻辑蕴涵以不等式 $x - M\alpha \leq 0$ 的形式逆转。直观上, 只要 M 选择满足 $M \geq \bar{x}$, 蕴涵就会自然成立。然而, 通过考虑系数 M 过大的值, 我们会增加搜索域的大小。因此, 想法是尽可能使 M 小 (但仍足够大以发挥其松弛作用)。在我们的简单示例中, 一个合适的选项是: $M “x > 0 \rightarrow \alpha = 1”$

¹ 它也被称为 “大-D” 或 “大-R”。² 由于物理限制。

蕴涵就会自然成立。然而, 通过考虑系数 M 过大的值, 我们会增加搜索域的大小。因此, 想法是尽可能使 M 小 (但仍足够大以发挥其松弛作用)。在我们的简单示例中, 一个合适的选项是: $M =$

$\max_{x \in \mathbb{X}} x$, 即 $M = \bar{x}$ 。♦

此外, 成本通过将每个成本变量 c_k 重写为乘积 $\gamma_{jk}y_{jk}$ 进行重新表述。同样, 条件 (1d) 以代数形式 $Ay \leq a$ 写出。

GDP相较于MIP的明显优势被削弱了, 因为所有现有的GDP专用求解器都是基于MIP重构的。通常来说, 假设直接对当前问题进行MIP建模能够得到一个更好、更紧凑的模型就足够了。尽管这种方式避开了GDP, 但仍有另一种可能从GDP中获益。问题可以建模为GDP, 然后使用(Williams, 2013)中详细描述的几种工具之一来解析得到的MIP重构。

2.2. MIP - 几何视角

复杂的控制综合或设计会导致约束优化问题, 每当需要处理非凸可行域时, MIP在数学上形式化析取约束的能力就可以被利用。一个活跃的研究课题是这些非凸区域的MIP高效描述。初步结果利用超平面排列来刻画这些区域。以下, 我们简要回顾这些涉及超平面排列和某些集合论概念的结果。本综述的范围之外, 无法对这些技术进行详尽介绍, 并建议感兴趣的读者参考本文献中引用的补充材料 (例如, Vielma and Nemhauser (2011), Vielma (2015)或Prodan et al. (2015))。

大多数参与MIP模型的集合都是多面体, 因为它们能够为线性约束提供几何表示。在以下内容中, 我们使用多面体的概念, 它是一个有界多面体集合, 并且可以通过半空间的交集或极端点的凸包来表示其对偶形式: $P = \{x : s_i^\top x \leq r_i \quad \forall i\} = \{x : x = \sum \alpha_i v_i, \sum \alpha_i = 1, \alpha_i \geq 0\}$, 。

考虑来自 $\mathbb{R}^d$ 的有限个超平面

$$\mathcal{H}_i = \{x \in \mathbb{R}^d : h_i x = k_i\}, \quad i \in \mathcal{I} \quad (3)$$

其中 $\mathcal{I} \triangleq \{1 \dots N\}$, 和 $(h_i, k_i) \in \mathbb{R}^{1 \times d} \times \mathbb{R}$ 。

这些超平面中的每一个都将空间分为两个不相交³ 区域:

$$\mathcal{R}_i^+ = \{x \in \mathbb{R}^d : h_i x \leq k_i\}, \quad (4a)$$

$$\mathcal{R}_i^- = \{x \in \mathbb{R}^d : -h_i x \leq -k_i\}. \quad (4b)$$

一个多面体 P 是这些半空间的⁴ 有界交集:

$$P = \bigcap_{i \in \mathcal{I}} \mathcal{R}_i^-, \quad (5)$$

它的补集 (至其相对内部, 另见脚注1) $\mathcal{C}(P) \triangleq \mathcal{C}(\mathbb{R}^d \setminus P)$ 在 $\mathbb{R}^d$ 上表示所有覆盖整个空间但不包括 P 的区域的并集。回想并集和交集运算在补运算下会互换, 我们写成:

³ 这些区域的相对内部是互不相交的, 但它们的闭包以超平面 $\mathcal{H}_i$ 作为公共边界。 “-” 符号的选择是为了简化记法, (4a)-(4b) 中的任何其他可行符号组合都可以用来描述多胞体 P 。

¹ It is also referred to as “big-D” or “big-R”.

² Due to the physical limitations.

$$\mathcal{C}(P) = \mathcal{C}\left(\bigcap_{i \in \mathcal{I}} \mathcal{R}_i^-\right) = \bigcup_{i \in \mathcal{I}} \mathcal{C}(\mathcal{R}_i^-) = \bigcup_{i \in \mathcal{I}} \mathcal{R}_i^+. \quad (6)$$

The region (6) is a union of convex sets⁵ and for a tractable characterization we may employ mixed-integer techniques. Hence, the binary variables $(\alpha_1 \dots \alpha_N) \in \{0, 1\}^N$ are introduced to obtain the representation of the polytopic set in the extended space of state and auxiliary binary variables:

$$h_i x \leq k_i + M\alpha_i, \quad i \in \mathcal{I}, \quad (7a)$$

$$\sum_{i \in \mathcal{I}} \alpha_i \leq N - 1, \quad (7b)$$

where M is a big-M coefficient as in Remark 1.

The requirement that a point is outside of the polytopic set P is converted to the condition that the point has to be within the complement of at least one of the half-spaces defining the polytope.

Remark 3. The conditions (7a)–(7b) describe the region (6) via a proper combination of binary variables. For instance, the region $\mathcal{R}_i^+$ is expressed by (7a) with the following binary variables:

$$(\alpha_1 \dots \alpha_N) = \left(1 \dots 1 \underbrace{0}_i 1 \dots 1\right). \quad (8)$$

When a binary variable takes the value “1” the associated inequality describes, for the limit case ($M \rightarrow \infty$) the entire domain $\mathbb{R}^n$. Hence, the condition (7b) is necessary in order to ensure that at least one of the binary variables is “0” and, thus, that at least an inequality remains active. ♦ The above reasoning is not restricted to the complement of one convex polytopic region and can be generalized for the case in which the interdicted region is a union of polytopes as in (Stoican, Prodan, & Olaru, 2013). Explicitly, the non-convex region is characterized by:

$$-h_{il}x \leq -k_{il} + M\alpha_{il}, \quad \forall l \in \mathcal{I}_l, \quad (9a)$$

$$\sum_{l \in \mathcal{I}_l} \alpha_{il} \leq \#\mathcal{I}_l - 1. \quad (9b)$$

with $\mathcal{I}$ in (3) redefined as a union of discrete intervals⁶: $\mathcal{I} = \bigcup_l \mathcal{I}_l$.

Remark 4. As stated in Remark 1, choosing the value of the “big-M” constant may lead to increasing complexity in terms of the computational effort. It is straightforward that the constraint (9a) cannot describe the entire $\mathbb{R}^n$ for $\alpha_i = 1$, but rather a domain including the region of interest, $\mathbb{X}$. Assuming, for instance, that the union of interdicted regions lies in a bounded cluttered environment $\mathbb{X}$, the value of M is obtained via the following LP:

$$M = \max_i M_i, \quad M_i = \max_{x \in \mathbb{X}} (k_i - h_i x).$$

Moreover, we may consider a different value of M for each half-space, that is, M_i . See the illustrative example in Fig. 1. The formalism can be further extended and the space is partitioned into cells using the hyperplane arrangement notion (Ziegler (2012)).

Definition 1. (Hyperplane arrangements) The collection $\mathbb{H}$ partitions the space into a union of disjoint cells $\mathcal{A}(\sigma)$, characterized by a sign tuple $\sigma \in \{-, +\}^N$:

⁵ In general (6) is non-convex with notable exceptions: the complement of an empty set, the complement of the unconstrained space or the complement of a half-space (possibly in a non-minimal representation).

⁶ Sets of indices.

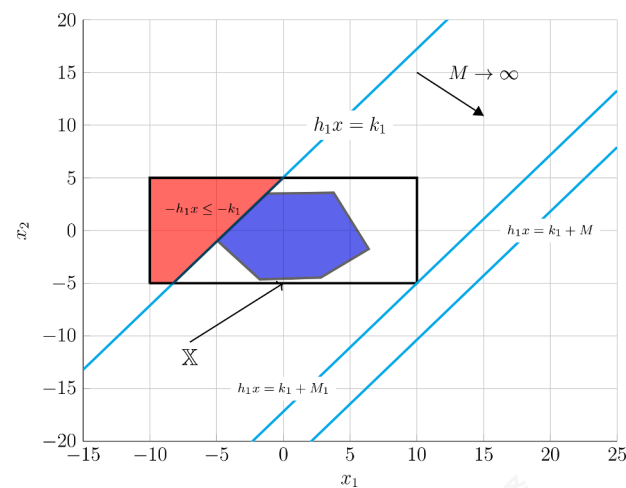


Fig. 1. Illustration “big-M”, M computed as in Remark 4. While the constraint is active (i.e., $\alpha_1 = 0$ in (9a)), the resulting inequality: $-h_1x \leq -k_1$ describes the highlighted region. Once $\alpha_1 = 1$ (inactive/relaxed constraint), the selection of M gives the measure of relaxation. Thus, the idea is to relax sufficiently such that the remainder of the constraints is not affected.

$$\mathcal{A}(\sigma) = \bigcap_{i \in \mathcal{I}} \mathcal{R}_i^{\sigma(i)}. \quad (10)$$

The hyperplane arrangement of cells covering the entire space is described by the collection of all feasible sign tuples:

$$\mathcal{A}(\mathbb{H}) = \bigcup_{l=1 \dots \gamma(N)} \mathcal{A}(\sigma_l), \quad (11)$$

where $\mathbb{H} = \{\mathcal{H}_i\}_{i \in \mathcal{I}}$, $\sigma_l \in \{-, +\}^N$ is the sign tuple resulting from a non-empty intersection of half-spaces and $\gamma(N)$ is the number of feasible cells.

♦ Consider a collection of obstacles (as in Fig. 2):

$$\mathbb{P} = \bigcup_{j=1}^N P_j$$

Briefly, gathering the collection of associated support hyperplanes defined as (3) we reach the hyperplane arrangement (11). Labeling the feasible cells (10) into interdicted $\Sigma_{\mathbb{P}} = \{\sigma : \mathcal{A}(\sigma) \cap \mathbb{P} \neq \emptyset\}$ or allowed $\Sigma_{\mathbb{X} \setminus \mathbb{P}} = \{\sigma : \mathcal{A}(\sigma) \cap \mathbb{P} = \emptyset\}$ one can obtain a mixed-integer characterization of the feasible domain:

$$h_i x \leq k_i + M(1 - \alpha_i), \quad (12a)$$

$$-h_i x \leq -k_i + M\alpha_i, \quad (12b)$$

$$\sum_{\sigma_l(i)=+} (1 - \alpha_i) + \sum_{\sigma_l(i)=-} \alpha_i > 0, \quad (12c)$$

$$\forall \sigma_l \in \Sigma_{\mathbb{P}} \quad (12d)$$

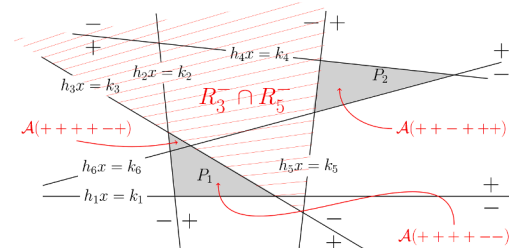


Fig. 2. Polytopic obstacles and hyperplane arrangements.

$$\mathcal{C}(P) = \mathcal{C}\left(\bigcap_{i \in \mathcal{I}} \mathcal{R}_i^-\right) = \bigcup_{i \in \mathcal{I}} \mathcal{C}(\mathcal{R}_i^-) = \bigcup_{i \in \mathcal{I}} \mathcal{R}_i^+. \quad (6)$$

该区域 (6) 是凸集的并集⁵, 为了便于描述, 我们可以采用混合整数技术。因此, 引入了二元变量 $(\alpha_1 \dots \alpha_N) \in \{0, 1\}^N$, 以在状态和辅助二元变量的扩展空间中获得多面体的表示:

$$h_i x \leq k_i + M\alpha_i, \quad i \in \mathcal{I}, \quad (7a)$$

$$\sum_{i \in \mathcal{I}} \alpha_i \leq N - 1, \quad (7b)$$

其中 M 是如注释 1 中所述的大 M 系数。

要求一个点在多面体集合 P 之外, 被转换为该点必须在定义多面体的至少一个半空间的补集中。

注释 3. 条件 (7a)–(7b) 通过对二元变量的适当组合来描述区域 (6)。例如, 区域 $\mathcal{R}_i^+$ 由 (7a) 和以下二元变量表示:

$$(\alpha_1 \dots \alpha_N) = \left(1 \dots 1 \underbrace{0}_i 1 \dots 1\right). \quad (8)$$

当二元变量取值为 “1” 时, 相关的不等式对于极限情况 ($M \rightarrow \infty$) 描述了整个域 $\mathbb{R}^n$ 。因此, 条件 (7b) 是必要的, 以确保至少有一个二元变量为 “0”, 从而至少有一个不等式保持有效。♦ 上述推理不仅限于一个凸多面体区域的补集, 并且可以推广到被禁止的区域是多个多面体的并集的情况, 如 (Stoican, Prodan, & Olaru, 2013) 中所述。明确地说, 非凸区域由以下特征描述:

$$-h_{il}x \leq -k_{il} + M\alpha_{il}, \quad \forall l \in \mathcal{I}_l, \quad (9a)$$

$$\sum_{l \in \mathcal{I}_l} \alpha_{il} \leq \#\mathcal{I}_l - 1. \quad (9b)$$

with $\mathcal{I}$ 在 (3) 中重新定义为离散区间的并集⁶: $\mathcal{I} = \bigcup_l \mathcal{I}_l$.

备注 4. 如备注 1 所述, 选择 “大-M” 常数的值可能会导致计算工作量的复杂性增加。很明显, 约束 (9a) 不能描述 $\alpha_i = 1$ 的整个 $\mathbb{R}^n$, 而只是包含感兴趣区域 $\mathbb{X}$ 的域。例如, 假设禁止区域的并集位于有界杂乱环境中 $\mathbb{X}$, 则通过以下线性规划获得 M 的值:

$$M = \max_i M_i, \quad M_i = \max_{x \in \mathbb{X}} (k_i - h_i x).$$

此外, 我们可以为每个半空间考虑不同的 M 值, 即 M_{i_0} 。参见图 1 中的说明性示例。该形式化方法可以进一步扩展, 并使用超平面排列概念将空间划分为单元 (Ziegler (2012))。

定义 1. (超平面排列) 集合 $\mathbb{H}$ 将空间划分为一系列不相交的单元 $\mathcal{A}(\sigma)$, 这些单元由一个符号元组 $\sigma \in \{-, +\}^N$ 表征

⁵ 通常情况下 (6) 是非凸的, 但有显著例外: 空集的补集、无约束空间的补集或半空间的补集 (可能以非最小表示形式存在)。⁶ 索引集。

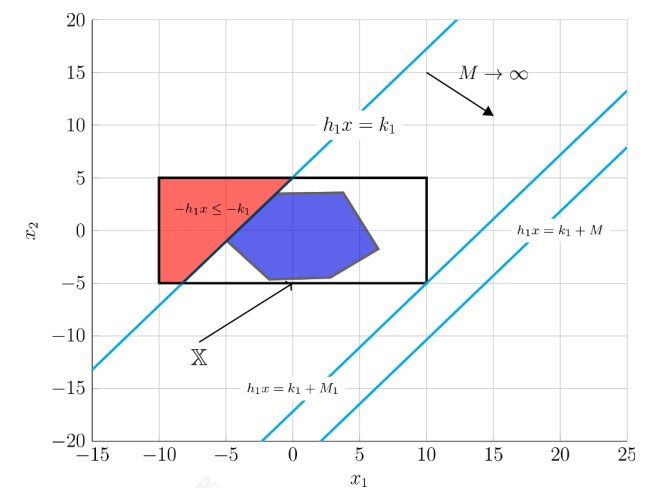


图1. “大M”方法说明, M 按注释4计算。当约束有效 (即 $\alpha_1 = 0$ 在 (9a) 中) 时, 所得不等式: $-h_1x \leq -k_1$ 描述了高亮区域。一旦 $\alpha_1 = 1$ (约束无效/松弛), M 的选择给出了松弛程度。因此, 思路是充分松弛, 以使剩余约束不受影响。

$$\mathcal{A}(\sigma) = \bigcap_{i \in \mathcal{I}} \mathcal{R}_i^{\sigma(i)}. \quad (10)$$

覆盖整个空间的所有单元的超平面排列由所有可行的符号元组的集合描述:

$$\mathcal{A}(\mathbb{H}) = \bigcup_{l=1 \dots \gamma(N)} \mathcal{A}(\sigma_l), \quad (11)$$

其中 $\mathbb{H} = \{\mathcal{H}_i\}_{i \in \mathcal{I}}$, $\sigma_l \in \{-, +\}^N$ 是由半空间非空交集产生的符号元组, $\gamma(N)$ 是可行单元的数量。♦ 考虑一组障碍物 (如图 2 所示):

$$\mathbb{P} = \bigcup_{j=1}^N P_j$$

简而言之, 通过定义 (3) 中关联支持超平面的集合, 我们得到超平面排列 (11)。将可行单元 (10) 标记为受限 $\Sigma_{\mathbb{P}} = \{\sigma : \mathcal{A}(\sigma) \cap \mathbb{P} \neq \emptyset\}$ 或允许 $\Sigma_{\mathbb{X} \setminus \mathbb{P}} = \{\sigma : \mathcal{A}(\sigma) \cap \mathbb{P} = \emptyset\}$, 即可获得可行域的混合整数特征:

$$h_i x \leq k_i + M(1 - \alpha_i), \quad (12a)$$

$$-h_i x \leq -k_i + M\alpha_i, \quad (12b)$$

$$\sum_{\sigma_l(i)=+} (1 - \alpha_i) + \sum_{\sigma_l(i)=-} \alpha_i > 0, \quad (12c)$$

$$\forall \sigma_l \in \Sigma_{\mathbb{P}} \quad (12d)$$

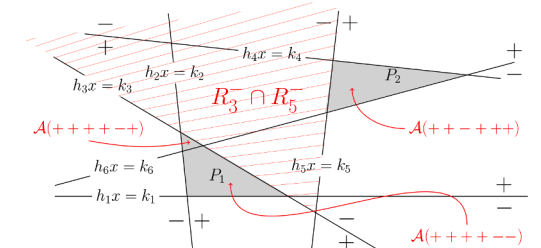


图2. 多面体障碍和超平面排列。

Even though the construction presented above is a generic one, the binary part of the representation is substantially large as we have a binary variable for each region (4b). There exists a variety of technical procedures like cell merging (Prodan et al., 2015) or logarithmic formulations (Vielma & Nemhauser, 2011) which may be employed to reduce the complexity of the formulation, a large number of obstacles and/or agents may still lead to an impractically large number of binary variables.

The representation of the interdicted domains is not limited to the polyhedral sets. For instance in Earl and D'Andrea (2005a), circular obstacles are considered as the one depicted in Fig. 3, each being determined by its radius R_o and by the coordinates of its center (x_o, y_o) :

$$\mathcal{O} = \{(x, y) : (x - x_o)^2 + (y - y_o)^2 \leq R_o^2\} \quad (13)$$

However, to incorporate the corresponding avoidance constraints, the obstacles as in (13) are approximated with polygons (polyhedra in $\mathbb{R}^2$, having p vertices). Hence, each interdicted zone is described by a set of K_p inequalities:

$$\tilde{\mathcal{O}} = \{(x, y) : (x - x_o) \sin \frac{2\pi m}{K_p} + (y - y_o) \cos \frac{2\pi m}{K_p} \leq R_o, \forall m = 1, \dots, K_p\} \quad (14)$$

Using the same reasoning as in (7) and the big-M technique, the avoidance constraints are formulated as:

$$(x - x_o) \sin \frac{2\pi m}{K_p} + (y - y_o) \cos \frac{2\pi m}{K_p} \geq R_o - M\beta_m, \forall m \quad (15a)$$

$$\sum_{m=1}^{K_p} \beta_m \leq K_p - 1. \quad (15b)$$

Remark 5. Besides the polytopic representation of the obstacles, some others representations are considered in the literature. For instance, Ioan, Prodan, Stoican, Oлару, and Niculescu (2019) considers additional structural constraints by the use of zonotopic sets (a particular class of polytopes with increased symmetry). The use of a such representation leads to a trade off between the complexity and the fidelity of the representation. ♦

It is worth to mention that restrictions within the state space can be directly treated in the dynamical system modeling. In particular is the case of the hybrid systems where switching (a disjunctive type of selection in between modes of functioning) include binary variables or discrete alternatives. The Piece-Wise Affine (PWA) models represent for

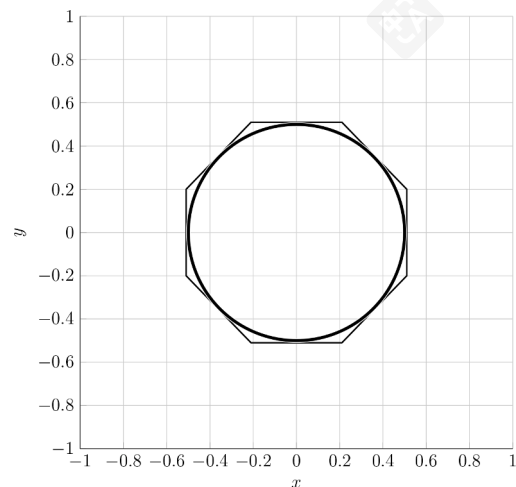


Fig. 3. A circular obstacle and its polytopic approximation.

example an integrated approach, where non-convex domains and local dynamics find a unitary representation. This unitary modeling needs to be differentiated from the classical motion planning framework where the dynamics are defined globally and the control decisions need to satisfy the non-convex feasible domain or disjunctive constraints. Because our review is focused rather on classical motion planning techniques, we mention some relevant papers, e.g. Bemporad and Morari (1999), Vielma and Nemhauser (2011), which provide an extensive presentation of the PWA modeling tools.

2.3. GDP formulation for region description

In order to highlight the relationship between GDP and MIP, consider the following illustrative example. We consider four supporting hyperplanes describing the interdicted region and use a GDP formulation.

Hence, corresponding to each hyperplane $i \in \{1, 2, 3, 4\}$ we have the next disjunction:

$$\left[\begin{array}{c} Y_{i1} \\ g_i(x) \leq 0 \end{array} \right] \vee \left[\begin{array}{c} Y_{i2} \\ g_i(x) \geq 0 \end{array} \right]$$

$$Y_{i1} \vee Y_{i2} \leftrightarrow Y_{i2} = \neg Y_{i1}$$

where $g_i(x) = x_1 \pm x_2 \pm 6$ and we associate: $Y_{i1} = \text{true}$ with $\mathcal{R}_i^-$ (the below half-space) and $Y_{i2} = \text{true}$ with $\mathcal{R}_i^+$ (the above half-space). The interdicted and the allowed region (both illustrated in Fig. 4) can be described using the following boolean function:

- convex region: $\Omega(Y) = \neg Y_{11} \wedge Y_{21} \wedge \neg Y_{31} \wedge Y_{41}$
- non-convex region: $\Omega(Y) = Y_{11} \vee \neg Y_{21} \vee Y_{31} \vee \neg Y_{41}$

The set-theoretic tools involved in MIP formulations were briefly sketched. For a more extensive presentation the interested reader is referred to the well-known works about polyhedral and hyperplane arrangement notions Ziegler (2012), Kuhn (1998) or to the more recent monographies, as Prodan et al. (2015), which offer a more detailed mathematical descriptions of the concepts presented here.

3. MIP in motion planning

The presence of MIP in motion planning can be originated by algebraic or geometrical approaches. The former relates to situations and circumstances, in which logical decisions are involved, e.g. the selection from a priori known set of possible alternatives. The later usually refers to the ability of MIP to describe non-convex constraints.

Prerequisites

Ahead of proceeding with typical MIP implementations in motion

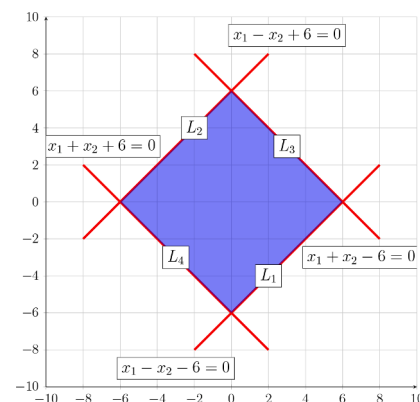


Fig. 4. The illustrative example for GDP formulation. $\{L_i\}_{i=1,4}$ hyperplanes.

尽管上述构建是通用的，但表示的二进制部分非常大，因为我们为每个区域都有一个二进制变量 (4b)。存在多种技术程序，如单元格合并 (Prodan 等人, 2015年) 或对数形式化 (Vielma & Nemhauser, 2011年)，可以用来降低公式的复杂性，但大量障碍和/或代理仍然可能导致不切实际数量的二进制变量。

对禁域的表示不仅限于多面体集合。例如在 Earl 和 D' Andrea (2005a) 中，圆形障碍物被视为图 3 所示的形状，每个障碍物由其半径 R_o 和中心坐标 (x_o, y_o) 确定

$$\mathcal{O} = \{(x, y) : (x - x_o)^2 + (y - y_o)^2 \leq R_o^2\} \quad (13)$$

然而，为了包含相应的避障约束，(13) 中的障碍物被近似为多边形 ($\mathbb{R}^2$ 中具有 p 个顶点的多面体)。因此，每个禁域区域由一组 K_p 个不等式描述：

$$\tilde{\mathcal{O}} = \{(x, y) : (x - x_o) \sin \frac{2\pi m}{K_p} + (y - y_o) \cos \frac{2\pi m}{K_p} \leq R_o, \forall m = 1, \dots, K_p\} \quad (14)$$

使用与 (7) 相同的推理和 big-M 技术，避障约束被表述为：

$$(x - x_o) \sin \frac{2\pi m}{K_p} + (y - y_o) \cos \frac{2\pi m}{K_p} \geq R_o - M\beta_m, \forall m \quad (15a)$$

$$\sum_{m=1}^{K_p} \beta_m \leq K_p - 1. \quad (15b)$$

备注 5. 除了障碍物的多面体表示外，文献中还考虑了其他一些表示方法。例如，Ioan, Prodan, Stoican, Oлару 和 Niculescu (2019) 通过使用 zonotopic 集合（一类具有更高对称性的多面体）来考虑额外的结构约束。使用这种表示方法会导致表示的复杂性和保真度之间进行权衡。

♦

值得提及的是，状态空间内的限制可以直接在动态系统建模中处理。特别是混合系统的情况，其中切换（一种在功能模式之间选择的不同断类型）包含二元变量或离散替代方案。分段仿射 (PWA) 模型代表

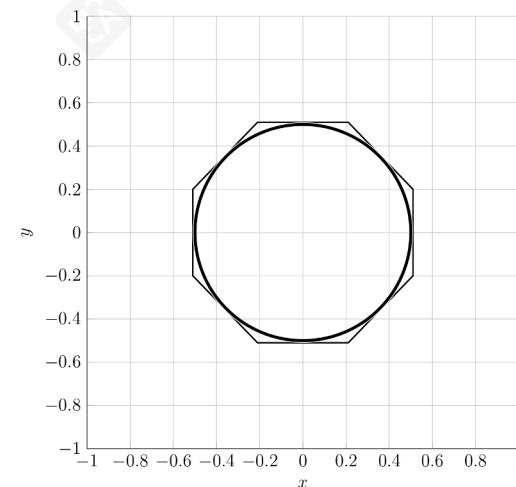


图3. 一个圆形障碍物及其多面体近似。

例如一种集成方法，其中非凸域和局部动力学找到一个统一的表示。这种统一建模需要与经典运动规划框架区分开来，在经典框架中，动力学是全局定义的，控制决策需要满足非凸可行域或不断约束。由于我们的综述更侧重于经典运动规划技术，我们提及了一些相关论文，例如 Bemporad 和 Morari (1999)、Vielma 和 Nemhauser (2011)，这些论文提供了对 PWA 建模工具的全面介绍。

2.3. 用于区域描述的 GDP 公式

为了突出 GDP 与 MIP 之间的关系，考虑以下示例。我们考虑四个支持超平面来描述被禁止的区域，并使用 GDP 公式。

因此，对于每个超平面 $i \in \{1, 2, 3, 4\}$ ，我们有以下析取式：

$$\left[\begin{array}{c} Y_{i1} \\ g_i(x) \leq 0 \end{array} \right] \vee \left[\begin{array}{c} Y_{i2} \\ g_i(x) \geq 0 \end{array} \right]$$

$$Y_{i1} \vee Y_{i2} \leftrightarrow Y_{i2} = \neg Y_{i1}$$

其中 $g_i(x) = x_1 \pm x_2 \pm 6$ ，我们将 Y_{i1} 与 $\mathcal{R}_i^-$ （下半个空间）关联，将 $Y_{i2} = \text{true}$ 与 $\mathcal{R}_i^+$ （上半个空间）关联。被禁止的区域和允许的区域（两者如图 4 所示）可以使用以下布尔函数来描述：

- convex region: $\Omega(Y) = \neg Y_{11} \wedge Y_{21} \wedge \neg Y_{31} \wedge Y_{41}$
- non-convex region: $\Omega(Y) = Y_{11} \vee \neg Y_{21} \vee Y_{31} \vee \neg Y_{41}$

MIP 公式中涉及的集合论工具简要介绍。对于更广泛的介绍，有兴趣的读者可以参考关于多面体和超平面排列概念的经典著作 Ziegler (2012)、Kuhn (1998)，或者参考更近期的专著，如 Prodan 等人 (2015)，这些专著提供了对这里介绍的概念更详细的数学描述。

3. MIP 在运动规划中的应用

MIP 在运动规划中的存在可以源于代数或几何方法。前者涉及涉及逻辑决策的情况和背景，例如从先验已知的可能替代集中进行选择。后者通常指的是 MIP 描述非凸约束的能力。

前提条件

在进行典型的运动规划 (MIP) 实现之前

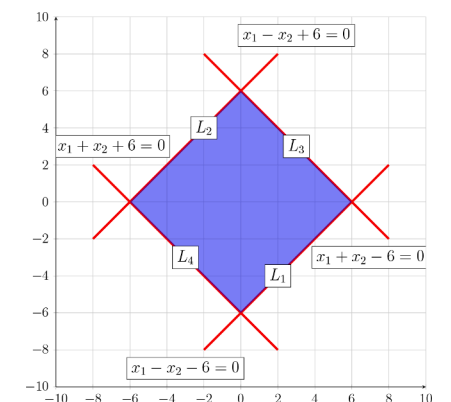


图4. GDP公式的说明示例。 $\{L_i\}_{i=1,4}$ 超平面。

planning, some clarifications are in order. Generally, for motion planning there exists plenty of applications that will be treated in detail in [Section 6](#). An important aspect is that all those applications have a dynamical behavior of a point of mass in 2D or 3D spaces governed by dynamical mathematical models (either linear or nonlinear).

Up front, let us consider a generic⁷ model of the controlled dynamical system designated by an ordinary differential equation (ODE):

$$\dot{x}(t) = f(x(t), u(t), w(t)) \quad (16)$$

where $x(t) \in \mathbb{R}^{n_x}$ denotes the state vector, $u(t) \in \mathbb{R}^{n_u}$ the control input vector, and $w(t) \in \mathbb{R}^{n_w}$ the disturbance.⁸ The mapping $f(\cdot, \cdot, \cdot) : \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \times \mathbb{R}^{n_w} \rightarrow \mathbb{R}^{n_x}$ is a continuous function admitting an equilibrium point (i.e., $f(\bar{x}, \bar{u}, 0) = 0$; without any loss of generality we may assume $f(0, 0, 0) = 0$).

Remark 6. The model (16) describes a dynamic in continuous time. As will be detailed below, a considerable part of the literature is based on discrete time dynamics. Hence, we need to consider, concurrently, keeping similar notations, the discrete time counterpart:

$$x(k+1) = f(x(k), u(k), w(k)) \quad (17)$$

The correlation between (16) and (17) is done using one of the various discretization techniques ([Sontag, 2013](#)). ♦ In what follows, a relevant feature of the previously-mentioned models is the way of selecting the input variables and consequently the decision domain. In the literature we can distinguish these choices with respect to the linearity/non-linearity of the models. Thus, for linear models the references concentrate towards the models in which the state is composed by the position and the speed of the agent and the input is represented by the acceleration of the agent, see, for example: [Bellingham, Richards, and How \(2002a\)](#); [Liu, Dai, Wu, and Lin \(2017\)](#); [Papen, Vandenhoeck, Bolting, and Defay \(2017\)](#); [Richards and How \(2002\)](#); [Yu, Wang, Bortoff, and Ueda \(2014\)](#). This type of model implies actuation on each direction and is usually based on the kinematics equations describing the behavior of the controlled systems. Indeed, there are several variations of this model derived from more complex modelisation techniques that do not represents the topic of this paper. Regarding the nonlinear models, the favored variant consists of models, in which the components of the state are the position and heading angle, and the input is represented by speed and steering angle, e.g., [Rey and Hijazi \(2017\)](#).

A usual approach is to do trajectory tracking which hides the nonlinearities induced by obstacle avoidance but moves the difficulty in the trajectory design step. This can be mitigated through the use of flatness (quite popular approach in this setting) have it. Still, computing offline a trajectory (or a path) assumes a known environment which may not be the case. Either linear or nonlinear, the models involved in motion planning are characterized by flatness. Generally speaking, a system is flat if there exists a set of variables (so-called, flat outputs) such that we can determine all states and inputs based on these variables without integration [Murray et al. \(2009\)](#). In [Martin, Murray, and Rouchon \(2002\)](#), the interested reader can find a detailed presentation of this type of system and how their properties are used in trajectory tracking or motion planning.

From the control perspective, the earliest methodologies have been based either on optimal control or on non-linear programming, but the controlled systems have operated in obstacle-free environment. Recently, this type of environment is no longer an appropriate assumption from practical reasons (especially, in a multi-agent context). Hence, the control community has had to propose some novel

techniques or to adapt the old ones. First adjustments consisted in restricting the controlled system to track predetermined trajectories around the obstacles, but the resulting control strategies were working only for particular systems and environments ([LaValle, 2006](#)).

MIP framework is *frequently* encountered within optimization-based control

While the focus of this manuscript is MIP and its various implementations, ultimately, the obtained formulation comes as the result of optimization-based control strategy. Thus, in order to provide a general overview of the MIP-based motion planning problems we briefly introduce some notions from optimal control theory.

Essentially, the optimal control problem ([Diehl, 2014](#); [Kirk, 2012](#)) consists in finding a control input $u^* \in \mathbb{R}^{n_u}$ such that the system (16) follows a trajectory $x^*(t) \in \mathbb{R}^{n_x}$, minimizing a specific criterion:

$$\min_{x(t), u(t)} J = \min_{x(t), u(t)} H(x(t_f)) + \int_{t_0}^{t_f} L(x(t), u(t)) dt \quad (18a)$$

$$s.t. \quad \dot{x}(t) - f(x(t), u(t), w(t)) = 0 \quad (18b)$$

$$x(t_0) - x_0 = 0 \quad (18c)$$

$$g(x(t), u(t)) \leq 0 \quad (18d)$$

where $t \in [t_0, t_f]$, $(t_f - t_0)$ denotes the time horizon length, x_0 the initial state, and $g(\cdot, \cdot)$ is a function incorporating the physical constraints (e.g., terminal or stage constraints of the controlled system). The OCP (optimal control problem) (18) can take various forms according to i) the choice of functions $H(\cdot)$ and $L(\cdot, \cdot)$, e.g. minimum time problem or terminal control problem [Kirk \(2012\)](#); ii) the selection of the horizon ($[t_0, t_f]$), e.g., [Garg et al. \(2011\)](#).

Standard MIP-based problems in motion planning

The motion planning problems ([LaValle, 2006](#)) are well established and have been studied in time optimal control, nonlinear control, stabilization, reachability and other related topics. To review all these topics is beyond the scope of this manuscript, but we retain those of them in which the presence of alternatives or discrete decisions making is explicitly considered. The emphasis is placed on the specific applications where non-convex feasible domains appear and are subsequently encoded by mixed-integer techniques.

In this section are presented the motion planning sub-areas, which can be efficiently formulated through a MIP approach:

- Task assignment (TA)** is the strategic decision of allocating one objective to a particular subsystem (*who-goes-where?*), these objectives can be interchanged and their suitability is measured by a specific criterion. TA is a discrete decision-making process where the number of alternatives within the pairs (resource objective) association is countable.
- (Collision-free) Path planning** is the construction of a route in the position space without an explicit parametrization in time and without explicitly considering the dynamics.
- (Collision-free) Trajectory planning** represents the problem of constructing a function which associates a time interval to a path. It takes into account the particularities of the agent's dynamics and the resulting function is generated in an open-loop manner.
- (Online) Obstacle and collision avoidance** is the problem of finding input control signals which minimize a performance criterion as in (18a) while simultaneously avoiding collisions and using a closed-loop strategy.

We have graphically illustrated [Fig. 5](#) the functional relationship among the items described above. A noteworthy aspect is that for the

需要做一些说明。通常，在运动规划中存在许多应用，将在第6节详细讨论。一个重要方面是，所有这些应用都具有一个由动力学数学模型（线性或非线性）控制的二维或三维空间中质点的动力学行为。

首先，让我们考虑一个通用的⁷控制动态系统模型，该系统由常微分方程 (ODE) 指定：

$$\dot{x}(t) = f(x(t), u(t), w(t)) \quad (16)$$

其中 $x(t) \in \mathbb{R}^{n_x}$ 表示状态向量， $u(t) \in \mathbb{R}^{n_u}$ 表示控制输入向量， $w(t) \in \mathbb{R}^{n_w}$ 表示干扰⁸ 映射 $f(\cdot, \cdot, \cdot) : \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \times \mathbb{R}^{n_w} \rightarrow \mathbb{R}^{n_x}$ 是一个连续函数，存在平衡点 (即, $f(\bar{x}, \bar{u}, 0) = 0$; 不失一般性，我们可以假设 $f(0, 0, 0) = 0$)。

注释 6. 模型 (16) 描述了连续时间动态。如下文将详细说明，文献中相当一部分基于离散时间动态。因此，我们需要同时考虑，保持相似的符号，离散时间的对应模型：

$$x(k+1) = f(x(k), u(k), w(k)) \quad (17)$$

(16) 和 (17) 之间的相关性是通过各种离散化技术之一 ([Sontag, 2013](#)) 完成的。♦ 在下文中，先前提到的模型的一个相关特征是选择输入变量的方式，以及随之而来的决策域。在文献中，我们可以根据模型的线性/非线性来区分这些选择。因此，对于线性模型，参考文献主要集中在状态由智能体位置和速度组成、输入由智能体加速度表示的模型中，例如：[Bellingham, Richards 和 How \(2002a\)](#); [Liu, Dai, Wu 和 Lin \(2017\)](#); [Papen, Vandenhoeck, Bolting 和 Defay \(2017\)](#); [Richards 和 How \(2002\)](#); [Yu, Wang, Bortoff 和 Ueda \(2014\)](#)。这种模型意味着每个方向上的驱动，并且通常基于描述受控系统行为的运动学方程。实际上，有几种这种模型的变体，它们是从更复杂的建模技术中推导出来的，但并不代表本文的主题。关于非线性模型，首选的变体包括状态分量是位置和航向角、输入由速度和转向角表示的模型，例如，[Rey 和 Hijazi \(2017\)](#)。

一种常见的方法是进行轨迹跟踪，这种方法会隐藏由避障引起的非线性，但将难度转移到了轨迹设计步骤。这可以通过使用平坦性（在此场景下是一种相当流行的做法）来缓解。然而，离线计算轨迹（或路径）假设环境是已知的，但这可能并不成立。无论是线性还是非线性，运动规划中涉及的模型都由平坦性特征。一般来说，如果一个系统存在一组变量（所谓的平坦输出），我们可以根据这些变量确定所有状态和输入，而无需积分 ([Murray 等人, 2009 年](#))。在 [Martin, Murray 和 Rouchon \(2002 年\)](#) 的著作中，感兴趣的读者可以找到关于这类系统的详细介绍，以及它们的特性如何在轨迹跟踪或运动规划中应用。

从控制角度来看，最早期的控制方法要么基于最优控制，要么基于非线性规划，但受控系统一直在无障碍环境中运行。最近，从实际原因来看（尤其是在多智能体环境中），这种环境不再是一个合适的假设。因此，控制领域不得不提出一些新颖的技术或对旧技术进行改进。最初的调整是将受控系统限制在绕过障碍物的预定轨迹上运行，但由此产生的控制策略仅适用于特定的系统和环境 ([LaValle, 2006](#))。

⁷ 文献中存在多种模型变体，但它们大多可以写成 (16) 指定的形式。⁸ 通常，扰动 $w(t) \in W$ 有界，且 $W \subset \mathbb{R}^{n_w}$ 是凸且紧的集合。

技术或对旧技术进行改进。最初的调整是将受控系统限制在绕过障碍物的预定轨迹上运行，但由此产生的控制策略仅适用于特定的系统和环境 ([LaValle, 2006](#))。

MIP 框架经常出现在基于优化的控制中控制

虽然本文的重点是 MIP 及其各种实现，但最终获得的公式是优化控制策略的结果。因此，为了概述基于 MIP 的运动规划问题，我们简要介绍一些最优控制理论的概念。

本质上，最优控制问题 ([Diehl, 2014](#); [Kirk, 2012](#)) 在于寻找一个控制输入 $u^* \in \mathbb{R}^{n_u}$ ，使得系统 (16) 遵循轨迹 $x^*(t) \in \mathbb{R}^{n_x}$ ，并最小化特定准则：

$$\min_{x(t), u(t)} J = \min_{x(t), u(t)} H(x(t_f)) + \int_{t_0}^{t_f} L(x(t), u(t)) dt \quad (18a)$$

$$s.t. \quad \dot{x}(t) - f(x(t), u(t), w(t)) = 0 \quad (18b)$$

$$x(t_0) - x_0 = 0 \quad (18c)$$

$$g(x(t), u(t)) \leq 0 \quad (18d)$$

其中 $t \in [t_0, t_f]$, $(t_f - t_0)$ 表示时间范围长度， x_0 为初始状态，而 $g(\cdot, \cdot)$ 是一个包含物理约束的函数（例如，受控系统的终端或阶段约束）。最优控制问题 (OCP) (18) 可根据 i) 函数 $H(\cdot)$ 和 $L(\cdot, \cdot)$ 的选择而呈现不同形式，例如最短时间问题或终端控制问题 ([Kirk, 2012](#))；ii) 时间范围的选择 ($[t_0, t_f]$)，例如 [Garg 等人 \(2011\)](#) 的研究。

基于标准MIP的运动规划问题

运动规划问题 ([LaValle, 2006](#)) 已非常成熟，并在最优控制、非线性控制、稳定化、可达性及其他相关主题中得到了研究。回顾所有这些主题超出了本文的范围，但我们保留了其中明确考虑了替代方案或离散决策制定的部分。重点放在那些非凸可行域出现并被混合整数技术编码的具体应用上。

本节介绍了可通过MIP方法高效表述的运动规划子领域：

- 任务分配 (TA) 是一种战略决策，将一个目标分配给特定的子系统（谁去哪里？），这些目标可以互换，其适用性由特定标准衡量。TA是一个离散决策制定过程，其中（资源目标）配对中的替代方案数量是可数的。
- (无碰撞) 路径规划是在位置空间中构建一条路线，不涉及显式的时间参数化，也不显式考虑动力学。
- (无碰撞) 轨迹规划代表构建一个函数的问题，该函数将时间区间与路径关联起来。它考虑了智能体动力学的特殊性，生成的函数以开环方式产生。
- (在线) 障碍物和碰撞避让是寻找输入控制信号的问题，这些信号在最小化性能指标（如18a）的同时，避免碰撞并采用闭环策略。

我们已在图5中图形化地说明了上述项目之间的功能关系。一个值得注意的方面是，对于

⁷ There exists in the literature several variation of models, but most of them can be written in the form designated by (16).

⁸ Usually, the disturbance is bounded $w(t) \in W$, and $W \subset \mathbb{R}^{n_w}$ is convex and compact set.

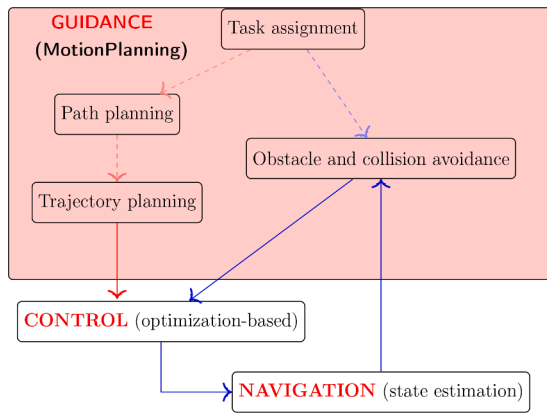


Fig. 5. Motion planning sub-areas and the link among them.

task assignment and path planning sub-areas the particularities of the dynamic have, generally, almost no influence in problem statement, in contrast with the other sub-areas. However, there are situations, in which the dynamic impacts on the definition of static costs. Besides, TA and path planning are not tackled in most of the motion planning literature as stand-alone topics, but together with or as part of the others sub areas. As depicted in Fig. 5 (the dotted lines), path planning is usually implicitly included in trajectory planning and TA is either a part of obstacle avoidance or included in path planning.

As well, collision-free trajectory planning and collision avoidance represent dual approaches, the difference between them being the manner of interaction with navigation and control. While the obstacle avoidance intervenes into the navigation-control loop, the trajectory planning only generates the reference (to be tracked) for this loop.

As a side remark, a considerable part of the control community includes also the formation control in motion planning sub-areas, although it represents, in fact, a control level which gathers all the above sub-areas of motion planning, and, as TA, it makes sense only in a multi-agent context. Therefore, we separately treat this topic in Section 4, where connectivity maintenance problems in multi-agent systems are presented.

Remark 7. Before detailing the motion planning problems, it is necessary to highlight the differences between the path and trajectory planning. There exists a wealth of manuscripts which treat this topic in detail (see, e.g., Beard and McLain (2012) or Yang, Qi, Xiao, and Yong (2014)). The choice between these two methodologies depends on particularities of the controlled systems, e.g. for a fixed wing UAV, a trajectory planning methodology can have some undesired consequences (Beard & McLain, 2012). These particularities are broadly presented and discussed in Section 6. However, there does not exist in the literature a generally accepted differentiation between these two concepts. Hence, for the ease of presentation we opted to consider the definitions as in ii), and iii) respectively. ♦

In Table 1 we depicted the classification of the references with respect to the motion planning problem predominantly treated therein.

Remark 8. In general, a pre-condition of the MIP approaches is the existence of an awareness map which entails the global knowledge about the environment. Thus, representation of the environment has a significant impact on the performances of the motion planning strategies (except the task assignment problem where the impact is not crucial). ♦

3.1. Task assignment

In what follows, we present a standard MIP formulation for the task assignment problem, emphasizing its importance in the context of motion planning for a multi-agent system.

Table 1
Classification of the research in motion planning using MIP.

Applications	References (alphabetical order)
Task assignment	Alighanbari et al. (2003); Bethke et al. (2008); Chen et al. (2016a); Chen and Wang (2005); Fayazi et al. (2017); Haghighi et al. (2016); Liu et al. (2017); Richards and How (2005); Wang et al. (2015b); Zhang et al. (2017a, 2017b)
Path planning	Janeček et al. (2017a); Ragi and Mittelmann (2017); Richards and How (2005, 2002); Schouwenaars (2006); Wang et al. (2015b)
Trajectory planning	Ballesteros-Tolosana et al. (2017); Cetin et al. (2007); Deits and Tedrake (2015b); Earl and D'Andrea (2002, 2005a, 2005b); Fayazi et al. (2017); Huang and Peng (2017); Zhang et al. (2017b)
Obstacles and collision avoidance	Bali and Richards (2018); Bellingham et al. (2002a); Haghighi et al. (2016); Janeček et al. (2017a); Mukai et al. (2017); Rey and Hijazi (2017); Richards and How (2002); Richards and Turnbull (2013); Wang et al. (2015b)

Task assignment refers to a strategic decision of allocating one objective to a particular subsystem (*it responds to the question “who-goes-where?”*).

At large, a task assignment algorithm must provide an assignment which optimizes a specific criterion. Consider, for instance, the undesired occurrence of an earthquake, which affects 10 buildings. An emergency committee must assign for each building a rescue team, in order to minimize the intervention time, taking into account the number of rescue teams and their necessary time to arrive at respective buildings. This example can be easily reformulated for the case of target tracking in a multi-agent system.

More generally, let us consider N agents and M targets. In order to minimize the amount of time to reach all the targets, we should optimally assign agents to targets. To this end, we define a binary variable x_{ij} , which equals one if the agent i visits the target j and zero otherwise. Also, we consider a cost associated to each combination agent-target: c_{ij} . We take into account that each agent must attain at least one target, and, at the same time, each target must be reached by an agent (constraints (19b) and (19c), respectively). Thus, we obtain a standard MIP formulation of task assignment problem, described in (19).

$$\min \sum_{i=1}^N \sum_{j=1}^M c_{ij} x_{ij} \quad (19a)$$

$$\text{s.t.} \sum_{i=1}^N x_{ij} \geq 1, \forall j \quad (19b)$$

$$\sum_{j=1}^M x_{ij} \geq 1, \forall i \quad (19c)$$

$$x_{ij} \in \{0, 1\}, \forall i, j \quad (19d)$$

Since we have an integer requirement on the variables x_{ij} , the problem (19) belongs to the class of ILP (integer linear programming) (a subclass of MIP, where all the variables are integer.) A noteworthy aspect is given by a special characteristic of (19): solving its LP relaxation⁹ allows obtaining the solution of ILP original problem. Nonetheless, if the task assignment problem is part of an enhanced control design, this property will be lost and the problem should be solved in the generic class of (M)ILP. For instance, let us define a variable t_{ij} which represents the necessary time for agent i to reach the target j . We impose that the total time to not exceed a certain value (20).

⁹ We replace (19d) with the relaxation $x_{ij} \in [0, 1]$

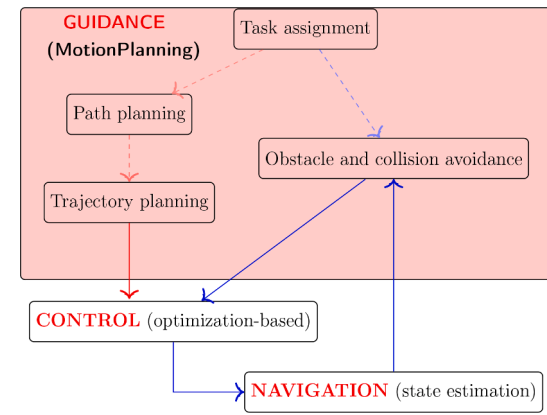


图5. 运动规划子领域及其相互关系。

对于任务分配和路径规划子领域, 动态特性的特殊性通常几乎不影响问题描述, 与其他子领域形成对比。然而, 在某些情况下, 动态特性会影响静态成本的定义。此外, 在大多数运动规划文献中, 任务分配和路径规划通常不被作为独立主题处理, 而是与其他子领域一起或作为其一部分。如图5 (虚线所示), 路径规划通常隐含在轨迹规划中, 而任务分配要么是避障的一部分, 要么包含在路径规划中。

同样地, 无碰撞路径规划和碰撞避免代表两种方法, 它们之间的区别在于与导航和控制交互的方式。虽然障碍物避免会干预导航-控制回路, 但路径规划仅为此回路生成参考 (待跟踪)。

作为旁注, 控制领域相当一部分也包括运动规划子领域的形成控制, 尽管它实际上代表了一个汇集了所有上述运动规划子领域的控制级别, 并且作为TA, 它只在多智能体环境中才有意义。因此, 我们在第4节中单独讨论这个主题, 在那里介绍了多智能体系统中的连接维护问题。

备注 7. 在详细阐述运动规划问题之前, 有必要强调路径规划和轨迹规划之间的区别。存在大量详细探讨此主题的手稿 (例如, 参见 Beard 和 McLain (2012) 或 Yang, Qi, Xiao 和 Yong (2014))。这两种方法之间的选择取决于受控系统的特性, 例如, 对于固定翼无人机, 轨迹规划方法可能会产生一些不希望的后果 (Beard & McLain, 2012)。这些特性在第六节中得到了广泛介绍和讨论。然而, 在文献中并不存在对这些两个概念的一般性区分。因此, 为了便于表述, 我们选择分别考虑 ii)、iii) 中的定义。♦

在表1中, 我们描绘了参考文献的分类, 这些参考文献主要涉及运动规划问题。

注释8. 一般来说, MIP方法的一个前提条件是存在一个意识地图, 该地图包含关于环境的全局知识。因此, 环境的表示对运动规划策略的性能有显著影响 (除了任务分配问题, 其影响不关键)。♦

3.1. 任务分配

在以下内容中, 我们针对任务分配问题提出了一种标准的MIP模型, 并强调了其在多智能体系统运动规划背景下的重要性。

表1
使用MIP的运动规划研究分类。

应用	参考文献 (按字母顺序)
任务分配	Alighanbari 等人 (2003); Bethke 等人 (2008); Chen 等人 (2016a); Chen 和 Wang (2005); Fayazi 等人 (2017); Haghighi 等人 (2016); Liu 等人 (2017); Richards 和 How (2005); Wang 等人 (2015b); Zhang 等人 (2017a, 2017b)
路径规划	Janeček 等人 (2017a); Ragi 和 Mittelmann (2017); Richards 和 How (2005, 2002); Schouwenaars (2006); Wang 等人 (2015b)
轨迹规划	Ballesteros-Tolosana 等人 (2017); Cetin 等人 (2007); Deits 和 Tedrake (2015b); Earl 和 D'Andrea (2002, 2005a, 2005b); Fayazi 等人 (2017); Huang 和 Peng (2017); Zhang 等人 (2017b)
障碍物与避障	Bali 和 Richards (2018); Bellingham 等人 (2002a); Haghighi 等人 (2016); Janeček 等人 (2017a); Mukai 等人 (2017); Rey 和 Hijazi (2017); Richards 和 How (2002); Richards 和 Turnbull (2013); Wang 等人 (2015b)

任务分配指的是将一个目标分配给特定子系统的战略性决策 (它回答了“谁去哪里?”的问题)。

总体而言, 任务分配算法必须提供一个优化特定标准的分配方案。例如, 考虑地震的发生, 它影响了10栋建筑。紧急委员会必须为每栋建筑分配救援队伍, 以最小化干预时间, 同时考虑救援队伍的数量及其到达各自建筑所需的时间。这个例子可以很容易地改写成多智能体系统中的目标跟踪情况。

更一般地, 让我们考虑 N 个智能体和 M 个目标。为了最小化所有目标的总到达时间, 我们应该优化地将智能体分配给目标。为此, 我们定义一个二元变量 x_{ij} , 当智能体 i 访问目标 j 时等于1, 否则等于0。此外, 我们考虑每个智能体-目标组合的成本: c_{ij} 。我们注意到每个智能体必须至少达到一个目标, 同时每个目标必须被一个智能体达到 (约束条件 (19b)和(19c)分别)。因此, 我们得到任务分配问题的标准MIP公式, 如(19)所述。

$$\min \sum_{i=1}^N \sum_{j=1}^M c_{ij} x_{ij} \quad (19a)$$

$$\text{s.t.} \sum_{i=1}^N x_{ij} \geq 1, \forall j \quad (19b)$$

$$\sum_{j=1}^M x_{ij} \geq 1, \forall i \quad (19c)$$

$$x_{ij} \in \{0, 1\}, \forall i, j \quad (19d)$$

由于变量 x_{ij} 有整数要求, 因此问题 (19) 属于整数线性规划 (ILP) 的范畴 (MIP 的一个子类, 其中所有变量都是整数)。(19) 的一个值得注意的方面是其特殊性质: 求解其 LP 松弛⁹ 可以得到原始 ILP 问题的解。然而, 如果任务分配问题是增强控制设计的一部分, 这种性质将丢失, 并且问题应该在通用的 (M)ILP 类中求解。例如, 让我们定义一个变量 t_{ij} , 它表示智能体 i 到达目标 j 所需的时间。我们要求总时间不超过某个值 (20)。

⁹ 我们用松弛变量 $x_{ij} \in [0, 1]$ 替换 (19d)

$$\sum_{i=1}^N \sum_{j=1}^M t_{ij} x_{ij} \leq T \quad (20)$$

In specific situations, LP relaxations may lead to an integer solution.

Furthermore there are many others variations of the task assignment problem (19). For example, in [Smith and Taskin \(2008\)](#) the authors considered a similar problem but they introduce in the cost a non-linear term corresponding to the case when an agent has to attain more than one target. Using some refinements (detailed therein) they posed the problem as a MILP, even if the formulation is somewhat more complicated compared with (19).

In [Alighanbari et al. \(2003\)](#) and in [Bethke, Valenti, and How \(2008\)](#) the task assignment problem for a fleet of UAV is extensively treated. The initial requirements are more complicated than the formulation in (19) in the sense that the targets are transformed in waypoints and each agent must visit a sequence of these. Moreover, some restrictions as the existence of coupled tasks or of tight timing constraints increase significantly the complexity of the formulation.

A further example is found in [Chen, Shih, and Tomlin \(2016a\)](#) where a variation of the task assignment problem is used to provide higher level control logic to synthesize a cooperative safety controller. Actually, the objective is to group the agents in pairs such that the corresponding avoidance maneuvers may not lead to a dangerous configuration for the other agents. Thus, the theoretical guarantees given by the lower level controller are extended for a greater number of agents.

Similar approaches are used for optimal scheduling at intersections where a prioritization among agents is necessary. The scheduling problem is a variation of the task assignment problem, while the tasks are assimilated to time slots or time instants. This variation is one of the few resource allocation problems in which the dynamic behaviour is not completely ignored. For example, in [Fayazi, Vahidi, and Luckow \(2017\)](#) the agent's dynamic impacts on the objective function by introducing a non-linearity in the cost. This non-linearity is eliminated using a MIP formulation as in [Smith and Taskin \(2008\)](#).

MILP formulations are NP-hard (Non-deterministic Polynomial-time hard) ([Van Leeuwen & Leeuwen, 1990](#)).

Given the fact that MILP formulations are NP-hard, the control community has a preference for solving this type of problems (task assignment or resource allocation) through heuristic methods, e.g., in [Alighanbari et al. \(2003\)](#) a comparison between the Tabu search and the MIP formulation is done, and, for higher dimensions the Tabu search exhibits better performance. However, the tangible theoretical advantage of the MIP formulation (19) is that it guarantees the global optimum. This has a substantial importance in some critical situations.

Remark 9. In a similar manner as the task assignment, the notorious travelling-salesman problem (TSP) appears first time in [Dantzig et al. \(1954\)](#) posed as MILP. For the classical TSP the formulation (19) is used by converting the constraints (19b) and (19c) into equalities and imposing that $N = 1$ and $M \geq 2$. ♦

Besides task assignment, a plethora of resource allocation problems can be readily formulated as MIP. Many of these problems are used in several applications within related motion planning domains such as aircraft maintenance (or repair) [Bajestani and Beck \(2013\)](#), crew scheduling (and flight retiming) [Mercier and Soumis \(2007\)](#) etc.

There exist approaches building on the idea of state space (work-space) organization in terms of available resources. More precisely, these approaches partition the environment in equally-shaped cells and the cells are viewed as a shared resource. In this manner, the collision and obstacle avoidance problems are transformed in resource allocation with mutual exclusion.

The collision avoidance problems can be viewed as resource allocation with mutual exclusion.

For instance, in [Wang, Kloetzer, Mahulea, and Silva \(2015b\)](#) the collision avoidance constraints are given by the fact that each cell cannot be visited by two distinct agents at the same time, i.e. the agents do not use the same resource simultaneously. The formulation is an extension of the task assignment problem (19), the novelty comes from the evaluation and resolution of the MIP in an online manner.

Another reference using this “resource allocation” approach is [Haghighi, Sadraddini, and Belta \(2016\)](#) where a robotic swarm has the objective of respecting specific and complex patterns in a bidimensional workspace. These patterns are described using the spatio-temporal logic (SpaTeL). SpaTeL consists in propositions indicating the number of agents which can access a cell at a certain time. These SpaTeL formulas are converted in mixed-integer linear constraints, resulting a formulation comparable with the task assignment (19) but more complex. A similar approach is employed in [Liu et al. \(2017\)](#), but the difference is given by the inclusion of the communication constraints in the motion planning problem under the form of signal temporal logic (STL).

3.2. Path planning

As stated earlier in [Section 3](#), a path planning strategy involves the generation of a route without an explicit parametrization¹⁰. Further, this route is called the planned path.

Path planning refers to the construction of a route in the position space without an explicit parametrization in time and explicitly considering the system dynamics.

Definition 2. A path between two points x_0, x_f from the navigation space is given by a map $\gamma : [0, 1] \rightarrow \mathcal{X}$ with

$$\gamma(0) = x_0, \gamma(1) = x_f \quad (21)$$

♦ There are in the literature two major classes of methods to describe the function γ . The first one, the explicit methodology, consists in providing the value of γ for the entire interval $[0, 1]$ [Janeček, Klaučo, Kalúz, and Kvasnica \(2017a\)](#). The second class of methods is the most used alternative, the function has to take their values from the given set of, so-called, waypoints [Afonso, Galvão, and Kienitz \(2013\)](#).

Remark 10. The waypoints represent either the nodes of graph in a dynamic-free setting, or the boundary conditions for intervals of time. ♦

For further use, we define the set of all waypoints describing the planned path: $\mathcal{W} = \{x_{w_0} \dots x_{w_{N_w}}\}$ and consider an additional condition to [Def. 2](#): $\forall x_{w_k}, \exists! \theta_k \in [0, 1]$ such that $\gamma(\theta_k) = x_{w_k}$.

The methodologies involved in the selection and determination of the waypoints are outside the scope of this paper, although they are succinctly presented in [Section 7](#). In the path planning problem, MIP is not employed in the generation of the path, but rather to address the problem of temporal distribution of the waypoints (i.e., their ordering along the path), satisfying some criteria. In other words, MIP plays an instrumental role for optimizing a path obtained with one of the well-established techniques from the robotic field (e.g., sampled based methods - see [Section 7](#)). The criterion of this optimization procedure has to include/cover costs given by the physical and “economical¹¹” limitation.

One of the most used approaches treating this type of problem is based on a particular case of the task assignment (19), the travelling-salesman problem (TSP). Under this form we can identify the MIP usage in path planning and a classical example in this sense is to be found in [Richards and How \(2002\)](#), [Schouwenaars \(2006\)](#). Therein, they

¹⁰ in the sense that the time is not explicitly associated

¹¹ In the sense of the economical MPC.

$$\sum_{i=1}^N \sum_{j=1}^M t_{ij} x_{ij} \leq T \quad (20)$$

在特定情况下，LP 松弛可能直接得到整数解。

此外，任务分配问题还有许多其他变体 (19)。例如，在 [Smith and Taskin \(2008\)](#) 中，作者们考虑了一个类似的问题，但他们引入了一个非线性项到成本中，对应于当智能体需要达成多个目标的情况。通过一些改进（详见文中），他们将问题表述为一个 MILP，尽管与 (19) 相比，该公式稍微复杂一些。

在 [Alighanbari 等人 \(2003\)](#) 和 [Bethke, Valenti 及 How \(2008\)](#) 中，针对无人机队的任务分配问题进行了广泛研究。初始要求比 (19) 中的公式更复杂，因为目标被转换为航点，并且每个智能体必须访问这些航点的一个序列。此外，一些限制条件，如耦合任务的存在或严格的时序约束，会显著增加公式的复杂性。

另一个例子见于 [Chen, Shih 和 Tomlin \(2016a\)](#)，其中任务分配问题的某种变体被用于为合成协同安全控制器提供更高层级的控制逻辑。实际上，目标是将智能体成对分组，以便相应的避障动作不会导致其他智能体处于危险配置。因此，由底层控制器给出的理论保证被扩展到更多智能体的情况。

类似的方法也用于交叉路口的最优调度，其中需要智能体之间的优先级排序。调度问题可以看作是任务分配问题的变体，而任务被整合为时间段或时间点。这种变体是少数几个在资源分配问题中未完全忽略动态行为的例子之一。例如，在 [Fayazi, Vahidi 和 Luckow \(2017\)](#) 中，智能体的动态行为通过在成本函数中引入非线性来影响目标函数。这种非线性通过 MIP 公式（如 [Smith 和 Taskin \(2008\)](#) 中所述）被消除。

MILP模型是NP难（非确定性多项式时间难）的（[Van Leeuwen & Leeuwen, 1990](#)）。

鉴于MILP模型是NP难的，控制领域倾向于通过启发式方法解决这类问题（任务分配或资源分配），例如，在[Alighanbari等人 \(2003\)](#)的研究中，对禁忌搜索和MIP模型进行了比较，并且，在更高维度下，禁忌搜索表现出更好的性能。然而，MIP模型 (19) 的显著理论优势在于它能保证全局最优。这在某些关键情况下具有重要意义。

注释9。与任务分配类似，臭名昭著的旅行商问题（TSP）首次出现在 [Dantzig等人 \(1954\)](#) 的研究中，并以MILP形式提出。对于经典TSP，使用模型 (19)，通过将约束 (19b) 和 (19c) 转换为等式，并要求 $N = 1$ 和 $M \geq 2$ 。♦

除了任务分配外，许多资源分配问题都可以方便地表述为混合整数规划（MIP）。其中许多问题被应用于相关的运动规划领域中的多个应用，例如飞机维护（或维修）[Bajestani和Beck \(2013\)](#)、机组调度（和航班重排）[Mercier和Soumis \(2007\)](#) 等。

存在一些基于可用资源的状态空间（工作空间）组织理念的方法。更精确地说，这些方法将环境划分为形状相同的单元格，并将单元格视为一种共享资源。通过这种方式，碰撞和避障问题被转化为具有互斥约束的资源分配问题。

避障问题可以被视为具有互斥的资源分配。

例如，在王、克洛策尔、马胡莱亚和席尔瓦（2015b）的研究中，避障约束是由每个单元格不能同时被两个不同代理访问这一事实给出的，即代理不会同时使用同一资源。该模型是任务分配问题（19）的扩展，其新颖之处在于以在线方式评估和解决MIP。

另一个使用这种“资源分配”方法的参考文献是[Haghighi, Sadraddini和Belta \(2016\)](#)，其中一群机器人有目标在二维工作空间中尊重特定的复杂模式。这些模式使用时空逻辑（SpaTeL）进行描述。SpaTeL由表示在特定时间有多少个智能体可以访问某个单元格的命题组成。这些SpaTeL公式被转换为混合整数线性约束，形成与任务分配（19）类似但更复杂的公式。类似的方法也应用于Liu等人（2017）的工作，但区别在于将通信约束以信号时序逻辑（STL）的形式纳入运动规划问题中。

3.2. 路径规划

如第3节所述，路径规划策略涉及生成一条无显式参数化的路径¹⁰。此外，该路径被称为规划路径。

路径规划是指在位置空间中构建一条路线，时间上无显式参数化，并明确考虑系统动力学。

定义2. 导航空间中两点 x_0, x_f 之间的路径由一个映射 $\gamma : [0, 1] \rightarrow \mathcal{X}$ 给出，满足

$$\gamma(0) = x_0, \gamma(1) = x_f \quad (21)$$

在文献中，描述函数 γ 的方法主要有两大类。第一种是显式方法，它为整个区间 $[0, 1]$ 提供 γ 的值 [Janeček, Klaučo, Kalúz, and Kvasnica \(2017a\)](#)。第二类方法是使用最广泛的选择，函数必须从给定的所谓航路点集合中获取其值 [Afonso, Galvão, and Kienitz \(2013\)](#)。

备注10。航点要么表示无动态设置中的图节点，要么表示时间区间的边界条件。♦

为后续使用，我们定义描述规划路径的所有航点的集合： $\mathcal{W} = \{x_{w_0} \dots x_{w_{N_w}}\}$ 并考虑对定义2的附加条件： $\forall x_{w_k}, \exists! \theta_k \in [0, 1]$ 使得 $\gamma(\theta_k) = x_{w_k}$ 。航点的选择和确定所涉及的方法论超出了本文的范围，尽管它们在第七节中简洁地介绍了。在路径规划问题中，MIP不用于路径生成，而是用于解决航点的时间分布问题（即它们沿路径的顺序），满足某些标准。换句话说，MIP在优化使用机器人领域的一种成熟技术（例如，基于采样的方法——见第七节）获得的路径时发挥着关键作用。该优化过程的准则必须包括/涵盖由物理和“经济¹¹”限制给出的成本。

处理这类问题最常用的方法之一是基于任务分配（19）的一个特例，即旅行商问题（TSP）。在这种形式下，我们可以识别出MIP在路径规划中的应用，而在这个意义上，一个经典的例子可以在[Richards和How \(2002\)](#)、[Schouwenaars \(2006\)](#) 中找到。在那里，他们

¹⁰ 在意义上，时间并未明确关联¹¹ 在经济学MPC的意义上。

deal with a robot which should visit N_w waypoints while minimizing the cost of the maneuver. Hence, the MI constraints which enforce the visiting of the considered waypoints are the following:

$$\forall i \in \{1, \dots, T\}, \forall k \in \{1, \dots, N_w\}, \quad (22a)$$

$$\|x_i - x_{w_k}\| \leq M(1 - b_{ik}), \quad (22b)$$

$$\sum_{i=1}^T b_{ik} = 1, \forall k \text{ and } b_{ik} \in \{0, 1\}, \quad (22c)$$

where T is the number of time steps, and b_{ik} is a binary variable that indicates whether or not the waypoint k was visited at the time step i . The constraint (22c) ensures that each of the waypoints is visited once by the agent. The ordering of the waypoints along the interval depends on the selected criterion (e.g., the minimum time to visit all the waypoints). In practice, the attainability of the waypoints is affected by various factor and, thus, in most of the situations, reaching a vicinity of the waypoint is a reasonable objective. In such cases, (22) needs to be adapted, e.g., the constraint (22b) is replaced by $H_k(x_i - x_{w_k}) \leq h_k + M(1 - b_{ik})$, where H_k and h_k are given by the supporting hyperplanes describing the considered vicinity region.

Remark 11. In the majority of sampled-based methods LaValle (2006), the problem of finding the shortest path is tackled using one of the various algorithms (Dijkstra, A* are common choices). However, in Taccari (2016) several MIP formulations for the elementary shortest path problem are introduced as extensions of TSP. ♦

Besides the approach relying on (22), there exist a few others works, which treat MIP-based path planning in a different manner. For instance, in Vitus, Pradeep, Hoffmann, Waslander, and Tomlin (2008), a Tunnel-MILP approach is employed. Basically, this algorithm divides the global motion planning problem in three main tasks. The first task consists in finding a path as in Def. 2, while ignoring the vehicle's dynamics constraints. Next, the path is used along with a convex decomposition of the space to generate a sequence of N_R convex polytopes from the start to the goal: $F_i = \{x \in \mathbb{R}^2 : A_i x \leq b_i\}$, $\forall i \in \{1, \dots, N_R\}$. In the third task, this sequence (a tunnel of polytopes) is used to constrain the position of the agent $p(t)$ to an optimal and dynamically feasible route from the initial point to the goal inside the tunnel. More precisely, a MILP problem is employed, forcing the agent to remain in one of the regions of the tunnel at all times. This leads to some OR-constraints¹², similar with the GDP formulation (1), which can be readily formulated using MI techniques:

$$\bigvee_{i \in \{1, \dots, N_R\}} A_i p(t) \leq b_i, \forall t \quad (23)$$

where $p(t)$ is the position of the considered agent.

The same idea is used in Afonso et al. (2013), where the procedure to find the path relies on generalized Voronoi diagrams. Here, a number of points is considered: on the facets of the obstacles (modeled as convex bounded sets) and on the boundaries of the navigation environment in order to obtain a Voronoi graph. Subsequently, the nodes of the graph that are inside the prohibited region are removed along with the edges that connected these nodes to the other ones in the graph. What then remains is a graph whose edges form paths that present the nice property of being equally between the obstacles which are the closest to an edge and the edges themselves. This property comes from the nature of the Voronoi graph construction, which maximizes the minimum distance to all points. The graph is augmented with the initial and final position, preserving its connectivity. After that, using a Delaunay triangulation the space is divided into triangles whose intersection is empty, except for their sides. This triangulation relies on the vertices of the obstacles and

of the known region. The triangles that are crossed by the planned path are computed and merged to form convex polytopes, attempting to have the minimal number of convex regions along the tunnel. These polytopes are used further to impose the obstacle avoidance constraints as in the tunnel-MILP formulation (23).

In a similar approach, Deits and Tedrake (2015b) treats the problem of path planning in a multi-obstacle environment for a rotary-wing UAV. The underlying idea is based on a mixed-integer optimization which assigns polynomial trajectories to convex regions (known to be obstacle free). The paths are defined as piece-wise polynomial functions in time with vector-valued coefficients. Having an a priori (offline) chosen degree of polynomials and number of pieces, the optimization problem returns the coefficients of each polynomial which ensure a collision-free trajectory. This polynomial parametrization is possible because the model of the considered system¹³ is characterized by differential flatness. Regarding the obstacle-free convex regions, they are obtained offline using IRIS (Iterative Regional Inflation by Semidefinite programming Deits & Tedrake (2015a)), a technique for greedy convex segmentation of the free space. The proposed approach is tested for various simulations scenarios with satisfactory results, although sometimes the convex segmentation does not cover the entire obstacle-free space. Moreover, a comparison with the classical¹⁴ MIP method is done, the second method may involve a more complex integer program. An interesting aspect of this work is that the generated paths are dependent on the considered convex safe region segmenting the space.

Besides the above presented works, there exists others which treat, but only marginally, the problem of path planning. For example, in Janeček et al. (2017a) or in Janeček, Klaučo, and Kvasnica (2017b) a toolbox is introduced and it includes various helpers to generate a reference path. Thus, a circular trajectory of a known radius or a polygonic reference passing through the given set of waypoints are provided, but the use of MIP is kept to a minimum.

3.3. Trajectory planning

The trajectory planning pertains to the problem of determining both the path and how to move along it. Thus, a trajectory planning strategy returns a path which is explicitly parametrized in time. In this section we consider first the works dealing with general problems and after that we focus on classical applications in the field of autonomous vehicles (i.e., autonomous overtaking, merging junctions or lane changing).

Trajectory planning refers to the construction of a function generated in open-loop, which associates a time to each point on the path and takes into account the particularities of the system dynamics.

Definition 3. A trajectory between two points x_0, x_f from the navigation space is given by the continuous function $\gamma : [t_0, t_f] \rightarrow \mathcal{X}$ with

$$\gamma(t_0) = x_0, \gamma(t_f) = x_f. \quad (24)$$

♦ As the path planning task, the specification of the continuous function describing the trajectory can be done in two different ways. For the second one, the set of all waypoints describing the trajectory is modified as follows: $\tilde{\mathcal{W}} = \{(x_{w_0}, t_{w_0}) \dots (x_{w_{N_w}}, t_{w_{N_w}})\}$ and an additional condition to the Def. 3 is considered: $\gamma(t_{w_k}) = x_{w_k}, \forall x_{w_k}$. Nevertheless, the waypoint method is rarely used in MIP-based trajectory planning (and in the related literature). Hence, whenever it is not specified otherwise, the trajectory is given explicitly as the continuous function $\gamma(t)$.

¹³ The nano-quadcopter Crazyflie 2.0.

¹⁴ From the authors point of view: use the obstacle faces ("hyperplanes") directly in the generation of safe region.

处理一个需要访问 N_w 个航点的机器人，同时最小化操作成本。因此，强制访问所考虑航点的MI约束如下：

$$\forall i \in \{1, \dots, T\}, \forall k \in \{1, \dots, N_w\}, \quad (22a)$$

$$\|x_i - x_{w_k}\| \leq M(1 - b_{ik}), \quad (22b)$$

$$\sum_{i=1}^T b_{ik} = 1, \forall k \text{ and } b_{ik} \in \{0, 1\}, \quad (22c)$$

其中 T 是时间步数， b_{ik} 是一个二元变量，表示在时间步 i 时航点 k 是否被访问。约束 (22c) 确保每个航点都被智能体访问一次。航点沿区间的顺序取决于所选标准（例如，访问所有航点的最短时间）。在实际中，航点的可达性受多种因素影响，因此，在大多数情况下，到达航点附近是一个合理的目标。在这种情况下，需要调整 (22)，例如，将约束 (22b) 替换为 $H_k(x_i - x_{w_k}) \leq h_k + M(1 - b_{ik})$ ，其中 H_k 和 h_k 是由描述所考虑的附近区域的支撑超平面给出的。

备注11. 在大多数基于采样的方法中（LaValle, 2006），寻找最短路径的问题通常使用各种算法之一来解决（Dijkstra, A* are是常见的选择）。然而，在Taccari (2016) 中，引入了几个用于基本最短路径问题的MIP公式，作为TSP的扩展。♦

除了依赖 (22) 的方法之外，还存在一些其他工作，它们以不同的方式处理基于MIP的路径规划。例如，在Vitus、Pradeep、Hoffmann、Waslander和Tomlin (2008) 中，采用了一种Tunnel-MILP方法。基本上，该算法将全局运动规划问题分为三个主要任务。第一个任务包括根据定义2寻找一条路径，同时忽略车辆的动力学约束。接下来，该路径与空间的一个凸分解一起使用，从起点到目标生成一个由 N_R 个凸多面体组成的序列： $F_i = \{x \in \mathbb{R}^2 : A_i x \leq b_i\} \forall i \in \{1, \dots, N_R\}$ 。在第三个任务中，该序列（一个多面体隧道）用于将智能体 $p(t)$ 的位置约束在隧道内从初始点到目标的最优且动态可行的路径上。更具体地说，采用了一个MILP问题，迫使智能体始终保持在隧道的某个区域内。这导致了一些与GDP公式 (1) 相似的OR约束¹²，可以使用MI技术轻松地表示：

$$\bigvee_{i \in \{1, \dots, N_R\}} A_i p(t) \leq b_i, \forall t \quad (23)$$

其中 $p(t)$ 表示所考虑智能体的位置。

Afonso 等人 (2013) 中也使用了相同的思想，其中寻找路径的步骤依赖于广义 Voronoi 图。这里，考虑了一系列点：在障碍物的面上（建模为有界凸集）以及导航环境的边界上，以获得 Voronoi 图。随后，将图中位于禁止区域内的节点以及连接这些节点与其他节点的边移除。剩下的图，其边形成路径，具有一个很好的特性：这些路径在距离最近障碍物和边本身之间保持等距。这一特性源于 Voronoi 图构建的本质，即最大化到所有点的最小距离。图中添加了初始和最终位置，同时保持其连通性。之后，使用 Delaunay 三角剖分将空间划分为交为空（除边以外）的三角形。这种三角剖分依赖于障碍物的顶点和

已知区域的顶点。被规划路径穿过的三角形被计算并合并，形成凸多面体，试图在隧道中沿路径保持最少的凸区域。这些多面体进一步用于施加障碍物避让约束，如隧道-MILP 公式 (23) 中所示。

采用类似方法，Deits 和 Tedrake (2015b) 研究了在多障碍环境中为旋翼无人机进行路径规划的问题。其核心思想基于混合整数优化，将多项式轨迹分配给凸区域（已知无障碍物）。路径被定义为时间上的分段多项式函数，具有向量值系数。通过预先（离线）选择多项式的阶数和分段数量，优化问题会返回确保无碰撞轨迹的多项式系数。这种多项式参数化是可能的，因为所考虑系统的模型¹³ 具有微分平坦特性。关于无障碍凸区域，它们通过离线使用 IRIS（迭代区域膨胀，基于半定规划，Deits & Tedrake (2015a)）技术获得，这是一种用于自由空间贪婪凸分割的技术。该方法在各种仿真场景中进行了测试，结果令人满意，尽管有时凸分割无法覆盖整个无障碍空间。此外，还与经典的¹⁴ MIP方法进行了比较，后者可能涉及更复杂的整数规划。这项工作的一个有趣之处在于，生成的路径取决于分割空间的凸安全区域段。

除了上述所介绍的研究工作外，还存在其他仅部分涉及路径规划问题的研究。例如，在Janeček等人的研究（2017a）或Janeček、Klaučo和Kvasnica（2017b）中，介绍了一种工具箱，其中包含多种辅助工具用于生成参考路径。因此，该工具箱提供了已知半径的圆形轨迹或穿过给定一系列航点的多边形参考路径，但MIP（混合整数规划）的使用被严格限制。

3.3. 轨迹规划

轨迹规划涉及确定路径以及如何沿该路径移动的问题。因此，轨迹规划策略会返回一个随时间显式参数化的路径。在本节中，我们首先考虑处理一般问题的研究工作，然后重点关注自动驾驶领域中的经典应用（即自动驾驶超车、汇入交叉路口或变道）。

轨迹规划是指构建一个在开环中生成的函数，该函数将时间与路径上的每个点关联起来，并考虑系统动力学的特性。

定义3. 给定导航空间中两点 x_0, x_f 之间的轨迹，由连续函数 $\gamma : [t_0, t_f] \rightarrow \mathcal{X}$ 给出

$$\gamma(t_0) = x_0, \gamma(t_f) = x_f. \quad (24)$$

♦ 作为路径规划任务，描述轨迹的连续函数的规范可以通过两种不同的方式完成。对于第二种方式，描述轨迹的所有航路点的集合按如下方式修改： $\tilde{\mathcal{W}} = \{(x_{w_0}, t_{w_0}) \dots (x_{w_{N_w}}, t_{w_{N_w}})\}$ ，并且考虑了定义3的一个附加条件： $\gamma(t_{w_k}) = x_{w_k}, \forall x_{w_k}$ 。然而，航路点方法在基于MIP的轨迹规划（以及相关文献）中很少使用。因此，除非另有说明，否则轨迹被明确地给出为连续函数 $\gamma(t)$ 。

¹³ 纳米四旋翼Crazyflie 2.0。¹⁴ 从作者的观点来看：直接在安全区域的生成中使用障碍面（“超平面”）。

¹² 解释为异或。

Trajectory planning in various tasks

One of the classical works treating the problem of trajectory planning is [Earl and D’Andrea \(2005a\)](#) which introduces an iterative MILP algorithm. More precisely, considering circular obstacles (see [Section 2.2, Eq. \(13\)](#)) and a traditional nonholonomic (car-like) vehicle, the authors propose an algorithm which guarantees obstacle avoidance over the entire trajectory and distributes avoidance times efficiently, resulting in smaller MILP formulations.

The proposed algorithm needs the discretization in time of the continuous nonholonomic model. An interesting aspect is that the non-uniform discretizations are exploited in order to bypass an extensive computational effort corresponding to the resolution of large MILPs. Support for non-uniform discretizations in time allows the use of intelligent time step selection algorithms for the generation of more efficient MILP formulations. Thus, the idea behind of the iterative MILP algorithm is to find a convenient distribution of the avoidance times, avoiding in this manner the growing of the specific MIP problem complexity (induced by the enumeration of the constraints at each sampling time). Therein, two different problems are considered: trajectory-generation with obstacle-avoidance requirements and minimum-time trajectory-generation problems.

Iterative MILP algorithms address the issues of MILP coping with large-scale models.

The work of [Earl and D’Andrea \(2002\)](#) studies the cooperative control of multi-vehicle systems. Building on the requirements of a robotic competition (but in a simplified version), the authors model these simplified competition rules using a hybrid system. Further, they put the control problem within an optimization framework, using MILP. The considered simplified competition involves two teams of robots, the attackers and the defenders, on a playing field with a region at its center called the defense zone. The attackers are drones directed toward the defense zone. The objective for the defenders is to block the attackers from entering the defense zone by intercepting each attacker before it enters the zone. Once an attacker enters the defense zone or is intercepted by a defender it remains stationary for the rest of the competition. While pursuing its objective defenders must avoid collisions with other defenders and obstacles as well as avoid entering the defense zone which is off limits to defending robots. The control strategy is implemented with a centralized controller with perfect knowledge of the system, perfect access to all states, and with the ability to transmit control signals to the defenders instantaneously. The controller needs to figure out the inputs to provide each defending robot so that the objective is achieved. Using MPC, they obtained a set of control inputs that minimize the number of attackers that enter the defense zone over the duration of the drill and, in addition, is consistent with the system dynamics (robot dynamics) and the constraints (no collisions, etc.). The obtained model is a system composed of continuous and discrete states (a hybrid system) with linear dynamics subject to inequality constraints and logical rules. In other words they obtained a MLD (mixed-logical dynamical) system ([Bemporad & Morari, 1999](#)). The resulting optimization problem is a MILP which can be readily solved using one of the state-of-the-art solvers (e.g., ILOG CPLEX).

Multi-robot systems can be modeled as hybrid (MLD) systems using MIP techniques.

Also, in [Earl and D’Andrea \(2005a\)](#) iterative MILP algorithms are presented, algorithms which address the issue of the MILP coping with large-scale models. The considered problem is trajectory-generation with obstacle-avoidance requirements and minimum-time trajectory-generation problems. These problems involve vehicles that are described by mixed logical dynamical [Bemporad and Morari \(1999\)](#) equations, a form of hybrid system. The algorithms use fewer binary variables than standard MILP methods, and require less computational effort. The iterative methods presented in that paper apply to MLD

systems, which are governed by a mixture of logical rules (or state machines) and linear dynamical equations and are validated using a vehicle in an obstacle field, a particular MLD system.

In [Earl and D’Andrea \(2005b\)](#) the results in [Earl and D’Andrea \(2002\)](#), and [Earl and D’Andrea \(2005a\)](#) are gathered and extended. Hence, the multi-vehicle control problems is tackled using MILP and MLD systems [Bemporad and Morari \(1999\)](#). As in [Earl and D’Andrea \(2002\)](#), the methods are motivated on problems derived from an adversarial game between two teams of robots. The strategy for one team is fixed and modeled by state machines, while for the other the behavior is controlled using the iterative MILP method. On a side note, the approach therein was developed independently from a similar one introduced in [Richards and How \(2002\)](#). One of the interesting aspects regarding this paper is that the considered environment involves an adversarial component whose intelligent behavior is modeled using finite state machines and, implicitly, MLD systems.

As well, the trajectory planning topic is treated in [Cetin, Bikdash, and Hadaegh \(2007\)](#). Therein, the addressed problem is the generation of collision-free trajectories for the reconfiguration of spacecraft formations, aiming also for an optimal fuel consumption. In order to model the spacecraft representation and their corresponding safety regions unrotated cubes (a particular polytopic representation) are used. Moreover, the trajectories to be followed are discretized in time using cubic splines and, thus, the generic problem is translated to an optimization problem. The resolution of this problem led to a trajectory parameterized by the spacecraft positions and velocities at a set of waypoints. To this end, the “big-M” technique is used to write the parametrized optimization as a MILP whose solution can be obtained either using standard MILP solvers (see [Section 6](#)) or using the concept of a sequential linear program. These two alternatives are compared therein, being applied on two standard validation tests whose aim is to swap the position within a spacecraft with minimum fuel consumption. The comparison leads to an ample discussion on the feasibility of MILP and on the methods necessary to shorten the resolution time.

Also, in [Richards and How \(2003\)](#) MILP has been used for open-loop vehicle trajectory design, enabling the inclusion of non-convex constraints such as plume impingement avoidance.

Trajectory planning in autonomous vehicles tasks

Classical applications in trajectory planning involve in most of the cases either merging junctions problem (e.g., [Bali and Richards \(2018\)](#); [Huang and Peng \(2017\)](#)) or autonomous overtaking (e.g., [Ballesteros-Tolosana, Oлару, Rodriguez-Ayerbe, Pita-Gil, and Deborne \(2017\)](#); [Molinari, Anh, and Re \(2017\)](#)).

Recent advances in the field of autonomous vehicles bring to light the issues like vehicle lane change and overtaking on highways in the assisted driving framework. In order to perform such maneuvers, it is fundamental to compute suitable and comfortable trajectories that take into account the vehicle limitations as well as safety restrictions.

Safety restriction and vehicle limitations require efficient MIP formulations.

For instance, in [Molinari et al. \(2017\)](#) an efficient MIP formulation for the autonomous overtaking problem is introduced. The considered vehicles have a simple dynamic model, the non-convex feasible region being represented using hyperplanes arrangement similar with [Prodan et al. \(2015\)](#)(as in [Section 2](#)). A complete formulation for trajectory generation with collision avoidance guarantees is presented for the case of an agent surrounded by a number of agents. The MPC criterion contains a desired reference state, which ensures that the overtaking takes place. They also considered two methods for reducing the numbers of binary variables: logarithmic formulation and cell merging ([Stoican et al., 2013](#)). The illustrative example therein and a comparative analysis between the two reduction methods from complexity point of view, are validated in IPG CarMaker (an accurate vehicle simulator).

各种任务中的轨迹规划

处理轨迹规划问题的一个经典工作是Earl和D’ Andrea (2005a), 该工作介绍了一种迭代MILP算法。更具体地说, 考虑圆形障碍物 (参见第2.2节, 公式(13)) 和传统的非完整 (汽车式) 车辆, 作者提出了一种算法, 该算法在整个轨迹上保证避障, 并有效地分配避障时间, 从而得到更小的MILP公式。

所提出的算法需要对连续非完整模型进行时间离散化。一个有趣的特点是, 利用非均匀离散化来绕过与求解大型MILPs相对应的大量计算工作。对非均匀时间离散化的支持允许使用智能时间步长选择算法来生成更高效的MILP公式。因此, 迭代MILP算法背后的想法是找到一个方便的避障时间分布, 以这种方式避免特定MIP问题复杂性的增长 (由在每个采样时间枚举约束引起)。在此, 考虑了两个不同的问题: 具有避障要求的轨迹生成问题和最小时间轨迹生成问题。

迭代MILP算法解决了MILP处理大规模模型的问题。

Earl和D’ Andrea (2002) 的工作研究了多车辆系统的协同控制。基于机器人比赛的要求 (但简化版), 作者使用混合系统对这些简化比赛规则进行建模。此外, 他们将控制问题置于优化框架内, 使用MILP。所考虑的简化比赛涉及两队的机器人, 即攻击者和防守者, 在一个有中心防御区的比赛场上。攻击者是朝向防御区的无人机。防守者的目标是阻止攻击者进入防御区, 方法是拦截每个攻击者, 在它进入区域之前。一旦攻击者进入防御区或被防守者拦截, 它在比赛剩余时间内保持静止。在追求其目标的同时, 防守者必须避免与其他防守者和障碍物发生碰撞, 以及避免进入防御区, 该区域对防守机器人来说是禁止区域。控制策略由一个集中式控制器实现, 该控制器对系统具有完美知识、对所有状态具有完美访问权限, 并且能够瞬间向防守者传输控制信号。控制器需要确定提供给每个防守机器人的输入, 以实现目标。使用MPC, 他们获得了一组控制输入, 这些输入在演练期间最小化了进入防御区的攻击者数量, 并且此外还与系统动力学 (机器人动力学) 和约束 (无碰撞等) 一致。获得的模型是一个由连续和离散状态组成的混合系统 (混合逻辑动态系统), 具有线性动力学, 受不等式约束和逻辑规则的约束。换句话说, 他们获得了一个MLD (混合逻辑动态) 系统 (Bemporad & Morari, 1999)。得到的优化问题是MILP, 可以使用最先进的求解器之一 (例如, ILOG CPLEX) 轻松求解。

多机器人系统可以使用MIP技术建模为混合 (MLD) 系统。

此外, 在Earl和D’ Andrea (2005a) 中介绍了迭代MILP算法, 这些算法解决了MILP处理大规模模型的问题。所考虑的问题是带避障要求的轨迹生成和最小时间轨迹生成问题。这些问题涉及由混合逻辑动力学Bemporad和Morari (1999) 方程描述的車輛, 这是一种混合系统。这些算法使用的二进制变量比标准MILP方法更少, 且计算量更小。该论文中提出的迭代方法适用于MLD系统, 这些系统由逻辑规则 (或状态机) 和线性动力学方程混合支配, 并使用一个在障碍场中的车辆 (一个特定的MLD系统) 进行验证。

系统, 这些系统由逻辑规则 (或状态机) 和线性动力学方程混合支配, 并使用一个在障碍场中的车辆 (一个特定的MLD系统) 进行验证。

在Earl和D’ Andrea (2005b)中, Earl和D’ Andrea (2002)以及Earl和D’ Andrea (2005a)中的结果被收集和扩展。因此, 多车辆控制问题使用MILP和MLD系统Bemporad和Morari (1999)得到解决。如在Earl和D’ Andrea (2002)中, 这些方法基于从两个机器人团队之间的对抗性游戏中推导出问题。一个团队的策略是固定的, 并由状态机建模, 而另一个团队的行为则使用迭代MILP方法进行控制。顺便一提, 该方法是在独立于Richards和How (2002)中介绍的一个类似方法的情况下开发的。关于这篇论文的一个有趣方面是, 所考虑的环境包含一个对抗性组件, 其智能行为使用有限状态机和隐式MLD系统进行建模。

此外, Cetin, Bikdash 和 Hadaegh (2007) 也研究了轨迹规划问题。其中, 所解决的问题是为航天器编队重构成无碰撞轨迹, 同时目标是最优燃料消耗。为了对航天器表示及其对应的安全区域进行建模, 使用了未旋转的立方体 (一种特殊的凸多面体表示)。此外, 要遵循的轨迹在时间上被离散化, 使用三次样条曲线, 从而将通用问题转化为优化问题。该问题的求解导致轨迹参数化为航天器在一系列航路点处的位置和速度。为此, 使用了“大-M”技术将参数化优化问题写成混合整数线性规划 (MILP), 其解可以通过标准的 MILP 求解器 (见第 6 节) 或使用顺序线性规划的概念获得。这两种方法在此进行了比较, 并在两个标准验证测试中应用, 目标是在最小燃料消耗的情况下交换航天器内的位置。比较结果引发了关于 MILP 可行性的广泛讨论, 以及缩短求解时间所需方法的讨论。

此外, 在Richards和How (2003) 的研究中, 混合整数线性规划 (MILP) 已被用于开环车辆轨迹设计, 从而能够包含非凸约束, 例如羽流冲击避免。

自动驾驶车辆任务的轨迹规划

轨迹规划的经典应用通常涉及合并交叉口问题 (例如, Bali 和 Richards (2018); Huang 和 Peng (2017)) 或自动驾驶超车 (例如, Ballesteros-Tolosana, Oлару, Rodriguez-Ayerbe, Pita-Gil 和 Deborne (2017); Molinari, Anh 和 Re (2017))。

自动驾驶领域最近的进展凸显了辅助驾驶框架下车道变换和高速公路超车等问题。为了执行这些操作, 计算适合且舒适的轨迹至关重要, 这些轨迹需要考虑车辆限制以及安全约束。

安全约束和车辆限制需要高效的 MIP 公式。

例如, 在Molinari等人 (2017) 的研究中, 介绍了一种用于自主超车问题的有效MIP模型。所考虑的车辆具有简单的动态模型, 非凸可行区域使用类似于Prodan等人 (2015) (如第2节所述) 的超平面排列来表示。针对被多个智能体包围的情况, 提出了包含碰撞避免保证的轨迹生成完整模型。MPC准则包含期望参考状态, 确保超车能够顺利进行。他们还研究了两种减少二元变量数量的方法: 对数形式化和单元合并 (Stoican等人, 2013)。其中的说明性示例以及从复杂度角度对两种减少方法的比较分析, 在IPG CarMaker (一个精确的车辆模拟器) 中得到了验证。

In Ballesteros-Tolosana et al. (2017), the lane change and overtaking maneuvers aim to generate trajectories able to ensure the comfort of passengers. Mathematically, the problem is formulated as an Optimal Control Problem (OCP) due to its handling kinematic and collision avoidance constraints (the last ones in terms of hyperplane arrangements as in Section 2). In order to mitigate the drawback of the substantial computational effort associated with that kind of representation, a pre-analysis step is presented therein, a step consisting in the enumeration of the all possible overtaking configurations and of their resulting compact MIP formulation. The so-obtained non-linear constrained optimal control problem is solved using a multiple-shooting approach which leads to improvements in the computational burden when compared to the pre-analysis step.

The presence of MIP in merging junctions problem can be found in a plethora of works. For instance, Fayazi et al. (2017) deal with the optimal scheduling of autonomous vehicle arrivals at intersection, eliminating the need of traffic signals. The idea is to design an intersection controller able to coordinate the flow of vehicles through intersection, scheduling the intersection access in safe conditions and receiving information from all subscribing vehicles. Moreover, each vehicle informs the controller about its movement and desired schedule. Therefore, the optimization-based controller should find the optimal sequence of vehicles crossing the intersection and their corresponding time of intersection access, minimizing the difference between the current time and the expected arrival time of the last vehicle passing the intersection and, also, the gap between assigned and desired access time of each vehicle. The resulting optimization problem has to take into account some physical constraints (speed limit and maximum acceleration) and a safety “window” between two consecutive vehicles accessing the intersection. These constraints lead to disjunctions and, respectively if-then statements, which are modeled through MIP (big-M formulation). In addition to this, the total optimization cost is composed of a min-max objective and a non-linear cost (an absolute value sign).

MIP may be employed to eliminate the non-linearity (Smith & Tas-kin, 2008).

Likewise, Huang and Peng (2017) tackled the problem of speed trajectory planning at signalized intersection. The idea is to optimize the vehicle speed trajectory over multiple intersections in order to minimize the fuel consumption and the travel time. In order to obtain a mathematical formulation of the problem, the authors assume that the signal traffic state is known and the influence of a lead vehicle is ignored. The main advantage of this formulation (with respect to, e.g., Fayazi et al. (2017)) is that they have considered the turnings at intersections (including a turning speed constraint). After a comprehensive description of the fuel consumption model, effect of turning and acceleration model, a MIP formulation is stated. The presence of integer variables models the crossing of the intersection without violating red light, indicating the active green phase window.

Also, Zhang, Su, Sun, and Zhang (2017b) presents a control strategy for a heterogeneous traffic network. Heterogeneity comes from the presence of both signalized and non-signalized¹⁵ intersections. After the validation of the proposed model of the heterogeneous traffic network, a control strategy is stated using a MIP formulation. Finally, some comparisons between homogeneous and heterogeneous systems are provided in order to leave the door open for developing a systematic planning approach on deciding what traffic junctions require signal control to ensure a good traffic control performance. Moreover, Zhang, Su, Gao, and Zhang (2017a) proposes a traffic signal scheduling strategy which takes into account both vehicles and pedestrians presence. After developing a mathematical model for the pedestrians flow and for the

vehicle traffic *network* (more than one intersection/junction), the problem of scheduling is stated as a MIQP (mixed-integer quadratic program).

3.4. Obstacle and collision avoidance

Obstacle and collision avoidance is the problem of finding input control signals which minimize a performance criterion as in (18a) while simultaneously avoiding collisions through a closed-loop strategy..

An adequate collision and obstacle avoidance strategy has become critical in order to ensure the safety and the integrality of both the system and the environment. In the last years, the control community has put the effort towards the obtaining of collision-free control strategies, which permit real-time operations. It is generally accepted that the collision avoidance problem is a challenging task due to the presence of the non-convex feasible domain. Describing this non-convex domain has computational and structural implications, generally leading to a trade-off between computational efficiency and control performance. Since MIP is able to explicitly model the non-convexity as in Section 2, it has become over time a proper approach for expressing collision avoidance problem.

The MIP approaches in the field of obstacle and collision avoidance problems can be distinguished in two main categories. The first class of MIP approaches is more often used, being a wide-spread approach in non-MIP (robot) motion planning literature. It consists in choosing a partitioning method which does not depend on the particularities of the environment (e.g the obstacles’ form). A standard partitioning is the one formed by equally-sized square cells (Haghighi et al., 2016; Wang et al., 2015b). Using this approach, the collision-avoidance problem becomes, in most of the cases, a resource allocation problem, as presented in the Section 3.1. The second MIP approach or the geometric approach relies on set theory and on the capability of mixed-integer techniques to efficiently encode the description of non-convex sets (see Section 2). Even though different, both approaches share a common characteristic: the workspace is decomposed in a certain kind of cells and in some manner the MIP is employed in modeling disjunctive constraints.

Remark 12. In the literature there are works which make a distinction between obstacle and collision avoidance. More precisely, the collision avoidance refers only to the mutual collision avoidance within multi-agent systems, and obstacle avoidance may refer to the collision avoidance either with stationary obstacles or with moving obstacles which have an a priori known trajectory, in most of the cases a periodic one. As well, some others references consider that obstacle avoidance refers to the static environment, while collision avoidance covers any kind of moving obstacles. ♦

Static obstacles

The work of Culligan (2006) presents a path planner using MILP to solve a receding horizon optimization problem for unmanned aerial vehicles (UAVs). The MILP formulation has two important components: the hard constraints for obstacle and multi-vehicle avoidance and an approximation of vehicle dynamics. The 3D case is, as well, considered and comprehensively presented. Moreover, some enhancements of the MILP framework are discussed in order to provide decreased resolution time and also increase the capability of the path planner. These improvements consist of various techniques like a variable time step size, linear interpolation points, and horizon minimization. A noteworthy aspect, the concept of variable time steps is extended to the receding horizon, non-iterative MILP formulation. Variable time step sizing allows the simulation horizon time to be lengthened without increasing resolution time dramatically. Horizon minimization decreases resolution time by removing unnecessary obstacle constraints from the problem (similar with Janeček et al. (2017a)).

In Ballesteros-Tolosana等人 (2017) 的研究中, 车道变换和超车操作旨在生成能够确保乘客舒适性的轨迹。从数学上讲, 由于需要处理运动学和碰撞避免约束 (后者以第2节所述的超平面排列形式表示), 该问题被表述为一个最优控制问题 (OCP)。为了减轻与这种表示方法相关的显著计算成本这一缺点, 研究中提出了一个预处理步骤, 该步骤包括枚举所有可能的超车配置及其产生的紧凑MIP模型。由此得到的非线性约束最优控制问题采用多射击方法求解, 与预处理步骤相比, 该方法在计算负担方面有所改进。

MIP 在合流交叉口问题中的存在可以在大量作品中找到。例如, Fayazi 等人 (2017) 处理了自动驾驶车辆在交叉口的最优调度, 消除了对交通信号的需求。该想法是设计一个能够协调车辆通过交叉口的流量、在安全条件下调度交叉口访问并从所有订阅车辆接收信息的交叉口控制器。此外, 每辆车都会向控制器报告其运动和期望的调度。因此, 基于优化的控制器应找到车辆穿过交叉口的最优序列及其相应的交叉口访问时间, 最小化当前时间与最后通过交叉口的车辆预期到达时间之间的差异, 以及每辆车分配访问时间与期望访问时间之间的差距。由此产生的优化问题必须考虑一些物理约束 (速度限制和最大加速度) 以及连续两辆车访问交叉口之间的安全 “窗口”。这些约束导致不相交和相应的 if-then 语句, 通过 MIP (大-M 公式) 进行建模。除此之外, 总优化成本由一个 min-max 目标和一个非线性成本 (绝对值符号) 组成。

MIP 可用于消除非线性 (Smith & Tas-kin, 2008)。

类似地, 黄和彭 (2017) 解决了信号交叉口的路径规划问题。其思路是通过优化多个交叉口的车辆速度轨迹, 以最小化燃油消耗和行驶时间。为了获得该问题的数学公式, 作者假设信号交通状态已知, 并忽略了前车的影响。该公式的优点 (相对于, 例如, Fayazi 等人 (2017)) 在于他们考虑了交叉口的转弯 (包括转弯速度约束)。在详细描述了燃油消耗模型、转弯影响和加速能力模型之后, 提出了一个 MIP 公式。整数变量的存在模拟了在不违反红灯的情况下通过交叉口, 表示活跃的绿灯相位窗口。

此外, 张、苏、孙和张 (2017b) 提出了一种异构交通网络的控制策略。异构性源于存在信号交叉路口和非信号交叉路口¹⁵。在验证了异构交通网络模型后, 使用MIP公式陈述了一种控制策略。最后, 提供了一些同质系统和异质系统之间的比较, 以便为开发一种系统规划方法留有余地, 该方法用于决定哪些交通交叉路口需要信号控制以确保良好的交通控制性能。此外, 张、苏、高和张 (2017a) 提出了一种交通信号调度策略, 该策略考虑了车辆和行人的存在。在开发行人流量和车辆交通网络 (多个交叉路口/节点) 的数学模型后, 调度问题被陈述为一个MIQP (混合整数二次规划)。

¹⁵ 无信号交叉路口遵循先到先得 (FCFC) 原则。

车辆交通网络 (多个交叉路口/节点) 的数学模型后, 调度问题被陈述为一个MIQP (混合整数二次规划)。

3.4. 障碍物和碰撞避免

障碍物和碰撞避免是寻找输入控制信号以最小化 (18a) 所示性能标准, 同时通过闭环策略避免碰撞的问题。

一个充分的碰撞和障碍物避障策略对于确保系统与环境的安全性和完整性至关重要。近年来, 控制领域一直致力于获取无碰撞控制策略, 以实现实时操作。人们普遍认为, 由于存在非凸可行域, 碰撞避障问题是一项具有挑战性的任务。描述这个非凸域具有计算和结构上的意义, 通常导致计算效率和控制性能之间的权衡。由于MIP能够像第2节那样显式地建模非凸性, 因此随着时间的推移, 它已成为表达碰撞避障问题的合适方法。

障碍物和碰撞避障问题领域的MIP方法可以分为两大类。第一类MIP方法更常用, 是非MIP (机器人) 运动规划文献中的一种广泛方法。它包括选择一种不依赖于环境特殊性的划分方法 (例如障碍物的形状)。一种标准的划分是由等大小的方形单元格组成的 (Haghighi等人, 2016年; Wang等人, 2015b年)。使用这种方法, 碰撞避障问题在大多数情况下变成资源分配问题, 如第3.1节所述。第二种MIP方法或几何方法依赖于集合论和混合整数技术有效编码非凸集描述的能力 (见第2节)。尽管不同, 这两种方法都有一个共同特征: 工作空间被分解成某种类型的单元格, 并且在某种程度上MIP被用于建模析取约束。

备注12。在文献中, 有些作品区分了避障和碰撞避免。更准确地说, 碰撞避免仅指多智能体系统内部的相互碰撞避免, 而避障可能指与静止障碍物的碰撞避免, 或与具有先验已知轨迹 (大多数情况下是周期性的) 的移动障碍物的碰撞避免。此外, 一些其他参考文献认为避障是指静态环境, 而碰撞避免涵盖任何类型的移动障碍物。♦

静态障碍物

Culligan (2006) 的工作提出了一种使用 MILP 解决无人机 (UAV) 回退地平线优化问题的路径规划器。MILP 公式包含两个重要组成部分: 用于障碍物和多车辆避让的硬约束以及车辆动力学的近似。3D 情况也被考虑并全面介绍。此外, 还讨论了 MILP 框架的一些改进, 以降低分辨率时间和提高路径规划器的能力。这些改进包括各种技术, 如可变时间步长、线性插值点和地平线最小化。一个值得注意的方面是, 可变时间步长的概念被扩展到回退地平线非迭代 MILP 公式。可变时间步长设置允许模拟地平线时间延长, 而不会显著增加分辨率时间。地平线最小化通过从问题中移除不必要的障碍物约束来减少分辨率时间 (类似于 Janeček 等人 (2017a) 的技术)。

¹⁵ A non-signalized intersection follows the first-come-first-serve (FCFC) principle.

In Schouwenaars (2006) the author presents a framework for safe online trajectory planning of unmanned vehicles through *partially unknown* environments. A MPC framework is employed, MILP playing an instrumental role in order to incorporate the collision avoidance constraints (similar with the techniques in Section 2). An interesting aspect is that the agents can be controlled either through a standard (velocity) control system, or by using a maneuver scheduler that allows the implementation of a maneuver from a discrete set of possible ones. This hybrid control architecture is applied and enhance for a particular type of dynamics, corresponding to a small-scale helicopter. As well, the problem of the feasibility of the proposed control strategy has been extensively treated. Therefore, a beneficial concept was considered: terminal feasible invariant set, a set in which an agent can remain for an indefinite period of time with anti-collision guarantees. Effectively, these sets are computed online, being represented as affine constraints on the last state of the horizon. Via these sets, it is provided an a priori known backup plan that is dynamically feasible and obstacle-free and, thus, the feasibility and safety can be guaranteed. The proposed strategy was tested on an unmanned Boeing aircraft using scalable loiter circles as feasible invariant sets. From the multi-agent perspective, the control strategy is a distributed one, each agent only computing its own trajectory while accounting for the latest planned behaviour of the agents from its proximity. The potential conflicts are solved in real-time such as to preserve the feasibility guarantees. In order to illustrate the benefits of the considered strategy, the algorithm was run over a scenario involving a fleet of small-scale helicopters which aimed at maintaining wireless connectivity in a cluttered environment.

Obstacle and collision avoidance constraints are usually imposed at the sampling time without regards to the intra-sample behavior of the agent. Hence, it is possible for an agent to “cut the corner” of an obstacle while apparently respecting the constraints. The idea employed in the literature is to consider additional constraints which ensure that the segment between two consecutive positions does not cut the obstacle Maia and Galvao (2009), Richards and Turnbull (2013). Stoican, Prodan, and Grøtli (2018) provides a treatment of the multi-obstacle case within a hyperplane arrangement setting with exact and over-approximated representations.

Mobile obstacles

The mobile obstacles in the MIP framework are not so common as the static environments due to the associated computational burden. In most of the references treating the time-varying environments, the interdicted mobile regions are modeled as rectangular/cubical exclusion regions Richards and How (2005, 2002). In fact, this representation is a particular case of the polytopic representation (see Section 2), but with a limited number of corresponding mixed-integer linear constraints. Thus, the exclusion region constraints in a MIP formulation for two moving obstacles within a 2D environment (x-y coordinates) are given as:

$$x_1 - x_2 \geq d - M\alpha_{d1} \quad (25a)$$

$$x_2 - x_1 \geq d - M\alpha_{d2} \quad (25b)$$

$$y_1 - y_2 \geq d - M\alpha_{d3} \quad (25c)$$

$$y_2 - y_1 \geq d - M\alpha_{d4} \quad (25d)$$

$$\sum_{i=1}^4 \alpha_{di} \leq 3 \quad (25e)$$

where d is the safety distance (length of the edge of square exclusion region), α_{di} are binary variables, while (25e) ensures that at least one of the above constraints is active. It is worth to mention that the rectangular exclusion zone can be used in order to model other non-convex constraints, e.g. Richards and How (2005), Culligan (2006)

Like trajectory planning, the collision avoidance problem can be

found/applied in tasks regarding the autonomous vehicles. For instance, in Mukai, Natori, and Fujita (2017) the problem of merging for vehicles on a motor way is tackled using MIP along with a receding horizon strategy (MPC). Thus, interdicted regions are described using logical statements (AND/OR), which are further modeled as mixed -integer constraints, using the “big-M” method.

Furthermore, Molinari et al. (2017) treat the autonomous overtaking problem using MPC in an efficient MIP formulation. The considered vehicles have a simple dynamic model, the non-convex feasible region being represented as in Section 2. The MPC criterion contains a desired reference state, which ensures that the overtaking take place.

As well, Bali and Richards (2018) propose a method for vehicle merging scenarios in junctions with relative cost prioritization. The method is based on MPC, employing MIQP optimization. The scheme provides optimal control properties while maintaining safety and recursive feasibility. The latter properties are ensured through positive control invariance of simple time headway constraints. For examples with two vehicles, tunable prioritization and gap acceptance are verified and presented on a decision graph. Priorities are then demonstrated to be respected in an example with four vehicles.

Equivalent MLD formulation for obstacle avoidance

The mixed logical dynamical (MLD) systems were introduced for the first time in Bemporad and Morari (1999) and represent a relevant framework for modeling and controlling systems which incorporates linear dynamic equations, logic rules, and operating constraints. These are described by linear dynamic equations subject to linear inequalities involving real and integer variables as in (26).

$$x_{k+1} = Ax_k + Bu_k + B_2\delta_k + B_3z_k \quad (26a)$$

$$y_k = Cx_k + Du_k + D_2\delta_k + D_3z_k \quad (26b)$$

$$E_2z_k + E_3\delta_k \leq E_1u_k + E_4x_k + E_5 \quad (26c)$$

where x_k, y_k, u_k denote state, output and, respectively, input vectors of the system. The $\delta_k \in \{0, 1\}^{n_\delta}$ and $z_k \in \mathbb{R}^{n_z}$ are auxiliary logic and continuous variables, respectively.

Let us consider a system described by a single agent which has to move in a non-convex domain. The agent is described by the dynamics¹⁶:

$$x_{k+1} = Ax_k + Bu_k \quad (27a)$$

$$y_k = Cx_k + Du_k \quad (27b)$$

We examine a basic example, the agent has to stay outside a polytope $P = \{x : Sx < R\}$:

$$x_{k+1} \notin P.$$

We aim to describe the considered system via a MLD formalism, more precisely, to rewrite the above condition in the form of (26c).

Firstly, it is straightforward that (27a) and (26a) are equivalent if we consider the matrices B_2 and B_3 null. Next, writing the imposed condition ($x_{k+1} \notin P$) in a similar manner as in (7a) and replacing x_{k+1} according to the dynamic, we compactly¹⁷ write:

$$\begin{bmatrix} -MI_N \\ \mathbf{1}^\top \end{bmatrix} \alpha_k \leq \begin{bmatrix} SA \\ \mathbf{0} \end{bmatrix} x_k + \begin{bmatrix} SB \\ \mathbf{0} \end{bmatrix} u_k + \begin{bmatrix} -R \\ N-1 \end{bmatrix}, \quad (28)$$

where M is a sufficiently large constant (according to the “big-M” formulation), N is the number of linear constraints describing the

¹⁶ For the clarity of the presentation, in what follows we will neglect the Eqs. (26b) and (27b) but the reasoning can be readily adapted.

¹⁷ “0” stands for the null matrix of dimensions appropriate to the place into which it is inserted.

在 Schouwenaars (2006) 中，作者提出了一种通过部分未知环境进行无人机安全在线轨迹规划的框架。采用 MPC 框架，MILP 在纳入避撞约束方面发挥着关键作用（类似于第 2 节中的技术）。一个有趣的特点是，智能体可以通过标准（速度）控制系统控制，或通过使用允许从一组可能的动作中选择一个动作的实现动作调度器进行控制。这种混合控制架构被应用于并增强于特定类型的动力学，对应于小型直升机。此外，所提出的控制策略的可行性已被广泛研究。因此，考虑了一个有益的概念：终端可行不变集，即智能体可以在其中无限期停留并具有防碰撞保证的集合。实际上，这些集是在线计算的，表示为地平线最后状态的仿射约束。通过这些集，提供了一种先验已知的备用计划，该计划动态可行且无障碍物，从而保证了可行性和安全性。所提出的策略在无人波音飞机上进行了测试，使用可扩展的盘旋圆作为可行不变集。从多智能体角度来看，控制策略是分布式的，每个智能体仅计算自己的轨迹，同时考虑其附近智能体的最新计划行为。潜在的冲突实时解决，以保持可行性保证。为了说明所考虑策略的好处，该算法在一个涉及小型直升机编队的场景中运行，该编队旨在在杂乱环境中保持无线连接。

障碍物和碰撞避免约束通常在采样时施加，而忽略智能体在采样内的行为。因此，智能体可能在表面上遵守约束的同时“绕过”障碍物的角落。文献中采用的方法是考虑额外的约束，以确保两个连续位置之间的段不会穿过障碍物 Maia 和 Galvao (2009), Richards 和 Turnbull (2013)。Stoican, Prodan 和 Grøtli (2018) 在超平面排列设置中处理了多障碍物的情况，使用精确和过度近似表示。

移动障碍物

MIP 框架中的移动障碍物不如静态环境常见，因为它们相关的计算负担较重。在大多数处理时变环境的参考文献中，禁止的移动区域被建模为矩形/立方体排除区域 Richards 和 How (2005, 2002)。事实上，这种表示是多面体表示的一个特例（见第 2 节），但具有较少的相应混合整数线性约束。因此，在 2D 环境 (x-y 坐标) 中，对于两个移动障碍物的 MIP 公式中，排除区域约束为：

$$x_1 - x_2 \geq d - M\alpha_{d1} \quad (25a)$$

$$x_2 - x_1 \geq d - M\alpha_{d2} \quad (25b)$$

$$y_1 - y_2 \geq d - M\alpha_{d3} \quad (25c)$$

$$y_2 - y_1 \geq d - M\alpha_{d4} \quad (25d)$$

$$\sum_{i=1}^4 \alpha_{di} \leq 3 \quad (25e)$$

其中 d 是安全距离（方形排除区域的边长）， α_{di} 是二元变量，而 (25e) 确保上述约束至少有一个处于激活状态。值得一提的是，矩形排除区可用于建模其他非凸约束，例如 Richards 和 How (2005)、Culligan (2006)

与轨迹规划类似，避碰问题可以应用于自动驾驶车辆相关的任务。

例如，在 Mukai, Natori 和 Fujita (2017) 的研究中，通过结合 MIP（混合整数规划）和回退地平线策略（MPC），解决了高速公路上车辆合并的问题。因此，受限区域被描述为逻辑语句（AND/OR），并进一步使用“大M”方法建模为混合整数约束。

此外，Molinari 等人 (2017) 使用 MPC 在高效的 MIP 公式中处理了自主超车问题。所考虑的车辆具有简单的动态模型，非凸可行区域表示如第 2 节所述。MPC 准则包含期望参考状态，这确保了超车能够发生。

同样地，Bali 和 Richards (2018) 提出了一种用于路口车辆合并场景的方法，该方法具有相对成本优先级。该方法基于 MPC，采用 MIQP 优化。该方案提供最优控制特性，同时保持安全性和递归可行性。后者特性通过简单时间头程约束的正控制不变性得到保证。对于两个车辆的情况，可调优先级和间隙接受性被验证并在决策图上展示。然后在四车辆示例中证明了优先级得到了尊重。

用于避障的等效 MLD 公式

混合逻辑动力学 (MLD) 系统首次由 Bemporad 和 Morari (1999) 提出，并代表了一个相关框架，用于建模和控制包含线性动态方程、逻辑规则和操作约束的系统。这些系统由线性动态方程描述，这些方程涉及实数和整数变量，如 (26) 所示。

$$x_{k+1} = Ax_k + Bu_k + B_2\delta_k + B_3z_k \quad (26a)$$

$$y_k = Cx_k + Du_k + D_2\delta_k + D_3z_k \quad (26b)$$

$$E_2z_k + E_3\delta_k \leq E_1u_k + E_4x_k + E_5 \quad (26c)$$

其中 x_k, y_k, u_k 分别表示系统的状态、输出和输入向量。 $\delta_k \in \{0, 1\}^{n_\delta}$ 和 $z_k \in \mathbb{R}^{n_z}$ 分别是辅助逻辑变量和连续变量。

考虑一个由单个智能体描述的系统，该智能体需要在非凸域内移动。该智能体由动力学方程¹⁶描述：

$$x_{k+1} = Ax_k + Bu_k \quad (27a)$$

$$y_k = Cx_k + Du_k \quad (27b)$$

我们考察一个基本示例，智能体必须停留在多面体 $P = \{x : Sx < R\}$ 外部

$$x_{k+1} \notin P.$$

我们旨在通过 MLD 形式化描述所考虑的系统，更精确地说，将上述条件以 (26c) 的形式重写

首先，如果我们考虑矩阵 B_2 和 B_3 为零矩阵，那么 (27a) 和 (26a) 是等价的。接下来，以 (7a) 类似的方式写出所施加的条件 ($x_{k+1} \notin P$)，并根据动态替换 x_{k+1} ，我们简洁地¹⁷ 写出：

$$\begin{bmatrix} -MI_N \\ \mathbf{1}^\top \end{bmatrix} \alpha_k \leq \begin{bmatrix} SA \\ \mathbf{0} \end{bmatrix} x_k + \begin{bmatrix} SB \\ \mathbf{0} \end{bmatrix} u_k + \begin{bmatrix} -R \\ N-1 \end{bmatrix}, \quad (28)$$

其中 M 是一个足够大的常数（根据“大 M”公式）， N 是描述多面体 $\{v1\}$ 的线性约束的数量，而 $\{v2\}$ 是单位矩阵。通过在输入 $\{v3\}$ 上添加约束，所考虑系统的 MLD 公式为：

¹⁶ 为了清晰起见，在以下内容中我们将忽略方程 (26b) 和 (27b)，但推理可以很容易地推广。¹⁷ “0”表示适当插入位置的零矩阵。

polytope $P = \{x \in \mathbb{R}^d | Sx \leq R, S \in \mathbb{R}^{N \times d}, R \in \mathbb{R}^N\}$, and $I_N \in \mathbb{R}^{N \times N}$ is the identity matrix. By adding constraints on input ($S_u u_k \leq R_u$), the MLD formulation of the considered system is:

$$x_{k+1} = Ax_k + Bu_k \tag{29a}$$

$$\begin{bmatrix} -MI_N \\ \mathbf{1}^\top \\ \mathbf{0} \end{bmatrix} \alpha_k \leq \begin{bmatrix} SA \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} x_k + \begin{bmatrix} SB \\ \mathbf{0} \\ -S_u \end{bmatrix} u_k + \begin{bmatrix} -R \\ N-1 \\ R_u \end{bmatrix} \tag{29b}$$

Remark 13. The MLD system (29) is *well posed* according to Definition 1 from Bemporad and Morari (1999). ♦

In Ritter, Euler, Ulbrich, and Stryk (2014) the equivalent MLD formulation is further extended. A multi-vehicle cooperating system is modelled and controlled such that the system reaches some target locations in a reasonable time avoiding any possible collision among the agents. Thus, MLD-system represents a valid compact formulation for modeling multi-vehicle system, because all existing constraints (dynamics, collision avoidance, measurement etc.) are embedded in a complete configuration, which allows to use existing methods to solve the control problem.

MIP reformulation for drift counteraction optimal control

Another form in which MIP is employed in motion planning is within DCOC (drift counteraction optimal control) or optimal exit-time control Zidek, Bemporad, and Kolmanovsky (2017a). The main objective in such problem is to satisfy prescribed constraints for as long as possible Zidek, Petersen, Bemporad, and Kolmanovsky (2017b). In other words, DCOC is a particular optimal control problem, aiming to determine a sequence of control inputs that maximizes the first exit-time from a given set. There exist various applications in motion planning where DCOC represents an useful tool, handling well the systems with finite resources (fuel or energy). In Zidek et al. (2017b) the MILP formulation of DCOC with application in spacecraft attitude control is fully addressed and for its resolution a LP-based iterative procedure is introduced.

As well, in a similar approach, Maia and Galvao (2009) present an implementation of the shifting prediction horizon for an MPC controller, tackling the obstacle avoidance problem, using binary variables and implicitly MIP. Moreover, Richards and How (2003) MILP optimization is used to effect a variable horizon length, leading to guaranteed finite-time completion.

4. MIP in multi-agent systems. Connectivity maintenance and formation control

In today’s complex and various environments, the vast majority of the activities is too difficult and time-consuming to be handled by only one agent/robot/entity. Thus, in order to perform these activities with increased accuracy, redundancy and in a reduced time, cooperative teams of robots/agents may be employed. In this manner, the key elements of risk for the safety and integrity of systems are mitigated at the expense of an extensive augmentation of the systems to be supervised and controlled. Factors like the large scale, geographical distribution, high failure rates and heterogeneity of network systems are becoming decisive in consideration of the suitable control architecture.

An important issue in the control of multi-agent systems is the coordination of a cooperative team of agents (robots) in order to accomplish a given “mission” in an efficient manner. In several cases, these teams of agents are required to converge (and maintain) a specific spatial configuration. Therefore, formation control is often a prerequisite for other applications in motion planning.

Connectivity maintenance refers to providing constraints which guarantee a connected communication graph among the agents at all times.

More precisely, connectivity maintenance for a formation of agents refers, generally, to the ensemble of feedback laws, path/trajectory planning, collision avoidance and task assignment which guarantee that agents can communicate between themselves (to ensure data flows for, e.g., control decisions, data gathering, distributed estimation, etc.).

Remark 14. Even though *connectivity maintenance* and *formation control* are not different labels for the same phenomenon, the underlying idea is similar, the difference being the global objective. A key difference between the two notions is the relevance of relative distance between agents: for connectivity maintenance the unobstructed communication is the key aspect, while the inter-distance or relative velocity are relevant only in their effect on the “line of sight” between agents (e.g., distance less than a communication range threshold). ♦

Formation control aims to provide constraints guaranteeing desired relative distances and velocities among a group of agents.

There are two main approaches in the literature which tackle the formation control problems Qu (2009): *the leader-follower design* and *leaderless approach*. The former consists in designating one of the agents as a leader moving in a certain way with the remaining agents tracking the leader in order to maintain formation around it. The later involves the coordination of the agents through a global consensus for attaining the global objective. Choosing between these two approaches depends on the particularities of the team/group of agents, i.e., their capability of communication and sensing. Moreover, these aspects affect further the control architecture, as will be detailed in Section 5. (Table 2)contains the classification of the references which employ MIP techniques for multi-agent systems motion planning strategies.

Corner cutting avoidance conditions

The corner cutting avoidance problem is an important but often overlooked part in motion planning. Obstacle and collision avoidance constraints are usually imposed at the sampling time without regards to the intra-sample behavior of the agent(s). Hence, it is possible for an agent to “cut the corner” of an obstacle while apparently respecting the constraints.

Corner cutting aims to provide constraints guaranteeing intra-sample collision avoidance.

The idea is conceptually simple: an agent avoids corner cutting iff its next position lies in its visibility region (the union of all rays spanned from the current position not intersecting any of the obstacles). Note that the dual notion (the under-shadow region – the region hidden from the viewpoint of an agent) may be used to enforce full coverage of the feasible space, e.g., the gallery problem.

Since describing exactly such regions is impractical, mixed-integer formulations (in either exact or over-approximated form) are used. Within the hyperplane arrangement framework presented earlier, these constraints employ (either implicitly or explicitly) three sign tuples: the

Table 2
MIP for multi-agent systems.

Connectivity maintenance	Afonso et al. (2020); Earl and D’Andrea (2005b); Richards and Turnbull (2013); Schouwenaars, De Moor, Feron, and How (2001a); Sun and Cassandras (2015)
Formation control	Bellingham, Tillerson, Alighanbari, and How (2002b); Papen et al. (2017); Prodan, Olaru, Stoica, and Niculescu (2012); Ritter et al. (2014)
Corner cutting avoidance	Afonso et al. (2016); Maia and Galvao (2009); Richards and Turnbull (2015); Stoican et al. (2018)

通过在输入 ($S_u u_k \leq R_u$) 上添加约束，所考虑系统的 MLD 公式为：

$$x_{k+1} = Ax_k + Bu_k \tag{29a}$$

$$\begin{bmatrix} -MI_N \\ \mathbf{1}^\top \\ \mathbf{0} \end{bmatrix} \alpha_k \leq \begin{bmatrix} SA \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} x_k + \begin{bmatrix} SB \\ \mathbf{0} \\ -S_u \end{bmatrix} u_k + \begin{bmatrix} -R \\ N-1 \\ R_u \end{bmatrix} \tag{29b}$$

注释 13. MLD 系统 (29) 根据 Bemporad 和 Morari (1999) 的定义 1 是适定的。♦

在 Ritter, Euler, Ulbrich 和 Stryk (2014) 中，等效的 MLD 公式得到了进一步扩展。一个多车辆协同系统被建模和控制，使得系统能在合理的时间内到达某些目标位置，同时避免代理之间发生任何可能的碰撞。因此，MLD-系统代表了一种有效的紧凑公式，用于对多车辆系统进行建模，因为所有现有的约束（动力学、避碰、测量等）都嵌入在完整的构型中，这允许使用现有方法来解决控制问题。

漂移补偿最优控制的 MIP 重新公式化

MIP 在运动规划中的另一种应用形式是 DCOC（漂移反作用最优控制）或最优退出时间控制，Zidek, Bemporad 和 Kolmanovsky（2017a）。这类问题的主要目标是在尽可能长的时间内满足规定的约束条件，Zidek, Petersen, Bemporad 和 Kolmanovsky (2017b)。换句话说，DCOC 是一种特定的最优控制问题，旨在确定一个控制输入序列，以最大化从给定集中的首次退出时间。在运动规划中存在多种应用，其中 DCOC 是一个有用的工具，能够很好地处理具有有限资源（燃料或能量）的系统。在 Zidek 等人 (2017b) 的研究中，DCOC 的 MILP 形式及其在航天器姿态控制中的应用得到了全面探讨，并引入了一种基于 LP 的迭代方法来解决该问题。

此外，在类似的方法中，Maia 和 Galvao (2009) 提出了一种 MPC 控制器的移动预测范围实现，用于解决避障问题，并使用二元变量和隐式 MIP。此外，Richards 和 How (2003) 使用 MILP 优化实现了可变预测范围长度，从而保证了有限时间完成。

4. 多智能体系统中的MIP。连接维护与编队控制

在当今复杂多样的环境中，绝大多数活动对于单个智能体/机器人/实体来说过于困难和耗时。因此，为了以更高的精度、冗余性和更短的时间完成这些活动，可以采用机器人/智能体合作团队。通过这种方式，系统的安全性和完整性风险要素得到缓解，但代价是监督和控制对象的系统大幅扩展。网络系统的大规模、地理分布、高故障率和异构性等因素，在考虑合适的控制架构时变得至关重要。

多智能体系统控制中的一个重要问题是协调一个合作智能体团队（机器人），以高效的方式完成给定的“任务”。在几种情况下，这些智能体团队需要收敛（并保持）特定的空间配置。因此，编队控制通常是运动规划中其他应用的一个先决条件。

连接性维护是指提供约束，以保证智能体之间始终存在一个连通的通信图。

更具体地说，智能体编队的连接维护通常指保证智能体之间能够相互通信（以确保数据流，例如控制决策、数据收集、分布式估计等）的反馈律、路径/轨迹规划、碰撞避免和任务分配的集合。

备注14. 尽管连接维护和编队控制不是同一现象的不同标签，但底层思想相似，区别在于全局目标。这两个概念之间的一个关键区别在于智能体之间相对距离的相关性：对于连接维护，无障碍通信是关键方面，而距离或相对速度仅在其对智能体之间“视线”的影响中相关（例如，距离小于通信范围阈值）。♦

编队控制旨在提供约束，以保证一组智能体之间的期望相对距离和速度。

文献中主要有两种方法处理编队控制问题（Qu，2009）：领导者-跟随者设计和无领导者方法。前者包括指定一个智能体作为领导者以特定方式移动，其余智能体跟踪领导者以在其周围保持编队。后者涉及通过全局共识协调智能体以实现全局目标。选择这两种方法中的哪一种取决于智能体团队/组的特性，即它们的通信和感知能力。此外，这些方面还会进一步影响控制架构，第5节将详细说明。（表2）包含了采用MIP技术进行多智能体系统运动规划策略的参考文献分类。

拐角规避条件

拐角规避问题是在运动规划中一个重要但常被忽视的部分。障碍物和碰撞规避约束通常在采样时施加，而忽略智能体（们）的采样内行为。因此，智能体有可能在表面上遵守约束的同时“拐弯抹角”地绕过障碍物。

拐角规避旨在提供约束，以保证样本内部碰撞避免。

这个想法在概念上是简单的：一个智能体避免走捷径，当且仅当它的下一个位置位于它的可视区域内（即从当前位置发出的所有射线与任何障碍物都不相交的区域的并集）。请注意，这个对偶概念（即下影区——即从智能体的视角被隐藏的区域）可用于确保可行空间的全覆盖，例如画廊问题。

由于精确描述此类区域不切实际，因此使用混合整数公式（无论是精确形式还是过度近似形式）。在之前介绍的超平面排列框架中，这些约束（无论是隐式还是显式地）采用三个符号三元组：

表2
多智能体系统的MIP。

连接性维护	Afonso 等人 (2020); Earl 和 D’Andrea (2005b); Richards 和 Turnbull (2013); Schouwenaars, De Moor, Feron 和 How (2001a); Sun 和 Cassandras (2015)
编队控制	Bellingham, Tillerson, Alighanbari 和 How (2002b); Papen 等人 (2017); Prodan, Olaru, Stoica 和 Niculescu (2012); Ritter 等人 (2014)
走捷径回避	Afonso 等人 (2016); Maia 和 Galvao (2009); Richards 和 Turnbull (2015); Stoican 等人 (2018)

current and future positions of the agent, $\sigma, \sigma^+ \in \Sigma_{X \setminus \mathbb{P}}$, and the coordinates of the obstacle(s), $\sigma^* \in \Sigma_{\mathbb{P}}$.

We are aware of results from Afonso, Galvão, and Kienitz (2016); Maia and Galvao (2009); Richards and Turnbull (2015) which discuss over-approximated corner cutting constraints and provide constructive details: Maia and Galvao (2009) provides the initial construction; Richards and Turnbull (2015) and Afonso et al. (2016) improve it by reducing the number of necessary constraints and by reducing the number of necessary binary variables, respectively. Stoican et al. (2018) goes further by providing exact and over-approximated descriptions of the under-shadow (and of its complement, the visible) region generated by multiple agents within a multi-obstacle environment.

Specifically, in Maia and Galvao (2009), to avoid cutting a single obstacle, there has to exist at least one half-space containing the obstacle which does not contain the current and successor positions of the agent:

$$\exists i \text{ s.t. } \sigma(i) = \sigma^+(i) = 0. \quad (30)$$

Since such conditions often appear in MPC problems where both current and successor sign tuples are decision variables, (30) is rewritten such as to avoid nonlinearities:

$$\sum_i [1 - \sigma^*(i)] \sigma^+(i) < N + \sum_i [1 - 2\sigma^*(i)] \sigma(i), \quad (31)$$

for all possible values $\sigma^* \in \{0, 1\}^N$. The idea is that from all constraints (31) at least one reduces to¹⁸ (30), i.e., the one for which $\sigma^* = \sigma^+$.

Richards and Turnbull (2015) improves on Maia and Galvao (2009) by reducing the number of constraints (31) from 2^N to a more manageable N . This is done by forcing two consecutive positions x, x^+ to respect the same constraint, with our notation: $-h_i^\top x \leq -k_i + M\sigma^+(i)$ and $-h_i^\top x^+ \leq -k_i + M\sigma^+(i)$ for all $i = 1 \dots N$. This implies that there exists at least one index i s.t. both x and x^+ lie on the same side of the hyperplane (and thus on the opposite side from the obstacle), similar with (30). Afonso et al. (2016) proposes a logarithmic scheme to reduce the number of binary variables involved in the selection of the active hyperplanes.

The common shortcoming of these approaches is that they do not easily handle the multi-agent multi-obstacle case. Stoican et al. (2018) generalizes (30) with:

$$\sum_j |\sigma^{*j}(i) - \sigma(i)| \cdot |\sigma^{*j}(i) - \sigma^+(i)| > 0, \forall \sigma^{*j} \in \Sigma_{\mathbb{P}}. \quad (32)$$

As stated earlier, when both σ, σ^+ are variables, (32) becomes nonlinear due the bi-linear term appearing in it. Hence, a relaxation such as in (31) may be used:

$$\sum_i |\sigma^{*j}(i) - \sigma^*(i)| \cdot |\sigma^{*j}(i) - \sigma^+(i)| > -|\sigma^* - \sigma|, \forall \sigma^{*j} \in \Sigma_{\mathbb{P}}, \quad (33)$$

for all $\sigma^* \in \Sigma_{X \setminus \mathbb{P}}$, i.e., (33) reduces to (32) when $\sigma^* = \sigma^+$.

Connectivity maintenance conditions

The optimal dynamic formation control is a popular topic in the multi-agent systems field, and MIP-based methods provide a suitable tool for the modeling of the corresponding constraints. In what follows, we briefly present the topics where the connectivity is induced by the internal factors (as, e.g., communication).

The mixed-integer constraints enforce that each agent is within the communication range of at least one other agent. Moreover, pairs isolation need to be avoided, i.e., if an agent is enforced to stay in a communication range of another, the latter cannot be constrained to

stay in the range of the former, but in the range of a third one. In order to formulate the corresponding mixed-integer constraints we proceed as in Afonso, Maximo, and Galvão (2020). Hence, we consider a binary variable $\xi_{i,j,k} \in \{0, 1\}$ which is 1, if the i th agent is inside of the communication range of j th agent at the timestep k , and 0 otherwise. Next, we can write the following constraint, ensuring the communication connectivity among the N agents:

$$\sum_{i=1}^{N-1} \sum_{j=1}^N \xi_{i,j,k} = N - 1, \quad 0 \leq k \leq T, \quad (34a)$$

$$\sum_{v_i, v_j} \xi_{i,j,k} \leq \#S - 1, \quad 0 \leq k \leq T, \quad \forall S \subset V, \quad (34b)$$

$$\xi_{i,j,k} = 0, \quad 1 \leq i \leq N, \quad 1 \leq j \leq i, \quad 0 \leq k \leq T. \quad (34c)$$

The constraint (34b) prevents the possible pairs isolation, imposing for each sub-graph S of the original communication graph V that at least one node has to be able to communicate with a node outside the graph. Moreover, alongside with (34a) it guarantees that only connected communication graph can be obtained (for a detailed discussion, see Theorem 2 from Afonso et al. (2020)), i.e. the connectivity is maintained.

Furthermore, Sun and Cassandras (2015) is focusing on the *coverage control problem*. The agent team works in a 2D mission space, maximizing coverage in this space. The formation control is approached from a leader-follower setting perspective, the leader moves on a given trajectory while the remaining agents must maintain the formation. Thus, the formation becomes dynamic as soon as the leader starts moving along the given trajectory, and it must adapt to any environmental change (a new mission, a new composition of the team or some obstacles detected). Firstly, it is presented a brief formulation for a general optimal formation problem as an optimization problem, whose objective function depends on the agent spatial positions. The constraints of the optimization problem are given by a feasible space for positions (included in the 2D mission space) and by the condition that an undirected graph must be connected. This graph is modeling the desirable links among agents. The feasible space can be either convex (without obstacles) or non-convex (with obstacles). This aspect complicates the way in which the graph constraint should be treated. Hence, these two cases must be tackled separately. For a convex feasible space, the optimal dynamic formation problem is rewritten as a MINLP, by introducing a set of flow variables over the undirected graph (associating to every link between two agents an integer flow amount). The problematic constraint is transformed into a set of mixed-integer inequalities. Also, the computationally demanding re-solving of the MILP for all time instants is avoided, providing a sufficient condition for maintaining optimal formation in a certain time interval. For a non-convex feasible space, the problem is solved in a different approach. At each moment, when the connectivity is lost, a new undirected connected graph is constructed such that the effort to maintain the initial formation is minimal. An algorithm is developed to construct the new graph, and this is used as an input for CPA (Connectivity Preservation Algorithm). An illustrative example of this second situation is considered therein, where the initial formation is computed using the MILP from the convex feasible space case, obstacles being considered afterwards.

Connectivity induced by external constraints

This subsection introduces the types of objectives/ goals encountered in the formation control, goals enforced by the external¹⁹ factors.

One of the classical examples where connectivity maintenance is externally induced is the problem of efficient coverage of a specific area

¹⁸ Both (30) and (31) assume a single obstacle ($\Sigma_{\mathbb{P}} = \{\sigma^{*1}\}$) where, by convention, $\sigma^{*1}(i) = 1, \forall i$. This implies that $\sum_i \sigma(i) \leq N - 1, \sum_i \sigma^+(i) \leq N - 1$.

¹⁹ W.r.t the multi-agent system.

智能体当前和未来的位置, $\sigma, \sigma^+ \in \Sigma_{X \setminus \mathbb{P}}$, 以及障碍物的坐标, $\sigma^* \in \Sigma_{\mathbb{P}}$.

我们了解 Afonso, Galvão 和 Kienitz (2016)、Maia 和 Galvao (2009)、Richards 和 Turnbull (2015) 的研究成果, 这些成果讨论了过度近似的角落切割约束, 并提供了构建细节: Maia 和 Galvao (2009) 提出了初始构建; Richards 和 Turnbull (2015) 以及 Afonso 等人 (2016) 分别通过减少必要约束的数量和减少必要二进制变量的数量来改进它。Stoican 等人 (2018) 进一步提供了多智能体在多障碍环境中生成的阴影区域 (及其补集, 即可见区域) 的精确和过度近似描述。

具体来说, 在 Maia 和 Galvao (2009) 中, 为了避免切割单个障碍物, 必须存在至少一个包含该障碍物但不包含智能体当前和后继位置的半空间:

$$\exists i \text{ s.t. } \sigma(i) = \sigma^+(i) = 0. \quad (30)$$

由于此类条件在MPC问题中经常出现, 其中当前和后继符号元组都是决策变量, 因此(30)被重写以避免非线性:

$$\sum_i [1 - \sigma^*(i)] \sigma^+(i) < N + \sum_i [1 - 2\sigma^*(i)] \sigma(i), \quad (31)$$

对于所有可能的值 $\sigma^* \in \{0, 1\}^N$, 其思想是所有约束(31)中至少有一个会减少 to¹⁸ (30), 即满足 $\sigma^* = \sigma^+$ 的那个。

Richards和Turnbull (2015) 在Maia和Galvao (2009) 的基础上进行了改进, 将约束(31)的数量从 2^N 减少到更易于管理的 N 。这是通过强制两个连续位置 x, x^+ 遵守相同的约束来实现的, 用我们的符号表示: $-h_i^\top x \leq -k_i + M\sigma^+(i)$ 和 $-h_i^\top x^+ \leq -k_i + M\sigma^+(i)$ 对于所有 $i = 1 \dots N$ 。这意味着存在至少一个索引 i , 使得 x 和 x^+ 位于超平面的同一侧 (因此与障碍物位于相反侧), 类似于(30)。Afonso等人 (2016) 提出了一种对数方案, 以减少在选择活动超平面时涉及的二进制变量的数量。

这些方法的一个常见缺点是它们难以处理多智能体多障碍物的情况。Stoican 等人 (2018) 对 (30) 进行了推广, 如下:

$$\sum_j |\sigma^{*j}(i) - \sigma(i)| \cdot |\sigma^{*j}(i) - \sigma^+(i)| > 0, \forall \sigma^{*j} \in \Sigma_{\mathbb{P}}. \quad (32)$$

如前所述, 当 σ, σ^+ 都是变量时, (32) 会因其中出现的二阶线性项而变为非线性。因此, 可以使用 (31) 中的松弛方法:

$$\sum_i |\sigma^{*j}(i) - \sigma^*(i)| \cdot |\sigma^{*j}(i) - \sigma^+(i)| > -|\sigma^* - \sigma|, \forall \sigma^{*j} \in \Sigma_{\mathbb{P}}, \quad (33)$$

对所有 $\sigma^* \in \Sigma_{X \setminus \mathbb{P}}$ 适用, 即当 $\sigma^* = \sigma^+$ 时, (33) 简化为 (32)。

连通性维护条件

最优动态编队控制是多智能体系统领域的一个热门课题, 基于MIP 的方法为建模相应的约束提供了合适的工具。以下, 我们简要介绍连通性由内部因素 (例如通信) 引起的主题。

混合整数约束要求每个智能体至少处于另一个智能体的通信范围内。此外, 需要避免对智能体进行配对隔离, 即如果某个智能体被强制保持在另一个智能体的通信范围内, 后者不能被约束为

¹⁸ 公式(30)和(31)均假设存在单个障碍物 ($\Sigma_{\mathbb{P}} = \{\sigma^{*1}\}$), 根据惯例, $\sigma^{*1}(i) = 1 \forall i$, 。这意味着 $\sum_i \sigma(i) \leq N - 1, \sum_i \sigma^+(i) \leq N - 1$, i

保持在前者所在的范围内, 而应保持在第三个智能体的通信范围内。为了建立相应的混合整数约束, 我们按照Afonso、Maximo和Galvão (2020) 的方法进行。因此, 我们考虑一个二元变量 $\xi_{i,j,k} \in \{0, 1\}$, 当第 i 个智能体在时间步 k 处于第 j 个智能体的通信范围内时, 该变量为1, 否则为0。接下来, 我们可以写出以下约束, 确保 N 个智能体之间的通信连通性:

$$\sum_{i=1}^{N-1} \sum_{j=1}^N \xi_{i,j,k} = N - 1, \quad 0 \leq k \leq T, \quad (34a)$$

$$\sum_{v_i, v_j} \xi_{i,j,k} \leq \#S - 1, \quad 0 \leq k \leq T, \quad \forall S \subset V, \quad (34b)$$

$$\xi_{i,j,k} = 0, \quad 1 \leq i \leq N, \quad 1 \leq j \leq i, \quad 0 \leq k \leq T. \quad (34c)$$

约束 (34b) 阻止了可能对隔离, 对原始通信图 V 的每个子图 S 施加至少一个节点必须能够与图外节点通信。此外, 连同 (34a) 它保证了只能获得连通通信图 (详细讨论见 Afonso 等人 (2020) 的定理 2), 即连通性得到保持。

此外, Sun 和 Cassandras (2015) 关注覆盖控制问题。代理团队在一个 2D 任务空间中工作, 最大化该空间的覆盖范围。编队控制从领导者-跟随者设置的角度来处理, 领导者沿着给定轨迹移动, 而其余代理必须保持编队。因此, 一旦领导者开始沿着给定轨迹移动, 编队就变得动态, 并且必须适应任何环境变化 (新的任务、团队的新组成或检测到的障碍物)。首先, 提出一个一般最优编队问题的简要公式, 将其作为优化问题, 其目标函数取决于代理的空间位置。优化问题的约束由一个可行位置空间 (包含在 2D 任务空间中) 和必须是无向图连通的条件给出。该图模拟了代理之间期望的链接。可行空间可以是凸的 (无障碍物) 或非凸的 (有障碍物)。这一点使处理图约束的方式变得复杂。因此, 这两种情况必须分别处理。对于凸可行空间, 最优动态编队问题被重写为 MINLP, 通过引入一组无向图上的流量变量 (将两个代理之间的每条链接关联到一个整数流量量)。问题约束被转换为一组混合整数不等式。此外, 避免了在所有时间点重新求解 MILP, 从而在特定时间间隔内保持最优编队提供了一个充分条件。对于非凸可行空间, 问题采用不同的方法解决。每当连通性丢失时, 构造一个新的无向连通图, 使得保持初始编队的努力最小。开发了一种构造新图的算法, 并将其用作 CPA (连通性保持算法) 的输入。文中考虑了这种情况的一个说明性示例, 其中初始编队使用凸可行空间案例的 MILP 计算, 之后考虑了障碍物。

外部约束引起的连接性

本小节介绍了在形成控制中遇到的各类目标/目标, 这些目标由外部¹⁹ 因素强制执行。在连接性维护由外部诱导的经典例子中, 有一个问题是高效覆盖特定区域

¹⁹ 关于多智能体系统。

by mobile sensors. For instance, [Ritter et al. \(2014\)](#) tackled this problem for a mobile sensors platform composed of a number of sensor-equipped autonomous vehicles. An adaptive observation strategy for on-line estimation of the state of a dispersion process is proposed, based on the model of the process and the multi-agent collaborative systems. The multi-vehicle cooperating system is modelled using MLD systems ([Bemporad & Morari, 1999](#)); and controlled such that the target location is reached in a reasonable time while simultaneously avoiding any possible collision among the agents. The target positions of the agents are received from the PDE²⁰-based state estimation, more precisely the optimal measurement locations. Thus, the MLD-systems represent a good alternative for modeling multi-vehicle systems, because all existing constraints (dynamics, collision avoidance, measurement etc.) are embedded in a complete configuration, which allows to use standard methods to solve the control problem.

As well, a standard problem is given by the rendezvous maneuver, a group of agents (usually, a fleet of spacecraft) has to achieve a specific formation in a minimum time. [Papen et al. \(2017\)](#) treat this problem for a fixed-wing-UAV fleet in an environment containing both static and dynamic obstacles. The control strategy is implemented through a distributed architecture (specifically, MPC applied for each agent), an agent knowing only its own trajectory (defined by its current position and velocity). By introducing binary variables to model the obstacles and to eliminate some nonlinear constraints (like velocity and acceleration constraints), the MPC optimization problem is transformed into a MILP. Noteworthy, the wake vortex of turbulent air formed behind every UAV is modeled as a dynamic obstacle because within these regions it is arduous to maintain control over the UAVs. To solve the resulting MILP is necessary a considerable time, but with some relaxation of the constraints and fine-tuning the complexity is limited.

5. Control architectures

There are three well-established classes of control architectures, and they have been extensively studied in various application domains: centralized, distributed and decentralized. The last two methods require the local controllers to optimize over only their local inputs having similar computational burden. The difference between these two is given by the impact of communication, decentralized control requires no communication among the agents.

To identify/discuss the control architectures weaknesses and strengths with respect to the motion planning field is not the scope of this paper, thus, we only focus on how they are employed along with MIP. In [Table 3](#), we delineate a classification of MIP-based motion planning references w.r.t. the control architectures and number of agents involved in the formulations.

It is worth mentioning that the control strategies employed in other architectures than the centralized way are optimization-based, with predilection MPC. Thus, in this section an emphasis is put on the specific MPC implementations. Nevertheless, we have not neglected the references where MIP is used specifically in distributed/decentralized non-MPC strategies due to its capability to formulate task allocation problems. These are extensively treated in [Section 3.1](#).

The control architectures involving MIP naturally evolve from centralized to decentralized and distributed strategies.

5.1. Centralized

Due to its theoretical simplicity, the centralized approach is the most used way of controlling a multi-agent system. In this architecture, the multi-agent system is treated as a whole, equating an extended single-

Table 3
Classification of MIP approaches in motion planning.

Agent(s)	Centralized	Decentralized	Distributed	References(e.g.)
$N = 3 / N \geq 3$, partially	✓	×	×	Chen et al. (2016a)
1... N	✓	×	×	Earl and D'Andrea (2005b) ; Haghighi et al. (2016) ; Janeček et al. (2017a) , Ritter et al. (2014) ; Sun and Cassandras (2015)
1... N	✓	✓	×	Wang et al. (2015b)
1... N	×	×	✓	Papen et al. (2017)
$N = 1$	✓	×	×	Molinari et al. (2017) ; Ragi and Mittelmann (2017)

Note: (✓) treated, (×) not treated, (1... N) any number of agents

agent system. The physical restrictions (e.g. communication limitations) are completely ignored, each agent having complete knowledge on the behavior/actions of the others, all information being available in the single global controller [Rawlings and Mayne \(2009\)](#). However, this methodology is limited, not only because of the undeniable physical constraints, but also due to numerical difficulties, induced by the substantial complexity of the extended system.

As stated in [Section 3](#), in the literature, MIP may appear at different levels of control. For instance, in [Chen et al. \(2016a\)](#) the problem of collision avoidance is treated for a multi-agent system (minimum $N = 3$ agents). They consider N agents, each having a similar dynamic and $N - 1$, so-called, danger zones: Z_{ij} . Every agent makes use of two controllers: a “liveness controller” helps to complete the agent own objective (e.g. reach a target) and a “safety controller” has to keep away the agent from the *danger zones* of the other ones. The architecture of the “safety controller” is centralized and it guarantees, for $N = 3$, that every agent avoids entering into any other danger zones. Therein, MIP is used to provide the higher level control logic to synthesize the cooperative “safety controller”. The objective is to group the agents in a pair such that the corresponding avoidance maneuvers may not lead to a dangerous configuration for the other agents.

For the generic obstacle and collision avoidance problem (in MI-MPC framework [Section 3.4](#)), in the corresponding OCP for the global system the dynamic behavior of the individual agent is coupled by means of the cost function and the constraints. Also, perfect knowledge of each agent dynamics (described by equations) is available to all the other agents. Consequently, the global model will be used in a predictive control context which permits the use of non-convex constraints for collision avoidance behavior.

A non-MPC example, [Wang et al. \(2015b\)](#) the problem studied in this paper is the collision avoidance in a multi-robot system. The approach is quasi-similar with one of previously-mentioned papers ([Haghighi et al., 2016](#)). The workspace is decomposed in equally-shaped cells and each cell cannot be visited by two robots at the same time. Each robot has to complete its own task by choosing from a set of possible trajectories. These trajectories are described by a sequence of adjacent cells and by the crossing time (in which an agent passes through respective cell). In this formulation the collision-avoidance problem becomes a resource allocation problem (the cells can be considered a shared resource). Assuming that the control of the agents is independent (i.e. they cannot pause their movement in order to give or to receive priority), the idea presented in this paper is to compute an initial delay time for each agent such that no collision to appear. In the centralized control strategy the trajectory is chosen by a central unit. Hence, the control algorithm has to return, besides the initial time delay, an “optimal” trajectory, the objective remaining unchanged. The associated optimization problem is

通过移动传感器。例如，Ritter 等人 (2014) 针对一个由多个配备传感器的自主车辆组成的移动传感器平台解决了这个问题。基于该过程模型和多智能体协作系统，提出了一种自适应观测策略，用于在线估计扩散过程的状态。多车协作系统使用 MLD 系统 (Bemporad & Morari, 1999) 进行建模；并控制，以便在同时避免智能体之间任何可能碰撞的情况下，在合理的时间内到达目标位置。智能体的目标位置是从基于 PDE²⁰ 的状态估计中接收的，更精确地说，是最佳测量位置。因此，MLD 系统为多车系统建模提供了一个很好的替代方案，因为所有现有约束（动力学、避碰、测量等）都嵌入在一个完整的配置中，这允许使用标准方法来解决控制问题。

此外，一个标准问题是会合机动，一组智能体（通常是一队航天器）需要在最短时间内达到特定队形。Papen等人（2017）研究了固定翼无人机队在包含静态和动态障碍物的环境中这一问题。控制策略通过分布式架构（具体为每个智能体应用模型预测控制）实现，每个智能体仅知道自己的轨迹（由当前位置和速度定义）。通过引入二元变量来建模障碍物并消除某些非线性约束（如速度和加速度约束），模型预测控制优化问题被转化为混合整数线性规划问题。值得注意的是，每个无人机后方形成的湍流涡流被建模为动态障碍物，因为这些区域难以保持对无人机的控制。求解所得混合整数线性规划问题需要相当长的时间，但通过适当放宽约束并进行微调，其复杂度是有限的。

5. 控制架构

存在三种成熟的控制架构类别，它们在各种应用领域中已被广泛研究：集中式、分布式和去中心化。后两种方法要求本地控制器仅优化其本地输入，具有相似的计算负担。这两种方法之间的区别在于通信的影响，去中心化控制不需要代理之间进行通信。

识别/讨论控制架构在运动规划领域的弱点和优势不属于本文的范围，因此，我们仅关注它们如何与MIP结合使用。在表3中，我们根据控制架构和公式中涉及的代理数量，对基于MIP的运动规划参考进行了分类。

值得一提的是，除集中式方法外，其他架构中采用的控制策略基于优化，偏爱MPC。因此，在本节中，我们将重点介绍具体的MPC实现。然而，我们没有忽视那些将MIP专门用于分布式/去中心化非MPC策略的参考，因为其能够构建任务分配问题。这些内容将在第3.1节中详细讨论。

涉及MIP的控制架构自然地从中式演变为分散式和分布式策略。

5.1. 集中式

由于其理论上的简单性，集中式方法是目前控制多智能体系统最常用的方式。在这种架构中，多智能体系统被视为一个整体，将扩展的单智能体系统等同。

表3
运动规划中MIP方法的分类。

智能体	集中式	分布式	分布式	参考文献（例如）
$N = 3 / N \geq 3$, 部分	✓	×	×	陈等人 (2016a)
1... N	✓	×	×	埃尔和达安德烈亚 (2005b) ; 哈吉吉等 (2016) ; 简切克等 (2017a) , Ritter 等人 (2014) ; Sun and Cassandras (2015)
1... N	✓	✓	×	Wang 等人 (2015b)
1... N	×	×	✓	Papen 等人 (2017)
$N = 1$	✓	×	×	Molinari 等人 (2017) ; Ragi 和 Mittelmann (2017)

注意: (✓) 已处理, (×) 未处理, (1... N) 任意数量的代理

物理限制（例如通信限制）被完全忽略，每个智能体对其他智能体的行为/动作拥有完全的知识，所有信息都在单个全局控制器中可用（[Rawlings 和 Mayne, 2009](#)）。然而，这种方法有限，不仅因为不可否认的物理约束，还由于扩展系统固有的复杂性引起的数值困难。

如第3节所述，在文献中，MIP可能出现在控制的不同层级。例如，在Chen等人（2016a）中，针对多智能体系统（最小 $N = 3$ 个智能体）的避碰问题进行了处理。他们考虑了 N 个智能体，每个智能体具有相似的动态特性，并存在 $N - 1$ 个所谓的危险区域: Z_{ij} 。每个智能体使用两个控制器：一个“生命控制器”帮助完成智能体自身的目标（例如，到达目标点），而一个“安全控制器”必须使智能体远离其他智能体的危险区域。‘安全控制器’的架构是集中式的，它保证对于 $N = 3$ ，每个智能体都能避免进入任何其他危险区域。在此，MIP被用于为合成协同‘安全控制器’提供高级控制逻辑。目标是分组智能体，使得相应的避碰动作不会对其他智能体造成危险配置。

对于通用障碍物和碰撞避障问题（在MI-MPC框架第3.4节中），在全局系统的相应OCP中，单个智能体的动态行为通过成本函数和约束条件相互耦合。此外，所有其他智能体都掌握了每个智能体动态（由方程描述）的完美知识。因此，全局模型将在预测控制环境中使用，该环境允许使用非凸约束条件来实现碰撞避障行为。

一个非MPC示例，Wang等人（2015b）本文研究的问题是在多机器人系统中的避碰问题。该方法与之前提到的某篇论文（[Haghighi等人, 2016](#)）的准相似。工作空间被分解为形状相同的单元格，并且每个单元格不能同时被两个机器人访问。每个机器人必须通过从一组可能的轨迹中选择来完成自己的任务。这些轨迹由一系列相邻单元格和穿越时间（即智能体通过相应单元格的时间）描述。在这种形式下，避碰问题变成了资源分配问题（单元格可以被视为共享资源）。假设智能体的控制是独立的（即它们不能暂停移动以给予或接收优先级），本文提出的方法是每个智能体计算一个初始延迟时间，以避免碰撞出现。在集中式控制策略中，轨迹由一个中央单元选择。因此，控制算法除了返回初始时间延迟外，还必须返回一条“最优”轨迹，目标保持不变。相关的优化问题是

²⁰ Partial Differential Equation.

²⁰ 偏微分方程。

MIP, binary variables modeling the disjunction, resulted from the corresponding resource allocation problem. Noteworthy, the objective is a minmax one because it is assumed that the robots work in parallel and the shortest (min) time for finishing the movement is the time of the slowest (max) robot. The objective can be reformulated as a standard minimization through the tools from, e.g., [Smith and Taskin \(2008\)](#).

While the complexity of MIP formulations increase in worst case situations exponentially w.r.t the number of binary variables, the reliability of MIP for real-time implementation decreases in the case of higher dimensions of the global systems. The same drawback is valid for the increase of the number of agents regardless of the formulation's efficiency.

5.2. Distributed

The underlying idea of distributed control approaches [Maestre, Negenborn et al. \(2014\)](#) is to divide the global control problem into a certain number of sub-problems, each involving a specific collection of local controllers or agents. Hence, each agent does not have access at the global information, but can be partially informed on the behavior of the other components of the local subsystem.

In large-scale multi-agent systems, where the agents are dispersed within a given workspace, it is more convenient to handle a set of smaller and/or simpler problems than treating the complex global system. The overall control strategy is given by the behavior of the local controllers, which may have a cooperative interaction. There are several advantages compared to centralized architecture: e.g., decrease of complexity and scalability. However, a loss of performance and of global stability may become apparent, being hard to be ensured compared to the centralized approach. Decoupling into independent nodes, design of robust control strategies, consensus seeking, all of these try to solve the problem, but still, with limited success [Cao, Yu, Ren, and Chen \(2012\)](#).

Within the MIP framework, most of the distributed control approaches using MPC strategy have been proposed. The features of MPC allow to handle explicitly the interactions between the different sub-systems/agents. For example, in [Schouwenaars \(2006\)](#), a distributed MPC strategy is employed for navigation of a fleet of vehicles through a partially unknown cluttered environment.

Due to the inherent problem of complexity, and implicitly the lack of scalability, MIP was not a popular method for distributed architectures. However, the control community has given a particular attention to the distributed MIP resolution algorithms. For instance, [Testa, Rucco, and Notarstefano \(2017\)](#) propose an algorithm for resolution of a MILP where the constraints are *distributed* among the agents. Likewise, [Vujanic, Esfahani, Goulart, Mariéthoz, and Morari \(2016\)](#) provides a decomposition method particularly useful for large scale MILPs, based on Lagrangian duality. Moreover, there are some works, which formulate the problem using MIP techniques, and for its resolution an heuristic method is employed, e.g., [Van Parys and Pipeleers \(2016\)](#) use Alternating Direction Method of Multipliers (ADMM).

5.3. Decentralized

During the last decades, decentralized control of multi-agent systems has gained a significant amount of attention due to the great variety of its applications, including multi-robot systems, transportation, multi-point surveillance and biological systems. An important topic of research is multi-agent navigation in both the robotics and the control communities, due to the need for autonomous control of multiple robotic agents in the same workspace. Important applications of multi-agent navigation arise also in the fields of air-traffic management and in autonomous driving by guaranteeing collision avoidance with other vehicles and obstacles.

As seen in the previous subsection, a main motivation for the modification of the control architecture is the computational burden of the centralized problem. A first alteration was to decompose

(“distribute”) the problem and solve the resulting sub-problems in expectation of achieving a consensus. Proceeding further and eliminating the aim of reaching consensus leads to a decentralized strategy (“everybody for itself”). Under this strategy, each agent owns a controller, which acts without taking into account any information about the behaviour of the other agents. Moreover, the exchange of information is limited (in most of the cases, reduced at a minimum).

A basic illustration for the decentralized approach is the merging junctions²¹ problem [Bali and Richards \(2018\)](#). In a centralized (or in a well constructed distributed) approach, whenever two (or more) agents arrive at an intersection, there is a prioritization w.r.t a well defined criterion. For the decentralized approach, with limited communication and with no guarantee of optimality, in the most probable scenario the system reaches a deadlock. In other words, the decentralized approach cannot provide, usually, the theoretical guarantees characterizing the centralized and distributed approaches. However, in practice, the decentralized methods may lead to efficient strategies, avoiding the computational demand of the others methods; but an understanding of the sensitive aspects is absolutely necessary.

For instance, recalling the previously-mentioned ([Section 5.1](#)) example from [Wang et al. \(2015b\)](#) – the problem of collision avoidance in a multi-robot system. Using a decentralized control strategy, they assume that each agent chooses its trajectory independently (from an a priori known set), without informing the rest of the systems. In this case the collision avoidance strategy should consider all possible trajectories combinations and has to return the initial time delays such that the movement to be finished in the shortest time without collisions. The resulting optimization problem is MIP, as in the centralized case.

6. Control applications using MIP

While previous sections deal with modeling and control issues, this section covers some critical details for both the computer simulations and the hardware implementation of the solutions and MIP approaches presented throughout this manuscript.

It is worth mentioning that there does not exist some clear and uncontested guidelines capable of generating the most efficient MIP formulation for a given problem. The performance²² of a formulation is customarily strongly dependent on the specific software tools or hardware platform. In the sequel, these aspects are briefly documented, the emphasis being placed on differences between the existing alternative implementations and their influence over the practical performance in motion planning.

6.1. Software for MIP

As was stated above, MIP is a powerful tool for planning and control problems due to its modeling capability and, additionally, the availability of specialized solvers. In the last decade, a consistent effort was put on developing MIP-specialized solvers in order to mitigate the numerical issues generated by the presence of integer/binary variables.

The optimization modeling languages are toolboxes which convert the mathematical formulation in a solvable form for the solvers.

Prior to proceeding further, we should emphasize the difference between optimization modeling languages and solvers. The former designates a toolbox/package/library which interacts with the latter: the mathematical model of the constrained optimization problem is put into an internal form which is then solved, and whose subsequent result is retrieved and displayed. Note that most modern tools do pre-processing steps (which may reduce the number of binary variables)

²¹ Unsignalized intersections.

²² Computational time, feasibility etc.

MIP, 二进制变量建模的析取, 来自相应的资源分配问题。值得注意的是, 目标是minmax型的, 因为它假设机器人并行工作, 完成移动的最短 (min) 时间是最慢 (max) 机器人的时间。目标可以通过, 例如, Smith和Taskin (2008) 中的工具重新表述为标准最小化形式。

虽然 MIP 公式的复杂性在最坏情况下相对于二进制变量的数量呈指数级增长, 但 MIP 在全局系统维度较高时的实时实现可靠性会下降。这种缺点同样适用于代理数量增加的情况, 无论公式的效率如何。

5.2. 分布式

分布式控制方法的基本思想 (Maestre, Negenborn 等人, 2014) 是将全局控制问题分解为若干个子问题, 每个子问题涉及一组特定的本地控制器或代理。因此, 每个代理无法获取全局信息, 但可以部分了解本地子系统中其他组件的行为。

在大型多代理系统中, 代理分散在给定的工作空间内, 处理一组更小和/或更简单的问题比处理复杂的全局系统更为方便。整体控制策略由本地控制器的行为决定, 这些控制器可能存在协同交互。与集中式架构相比, 分布式架构有几个优点: 例如, 复杂性和可扩展性降低。然而, 性能和全局稳定性的损失可能变得明显, 与集中式方法相比, 很难确保。解耦为独立节点、设计鲁棒控制策略、寻求共识等, 都试图解决这个问题, 但仍然效果有限 (曹、余、任和陈, 2012)。

在MIP框架内, 大多数采用MPC策略的分布式控制方法已被提出。MPC的特性允许显式处理不同子系统/代理之间的交互。例如, 在 Schouwenaars (2006) 中, 采用了一种分布式MPC策略, 用于在部分未知且杂乱的环境中导航一组车辆。

由于固有的复杂性问题, 以及隐含的可扩展性不足, MIP并非分布式架构的流行方法。然而, 控制领域特别关注分布式MIP求解算法。例如, Testa、Rucco和Notarstefano (2017) 提出了一种用于求解MILP的算法, 其中约束分散在各个代理之间。同样, Vujanic、Esfahani、Goulart、Mariéthoz和Morari (2016) 提供了一种基于拉格朗日对偶性的分解方法, 特别适用于大规模MILP。此外, 一些工作采用MIP技术构建问题, 并使用启发式方法求解, 例如, Van Parys和Pipeleers (2016) 使用了乘子方向交替法 (ADMM)。

5.3. 去中心化

在过去的几十年里, 多智能体系统的去中心化控制因其广泛的应用而获得了大量关注, 包括多机器人系统、交通、多点监控和生物系统。在机器人技术和控制领域, 多智能体导航是一个重要的研究课题, 因为需要在同一工作空间中对多个机器人代理进行自主控制。多智能体导航在空中交通管理和自动驾驶领域也有重要的应用, 通过保证与其他车辆和障碍物的碰撞避免来实现。

如前一节所述, 修改控制架构的主要动机是集中式问题的计算负担。第一次变更是对分解

(“分布式”)的问题, 并在预期达成共识的情况下解决由此产生的子问题。进一步消除达成共识的目标会导致去中心化策略 (“各顾各”)。在此策略下, 每个智能体都拥有一个控制器, 该控制器在不考虑其他智能体行为信息的情况下独立运行。此外, 信息交换有限 (在大多数情况下, 减少到最低限度)。

去中心化方法的一个基本示例是合并交叉路口问题²¹, Bali和Richards (2018)。在集中式 (或结构良好的分布式) 方法中, 每当两个 (或更多) 代理到达交叉路口时, 都会根据一个明确定义的标准进行优先级排序。对于去中心化方法, 由于通信有限且没有最优性保证, 在最可能的情况下, 系统会达到死锁。换句话说, 去中心化方法通常无法提供集中式和分布式方法所具有的理论保证。然而, 在实践中, 去中心化方法可能会带来高效的策略, 避免其他方法的计算需求; 但必须充分理解其敏感方面。

例如, 回忆之前提到的 (第5.1节) 王等 (2015b) 的例子——多机器人系统中的避碰问题。他们采用分布式控制策略, 假设每个智能体独立选择其轨迹 (从一个先验已知的集合中), 而无需通知其他系统。在这种情况下, 避碰策略应考虑所有可能的轨迹组合, 并必须返回初始时间延迟, 以确保在最短的时间内无碰撞完成运动。由此产生的优化问题是MIP, 与集中式情况相同。

6. 使用MIP的控制应用

虽然之前的章节处理建模和控制问题, 但本节涵盖了贯穿本文提出的解决方案和MIP方法在计算机仿真和硬件实现方面的关键细节。

值得提及的是, 目前尚不存在一些明确且无争议的指南, 能够为给定问题生成最有效的MIP公式。公式的性能²²通常强烈依赖于特定的软件工具或硬件平台。在后续内容中, 这些方面将简要记录, 重点在于现有替代实现之间的差异及其对运动规划实际性能的影响。

6.1. MIP相关软件

如前所述, 由于MIP具备建模能力, 并且此外还有专门的求解器可用, 因此它是一种强大的规划与控制问题工具。在过去十年中, 人们持续投入努力开发MIP专用求解器, 以缓解整数/二进制变量存在所引发的数值问题。

优化建模语言是工具箱, 它们将数学公式转换为求解器可处理的格式。

在继续前进之前, 我们应该强调优化建模语言和求解器之间的区别。前者指的是一个工具箱/包/库, 它与其他者交互: 约束优化问题的数学模型被转换成内部形式, 然后被求解, 其后续结果被检索并显示。请注意, 大多数现代工具会进行预处理步骤 (这可能减少二元变量的数量)

²¹ 无信号交叉口。 ²² 计算时间、可行性等。

or may even, automatically, put the problem in a MI form (e.g., in YALMIP when specifying a complementarity condition or a bilevel program).

Diverse programming languages and online resources facilitate the specification and resolution of MIP problems. In addition to classical Matlab, recently the attention of the optimization-based control community is moving towards other advanced programming languages as Python or Julia, which have become more accessible to the broader scientific and engineering community. In Table 4 we summarize these programming languages, the modeling languages and the solvers which may be joined in order to solve MIP.

Without being exhaustive, there are some popular optimization modeling tools: YALMIP (Lofberg, 2004), MPT (Herceg, Kvasnica, Jones, & Morari, 2013), AMPL (Fourer, Gay, & Kernighan, 1987), CVXPY (Diamond & Boyd, 2016), PYOMO (Hart et al., 2017) or JuMP (Dunning, Huchette, & Lubin, 2017). All²³ of them are open-source modeling languages which allows users to express a wide range of optimization problems (not exclusively, MIP) in a high-level (almost algebraic or pseudo-code) syntax. As depicted in Table 4, each modeling language is developed taking into account the specific features of a programming language. Note that an hierarchy among these modeling tools is strongly dependent on the experience of the user/researcher, the preference for one of them has a negligible impact regarding the resolution performances (e.g., computational burden). These performances are influenced rather by the choice of the solver w.r.t. the formulation.

Regarding the solvers, there exists a broad variety of options, we mention here that the most used in the field of MIP-based motion planning are CPLEX (2009), Inc. (2014) or Mosek (2015). For instance, Huang and Peng (2017); Zhang et al. (2017b) or Mukai et al. (2017), use GUROBI, while Richards and How (2002), Schouwenaars, Richards, Feron, and How (2001b) or Earl and D’Andrea (2005a) utilize CPLEX.

Remark 15. The majority of the solvers (mentioned in Table 4) is able to deal with quadratic objectives and/or constraints, elements which are influential in many control strategies and/or applications, see, e.g., (18). ♦

While a review of how the resolution techniques employed by the solvers is well beyond the scope of this paper, we mention only the core techniques: branch-and-bound and cutting-plane algorithms. There exists a multitude of variations, each with its strengths and weaknesses. For instance, branch-and-cut method (Stubbs & Mehrotra, 1999) gathers the advantages of branch-and-bound and cutting-plane algorithms, decreasing the number of nodes to be explored in the search tree by

Table 4
Software for MIP implementation-examples.

Software		References(e.g.)
Programming Language	MATLAB	Bemporad and Mignone (2000); Culligan (2006); Janeček et al. (2017a)
	Python	Welder et al. (2018)
	Julia	Lubin, Yamangil, Bent, and Vielma (2018)
	Yalmip	Mukai et al. (2017); Zhang et al. (2017b)
Modelling Language	GAMS	Lee and Grossmann (2000)
	CVX/ CVXPY	Dueri et al. (2017)
	Pyomo	Legg et al. (2012)
	JuMP	Welder et al. (2018)
Solvers	AMPL	Richards and How (2005, 2002)
	CPLEX	Janeček et al. (2017a); Stoican et al. (2013)
	GUROBI	Janeček et al. (2017a)
	SCIP	Berthold et al. (2012)

iteratively introducing constraints to cut the feasible region.

In general, the solvers may be classified using different criteria, e.g. convex/non-convex, heuristic/deterministic. There are in the literature more detailed surveys, e.g. Belotti et al. (2013), treating this topic, but what is relevant from the current paper’s perspective is the following aspect. Some of the currently available and reliable solvers may employ heuristics in order to accelerate the standard algorithms. This is a necessary requirement especially for complex (large) problem formulations and real-time resolution. As a word of caution, the performances (particularly, computational times) may vary considerably from a solver to another due to the use of heuristics. Hence, the concept of “the best MIP solver” is pointless, in our opinion. On a more positive note, we have observed that for any given problem there can be found a solver, capable of handling it.

Besides these powerful commercial solvers, there exists a variety of non-commercial/open-source solvers able to provide reasonable performances, in some cases better than commercial ones. An important characteristic of this kind of solvers is that their adjustment and adaptation to the challenges encountered in the real-world applications and to the real-time control requirements can be done in a faster and more straightforward manner than commercial ones which, in addition, have to take into account commercial considerations, balance between free/payed functionalities, etc. For instance, SCIP (https://scip.zib.de) (Achterberg, 2009) was at the beginning a MILP solver which implemented the branch-and-bound algorithm with various heuristics, while the later versions were able to solve MINLPs with quadratic objective, non-convex MINLPs or MISDPs (mixed integer semidefinite programs).

Moreover, a part of the control community has concentrated its attention on (as in Table 5) techniques to adapt the standard MIP resolution algorithms. For instance, Bemporad, Mignone, and Morari (1999) propose an efficient branch-and-bound algorithm, enhancing the tree exploring strategy. The application therein involves the control and state-estimation for a MLD-system (Bemporad & Morari, 1999). Similarly, Bemporad (2015) provides an algorithm which combines the classical branch-and-bound with nonnegative-least-squares (NNLS) methods, in order to solve MIQP problems generated by the hybrid MPC applications. The idea is further developed in Naik and Bemporad (2017) where the NNLS is replaced by accelerated-dual-gradient projection algorithm.

There are also some other works which exploit the particularities of the problem structure within branch-and-bound algorithm. For example, Feng, Villanueva, Chachuat, and Houska (2017) propose a variation of the branch-and-bound algorithm, branch-and-lift algorithm, which has better performances in resolution of a classical obstacle avoidance problem. As well, Hespanhol, Quirynen, and Di Cairano (2019) provide an iterative variation branch-and-bound algorithm which exploits the block-sparse optimal control structure of the problem and also the information at previous time steps.

As a side remark, besides the standard MIP resolution algorithms,

Table 5
Alternative and heuristic techniques for MIP resolution.

Alternative	References
Branch-and-bound variation	Bemporad (2015); Bemporad and Mignone (2000); Bemporad et al. (1999); Feng et al. (2017); Hespanhol et al. (2019); Naik and Bemporad (2017)
ADMM	Abboud et al. (2015); Kanno and Kitayama (2018); Scott and Thiébaux (2014); Takapoui et al. (2020); Timotheou et al. (2014)
FP	Fischetti et al. (2005); Fischetti and Salvagnin (2009); Miertoiu and Dumitrescu (2019)

或者甚至可以自动将问题转换为MI形式（例如，在指定互补条件或双层程序时，在YALMIP中）

不同的编程语言和在线资源促进了MIP问题的规范化和解决。除了经典的Matlab之外，近年来基于优化的控制社区的关注点正转向其他更高级的编程语言，如Python或Julia，这些语言对更广泛的科学和工程社区来说变得更加容易获取。在表4中，我们总结了这些编程语言、建模语言以及求解器，它们可以组合起来以解决MIP问题。

Without being exhaustive, there are some popular optimization modeling tools: YALMIP (Lofberg, 2004), MPT (Herceg, Kvasnica, Jones, & Morari, 2013), AMPL (Fourer, Gay, & Kernighan, 1987), CVXPY (Diamond & Boyd, 2016), PYOMO (Hart et al., 2017) or JuMP (Dunning, Huchette, & Lubin, 2017). All²³ of them are open-source modeling languages which allows users to express a wide range of optimization problems (not exclusively, MIP) in a high-level (almost algebraic or pseudo-code) syntax. As depicted in Table 4, each modeling language is developed taking into account the specific features of a programming language. Note that an hierarchy among these modeling tools is strongly dependent on the experience of the user/researcher, the preference for one of them has a negligible impact regarding the resolution performances (e.g., computational burden). These performances are influenced rather by the choice of the solver w.r.t. the formulation.

关于求解器，存在多种选择，我们在此提到，在基于MIP的运动规划领域中最常用的有CPLEX（2009）、Inc.（2014）或Mosek（2015）。例如，Huang和Peng（2017）；Zhang等人（2017b）或Mukai等人（2017）使用GUROBI，而Richards和How（2002）、Schouwenaars、Richards、Feron和How（2001b）或Earl和D’Andrea（2005a）则使用CPLEX。

注释15. 表4中提到的多数求解器能够处理二次目标函数和/或约束条件，这些元素对许多控制策略和/或应用具有重要影响，例如（18）。♦

尽管求解器所采用的求解技术方法的回顾已远远超出本文的范围，但仅提及核心技术：分支定界法和割平面算法。存在多种变体，每种都有其优缺点。例如，分支定界割方法 (Stubbs & Mehrotra, 1999) 结合了分支定界法和割平面算法的优点，通过迭代引入约束来切割可行域，从而减少搜索树中需要探索的节点数量。

表4
用于MIP实现的软件示例。

软件		参考文献（例如）
编程语言	MATLAB	Bemporad 和 Mignone (2000); Culligan (2006); Janeček 等人 (2017a)
	Python	Welder 等人 (2018)
	朱莉娅	鲁宾、亚曼吉尔、本特和维埃拉马 (2018)
建模语言	Yalmip	村上等人 (2017) ； 张等人 (2017b)
	GAMS	Lee 和 Grossmann (2000)
求解器	CVX/	Dueri 等人 (2017)
	CVXPY	
	Pyomo	Legg 等人 (2012)
	JuMP	Welder 等人 (2018)
	AMPL	Richards和How (2005, 2002)
	CPLEX	Janeček等人 (2017a)； Stoican等人 (2013)
	GUROBI	Janeček等人 (2017a)
SCIP	Berthold 等人 (2012)	

迭代引入约束以切割可行域。

通常，求解器可以根据不同标准进行分类，例如凸/非凸、启发式/确定性。文献中有更详细的综述，例如Belotti等人（2013）对此主题进行了处理，但从当前论文的角度来看，重要的是以下方面。目前可用且可靠的某些求解器可能会采用启发式方法来加速标准算法。这对于复杂的（大型的）问题公式化和实时求解来说是一个必要要求。需要注意的是，由于使用启发式方法，不同求解器的性能（尤其是计算时间）可能会有很大差异。因此，我们认为“最佳MIP求解器”的概念是没有意义的。从更积极的角度来看，我们观察到对于任何给定问题，都可以找到一个能够处理它的求解器。

除了这些强大的商业求解器之外，还存在多种非商业/开源求解器，它们能够提供合理的性能，在某些情况下甚至优于商业求解器。这类求解器的一个重要特征是，它们能够比商业求解器更快、更直接地调整 and 适应实际应用中遇到的挑战以及实时控制要求。此外，商业求解器还需要考虑商业因素、免费/付费功能的平衡等。例如，SCIP（https://scip.zib.de）（Achterberg，2009）最初是一个MILP求解器，它实现了带有各种启发式的分支定界算法，而后续版本则能够求解具有二次目标函数的MINLPs、非凸MINLPs或MISDPs（混合整数半定规划）。

此外，控制领域的一部分研究人员已经集中注意力（如表5所示）于改进标准MIP求解算法的技术。例如，Bemporad、Mignone和Morari（1999）提出了一种高效的分支定界算法，增强了树探索策略。该应用涉及对MLD系统（Bemporad & Morari，1999）的控制和状态估计。类似地，Bemporad（2015）提供了一种结合经典分支定界和非负最小二乘（NNLS）方法的算法，用于求解混合MPC应用生成的MIQP问题。该思想在Naik和Bemporad（2017）的研究中得到了进一步发展，其中NNLS被替换为加速对偶梯度投影算法。

此外，还有一些利用分支定界算法中问题结构的特殊性的其他研究。例如，Feng、Villanueva、Chachuat和Houska（2017）提出了一种分支定界算法的变体——分支提升算法，该算法在解决经典避障问题方面表现更优。同样地，Hespanhol、Quirynen和Di Cairano（2019）提供了一种迭代式分支定界算法变体，该算法利用了问题的块稀疏最优控制结构，并利用了先前时间步的信息。

作为旁注，除了标准的MIP求解算法之外，

表 5
MIP 求解的替代与启发式技术。

替代	参考文献
分支定界法变体	Bemporad (2015); Bemporad and Mignone (2000); Bemporad 等人 (1999); Feng 等人 (2017); Hespanhol 等人, (2019); Naik 和 Bemporad (2017)
ADMM	Abboud 等人 (2015); Kanno 和 Kitayama (2018); Scott 和 Thiébaux (2014); Takapoui 等人 (2020);
FP	Timotheou 等人 (2014) Fischetti 等人 (2005); Fischetti 和 Salvagnin (2009); Miertoiu 和 Dumitrescu (2019)

²³ Except AMPL.

²³ 除AMPL外。

there exist in the literature some heuristic (see Table 5) techniques. Without being exhaustive, we mention only two²⁴ of these:

- i) ADMM (alternating direction method for multipliers)-based methods Kanno and Kitayama (2018), Takapoui, Moehle, Boyd, and Bemporad (2020). Although an algorithm for solving convex optimization problems, ADMM turns out to be an effective method to approximately solve some nonconvex problems as well. The idea behind the heuristic method is to use multiple restarts of ADMM with random initial points, in most of cases this provides an acceptable solution with small computational cost. This technique is frequently used in optimal (power) flow problems, e.g. traffic signal control Timotheou, Panayiotou, and Polycarpou (2014) or network Abboud, Bařtuę, Hamidouche, and Debbah (2015); Scott and Thiébaux (2014). However, Van Parys and Pipeleers (2016) employ ADMM in motion planning for a multi-vehicle systems. As well, for this technique there exist a multitude of variations and adaptations. For instance, Stellato, Naik, Bemporad, Goulart, and Boyd (2018) develop a novel branch-and-bound algorithm, tailored for a ADMM-based solver.
- ii) FP (Feasibility Pump) Fischetti, Glover, and Lodi (2005). A heuristic method for finding a feasible solution of a given MIP, FP aims to minimize the difference between the solution of the LP-relaxation and the one of the original MIP. For instance, Miertoiu and Dumitrescu (2019) use and adapt the algorithm for sparse representation.

Remark 16. Apart of solvers and modeling languages, there exist in the literature works as Janeček et al. (2017a) which provide a toolbox for MPC-based control for obstacle avoidance problem. The toolbox is developed in an object-oriented manner, allowing to readily set up the associated MPC problem and, afterwards, providing an efficient formulation of the underlying optimization problem for any inexperienced user. In other words, the OPTIPLAN toolbox encapsulates all aspects related to formulation and resolution of the MIP problem. ♦

6.2. Hardware platforms

Before detailing, it is worth mentioning that some of the existing works takes into account the particularities of the hardware platforms developing specific methods, whereas most of the remaining presents generic methods able (or not) to adapt to the constructive constraints.

There exists a broad variety of robotic platforms which are used in academic and/or commercial applications: aerial, surface or underwater vehicles. Frequently, these robots are involved in activities which are unsafe or troublesome for humans. Even though different levels of autonomy are possible, we can easily remark that the control community has considered with predilection the unmanned vehicles. The main justification comes from the elimination/mitigation of human risk. This aspect has beneficial consequences on cost effectiveness and, in most of the cases, on accuracy of the operations.

In Table 6, we depicted the classes of unmanned vehicles which have been used as hardware platforms for MIP-based motion planning problems: UAV (unmanned aerial vehicle), USV (unmanned surface vehicle) or UGV (unmanned ground vehicle), UUV (unmanned underwater vehicle). The specific characteristics of each class (or, sometimes, subclass) of vehicles lead to various challenges in the design of the motion planning strategy. For instance, the USVs/UGVs are moving in a bidimensional workspace, whereas the UAVs and UUV in a 3D workspace thus leading to a higher complexity of the control problem. Another classification is given by the ability to stop and/or go backwards, for example, fixed-wing UAV need to maintain a minimum

Table 6
Classification of the hardware platform.

Experimental platform	References
UAVs	Bellingham et al. (2002a, 2002b); Chen et al. (2016a); Culligan (2006); Papen et al. (2017); Ragi and Mittelmann (2017); Rey and Hijazi (2017); Richards and How (2002); Schouwenaars (2006)
UGVs	Bali and Richards (2018); Ballesteros-Tolosana et al. (2017); Fayazi et al. (2017); Molinari et al. (2017)

velocity (to avoid stalling), but the quadcopters/helicopters (rotary wing UAVs), having more degrees of freedom, can maintain an arbitrary velocity (up to being stationary in midair, i.e. hovering).

For instance, in Culligan (2006) an experimental validation of the MILP framework is done: test flights on an indoor quadrotor testbed demonstrate the reliability of the approach for the optimal path planner. For example, using the MILP path planner to create a plan ten seconds into the future, the quadrotor can navigate through an obstacle-rich field with MILP solve times under one second. Simple plans in obstacle-spare environments are solved in less than 50ms. A multi-vehicle test is also used to show non-communicating deconfliction trajectory planning using MILP.

Many applications in precision agriculture, disaster management and target tracking assume a collaboration between an UAV and ground-based sensors. The UAV serves as a mobile sink: it prolongs the sensors’ lifetime (by canceling their need to communicate with a base station Xu, Xiao, Yang, Cuthbert, & Wang (2018)) and reduces operational costs (by canceling the need of direct human supervision Jawad, Nordin, Gharghan, Jawad, & Ismail (2017)). Such applications impose energy-based limitations in the motion planning procedure, either induced by path length Khan and Kumar (2016) or by communication requirements Xu et al. (2018). Furthermore, many works simplify the motion planning by assuming predefined path primitives (e.g., the UAV is constrained to move in straight, parallel lines Wang, Ma, Yan, De, and Das (2015a) or in spirals Yue and Jiang (2018)). Not in the least, when the environment is cluttered or uneven, the communication links may be weakened or lost as a result of signal attenuation Jawad et al. (2017). Thus, bounds on the communication time at a waypoint have to be considered, which are difficult to handle by fixed-wing UAVs. As stated elsewhere, the result is a nonlinear (in cost and constraints) constrained optimization problem which is often impractical to solve. Even a relatively simple requirement as ensuring hovering at the waypoint leads to a MINLP formulation (Mathur, Nielsen, Prasad, & Prasad, 2016).

7. Alternatives to MIP

This section briefly presents the existing alternatives to MIP which are extensively used in motion planning problems. We delineate in Table 7 the state-of-the-art references for each of these alternatives.

Table 7
Classification of the alternatives of MIP in motion planning.

Method/Technique	References
Graph-based algorithms	Hsu et al. (2007); Karaman and Frazzoli (2011); Ladd and Kavraki (2004); LaValle (2006); Lozano-Pérez and Wesley (1979); Weiss et al. (2017)
Potential field formulation	Chen and Wang (2005); Filotheou et al. (2018); Rimón and Koditschek (1992); Vlantis, Vrohidis, Bechlioulis, and Kyriakopoulos (2018); Vrohidis, Vlantis, Bechlioulis, and Kyriakopoulos (2018)
Convex relaxation	Liu and Lu (2014); Mao et al. (2017, 2016) Harris and Açıkmęşe (2014), Rey and Hijazi (2017) Frasch et al. (2013); Janeček et al. (2017a); Yu et al. (2014)

文献中还存在一些启发式（参见表5）技术。不追求详尽，我们仅提及其中两种²⁴：

- i) 基于ADMM（交替方向乘法）的方法 Kanno和Kitayama (2018), Takapoui, Moehle, Boyd, 和 Bem- porad (2020)。尽管ADMM是一种用于求解凸优化问题的算法，但它也被证明是一种有效的近似求解某些非凸问题的方法。启发式方法背后的思想是使用具有随机初始点的ADMM多次重启，在大多数情况下，这能以较小的计算成本提供可接受的解。该技术经常用于最优（电力）流问题，例如交通信号控制 Timotheou, Panayiotou, 和 Polycarpou (2014) 或网络 Abboud, Bastuę, Hamidouche, 和 Debbah (2015); Scott 和 Thiébaux (2014)。然而, Van Parys 和 Pipeleers (2016) 将ADMM应用于多车辆系统的运动规划。此外，这种技术还存在许多变体和改进。例如, Stellato, Naik, Bemporad, Goulart, 和 Boyd (2018) 开发了一种新型分支定界算法，专为基于ADMM的求解器设计。

- ii) FP (可行性泵) Fischetti, Glover, 和 Lodi (2005)。这是一种启发式方法，用于寻找给定MIP的一个可行解，FP旨在最小化LP松弛解与原始MIP解之间的差异。例如, Miertoiu 和 Dumitrescu (2019) 使用并调整了该算法用于稀疏表示。

注释16. 除了求解器和建模语言之外，文献中还存在像Janeček等人 (2017a) 那样提供基于MPC的避障控制工具箱的研究。该工具箱采用面向对象的方式开发，允许用户方便地设置相关的MPC问题，之后为任何不熟悉优化的用户提供高效的问题表述。换句话说，OPTIPLAN工具箱封装了与MIP问题表述和求解相关的所有方面。♦

6.2. 硬件平台

在详细说明之前，值得提及的是，一些现有工作考虑了硬件平台的特殊性，并开发了特定方法，而其余大部分则提出了通用的方法，这些方法能够（或不能）适应构造约束。

存在多种多样的机器人平台，这些平台被用于学术和/或商业应用：空中、水面或水下车辆。通常，这些机器人会参与对人类来说不安全或麻烦的活动。尽管可以实现不同级别的自主性，但我们很容易注意到，控制领域更倾向于研究无人驾驶车辆。主要理由在于消除或减轻人类风险。这一方面对成本效益有积极影响，并且在大多数情况下，对操作的准确性也有积极影响。

在表6中，我们展示了作为基于MIP的运动规划问题的硬件平台所使用的无人车辆类别：UAV（无人机）、USV（无人水面艇）或UGV（无人地面车辆）、UUV（无人水下航行器）。每种车辆类别（或有时是子类别）的具体特性导致运动规划策略设计时面临各种挑战。例如，USV/UGV在二维工作空间中移动，而UAV和UUV则在三维工作空间中移动，因此导致控制问题更加复杂。另一种分类是根据能否停止和/或后退的能力，例如固定翼UAV需要保持最低速度（以避免失速），但四旋翼/直升机（旋翼UAV）具有更多自由度，可以保持任意速度（甚至能在空中悬停）。

²⁴ 最常见的。

表6
硬件平台的分类。

实验平台	参考文献
UAVs	Bellingham 等人 (2002a, 2002b); Chen 等人 (2016a); Culligan (2006); Papen 等人 (2017); Ragi and Mittelmann (2017); Rey and Hijazi (2017); Richards and How (2002); Schouwenaars (2006)
UGVs	Bali and Richards (2018); Ballesteros-Tolosana 等人 (2017); Fayazi 等人 (2017); Molinari 等人 (2017)

速度（以避免失速），但四旋翼飞行器/直升机（旋翼无人机），由于拥有更多自由度，可以保持任意速度（甚至能在空中悬停，即悬停）。

例如，在Culligan（2006）中，对MILP框架进行了实验验证：在室内四旋翼测试平台上进行的测试飞行展示了该方法对于最优路径规划器的可靠性。例如，使用MILP路径规划器创建一个未来十秒的计划，四旋翼可以在不到一秒的MILP求解时间内通过一个障碍物丰富的区域。在障碍物稀疏的环境中，简单的计划可以在不到50毫秒内求解。还使用多车辆测试来展示使用MILP进行非通信解耦轨迹规划。

精准农业、灾害管理和目标跟踪中的许多应用都假设无人机与地面传感器之间的协作。无人机充当移动汇聚点：它通过取消传感器与基站通信的需求来延长其寿命（Xu, Xiao, Yang, Cuthbert, & Wang (2018)），并通过取消直接人工监督的需求来降低运营成本（Jawad, Nordin, Gharghan, Jawad, & Ismail (2017)）。此类应用在运动规划过程中引入基于能量的限制，要么由路径长度引起（Khan and Kumar (2016)），要么由通信需求引起（Xu et al. (2018)）。此外，许多研究通过假设预定义的路径基元来简化运动规划（例如，无人机被限制在直线、平行线（Wang, Ma, Yan, De, and Das (2015a)）或螺旋线（Yue and Jiang (2018)）中移动）。尤其是在环境杂乱或不均匀时，通信链路可能因信号衰减而减弱或丢失（Jawad et al. (2017)）。因此，必须考虑航路点的通信时间限制，这对固定翼无人机来说难以处理。如前所述，结果是一个非线性（在成本和约束方面）的约束优化问题，通常难以求解。即使是一个相对简单的需求，如确保在航路点悬停，也会导致MINLP公式（Mathur, Nielsen, Prasad, & Prasad, 2016）。

7. MIP的替代方案

本节简要介绍了在运动规划问题中广泛使用的 MIP 的现有替代方案。我们在表 7 中列出了这些替代方案的最新参考文献。

表 7
运动规划中 MIP 方案的分类。

方法/技术	参考文献
基于图的算法	Hsu 等人 (2007); Karaman 和 Frazzoli (2011); Ladd 和 Kavraki (2004); LaValle (2006); Lozano-Pérez 和 Wesley (1979); Weiss 等人 . (2017)
势场公式化	陈和王 (2005); Filotheou 等人 (2018); Rimón 和 Koditschek (1992); Vlantis, Vrohidis, Bechlioulis, 和 Kyriakopoulos (2018); Vrohidis, Vlantis, Bechlioulis, 和 Kyriakopoulos (2018)
凸松弛	刘和陆 (2014); 毛等人 (2017, 2016) Harris 和 Açıkmęse (2014), Rey and Hijazi (2017) Frasch 等人 (2013); Janeček 等人 (2017a); 余等人 (2014)

²⁴ The most encountered.

7.1. Alternative formulations

Alternative formulations for MIP range from graph-based approaches, potential field formulations to the corresponding optimization problems relaxations.

Graph-based approaches

In contrast to the MIP approaches where the discrete decisions are encoded in a mathematical formalism and are solved as such, the graph-based approaches reduce these discrete decision to the search of the shortest path between nodes in a graph. Although these techniques can be applied for any MIP-based motion planning problems (see Section 3), in this subsection we concentrate on collision-free path/trajectory planning.

Remark 17. For finding of the shortest path in a graph there exists in the literature various algorithms (e.g., LaValle (2006), Latombe (2012)). The most influential ones are Dijkstra's, greedy or A* search algorithms. ♦

The graph-based approaches in path planning are classified by graph construction. A first category builds the graph based on the cell decomposition methods, considering an explicit representation of the (multi-obstacle) environment. Thus, the similarity with the MIP approach is undeniable, polytopic sets being a popular decomposition primitive. For instance, polytopic sets were already used in Lozano-Pérez and Wesley (1979). More precisely, the graph nodes are defined by the vertices of the polytopic obstacles, and by the initial and the final point of the path. This graph is named the *visibility graph* because the link between the nodes is given by a straight line which does not intersect any obstacles. In other words two linked nodes can “see” each other. Thus, the collision-free path is the shortest path through this graph between the initial and final position of the agent. This approach performs well from a theoretical perspective if the considered agent is a point. For the general case when the dimensions are not negligible, the above method is extended by considering an artificial “growing” of the obstacles (commonly, via Minkowski sum). The main drawback of the above presented method is the computational burden, specifically in a complex environment (large number of obstacles with complicated forms) where the number of vertices is exhaustive.

The second category tries to mitigate this issue avoiding an explicit representation of the environment. This type of approach is also referred in the literature as *sampled-based algorithms* (see LaValle (2006)) for a complete literature review on this topic). Hence, the explicit representation is eliminated and the focus is put on a collision checking module. This module checks the feasibility of the trajectories connecting random sampled points from the obstacle-free space. The graph is constructed such as the link between two nodes represents a feasible path in the (multi-obstacle) environment.

Remark 18. An interesting notion for sampled-based algorithms is the *probabilistic completeness* (Barraquand et al., 1997; Ladd & Kavraki, 2004). That is, the probability that the algorithm returns a feasible solution tends to one if the number of sample points is sufficiently large ($\rightarrow \infty$). This is empirically demonstrated in Hsu, Latombe, and Kurniawati (2007). ♦

In the literature, there are two important sampled-based algorithms: PRM (Probabilistic RoadMaps) (Hsu et al., 2007) and RRT (Rapidly-exploring Random Tree) (Weiss, Danielson, Berntorp, Kolmanovsky, & Cairano, 2017). The difference between them is given by the method of constructing the graph. The former (PRM) is a multiple-query method in the sense that after the construction of the roadmap (a rich set of feasible paths) it answers queries by computing an optimal path through the graph. Henceforth, the PRM is an useful method if an awareness map of the environment is available (Karaman & Frazzoli, 2011). There are a plethora of variants for the PRM, each representing a

valuable improvement Ladd and Kavraki (2004), Karaman and Frazzoli (2011).

For the case when the environment is not a priori known, the RRT method is more suitable. In this approach the construction of the graph is incremental, the algorithm stops when a large enough set of collision-free paths is attained. Thus, a collision-free sample is added as a node in the graph and is linked with the surrounding nodes. The obtained graph is actually a tree. As PRM, RRT has a variety of versions. Some take into account the equations of motion and generate attainable paths LaValle and Kuffner (2001), others generate only geometric paths which became reference trajectories for a lower level controller. Moreover, some versions are tailored for complex and/or unstable dynamics Leonard et al. (2008) or for uncertain dynamics Weiss et al. (2017).

Besides the classical graph-based approach, there exists in the literature a multitude of methods which combine the standard graph algorithms with advanced control strategies. For instance, Berntorp, Danielson, Weiss, and Di Cairano (2018) presents a method for real-time integrated motion planning which uses feedback control, positive invariant sets, and equilibrium trajectories of the closed-loop system. In order to generate the collision-free trajectories the method employs, in an offline manner, a graph search over reference paths, each being associated with a constraint admissible positive invariant set. Next, they use pre-designed unconstrained linear quadratic controllers to track the reference paths.

As well, Althé and Fortelle (2017) addresses the problem of trajectory planning using an approach which consists in partitioning the feasible (“collision-free”) region, while allowing to decompose the NP-hard problem as a path-finding problem in a well-designed graph followed by a (simple) optimization phase (“in MPC fashion”) for a quadratic convex cost function. Also, Franzè and Lucia (2015) deals with the problem of obstacle avoidance in an unknown environment (considered agents are UGVs - autonomous ground vehicles). The proposed approach consists of two parts: an offline part which computes the ellipsoidal approximation of the one-step controllable sets for all possible scenarios (these approximations guarantee the existence of a feasible path through multi-obstacles environment) and a online part which involves the development of a MPC-based strategy in order to keep the agent in that sequence of ellipsoidal sets.

Potential field formulation

While MIP-based methods take explicitly into account the constraints and lead to a constrained optimization problem, the potential field-based formulations Chen, Luo, Mei, Yu, and Su (2016b) relax the constraints by adding penalty terms in the cost. Essentially, the potential field approach relies on construction of a scalar function (so-called, the potential). This function takes high values when the agent stays within the interdicted zones. In the collision-free workspace the function is decreasing towards the goal configuration (i.e., the potential associated to the destination point is minimal). Thus, the agent may attain the final point moving in the direction of the negative gradient of the potential. Rimon and Koditschek (1992) provides an historical (and more detailed) review on the potential field formulation and how this approach is involved in motion planning. An interesting characteristics is that the potential field formulation is frequently used in decentralized or distributed control strategies Filotheou, Nikou, and Dimarogonas (2018).

7.2. Optimization problem relaxation

As was stated above, one of the most significant capabilities of MIP is to handle non-convex constraints in non-convex optimal control problems. A natural way to tackle this type of problems is by extending the methods and techniques used for the convex optimal control. Often, the MI formulations are solved heuristically (e.g., Quaritsch et al. (2010) applies genetic algorithms) or relaxed by iterative solving (the optimization problem is broken into “reasonable” sub-problems which are

7.1. 其他公式

MIP 的替代公式范围从基于图的方案、势场公式到相应的优化问题松弛。

基于图的策略

与MIP方法不同，后者将离散决策编码为数学形式并以此求解，基于图的策略将离散决策简化为在图中节点之间搜索最短路径。尽管这些技术可应用于任何基于MIP的运动规划问题（见第3节），在本小节中我们专注于无碰撞路径/轨迹规划。

注释17. 在图中寻找最短路径的文献中存在多种算法（例如，LaValle (2006), Latombe (2012)）。最具影响力的是Dijkstra算法、贪婪算法或A*搜索算法. ♦

路径规划中的基于图的策略根据图构建方式分类。第一类基于单元分解方法构建图，考虑（多障碍）环境的显式表示。因此，与MIP方法的相似性不可否认，多边形集是一种流行的分解原始。例如，多边形集已在Lozano-Pérez和Wesley (1979)中使用。更精确地说，图节点由多边形障碍物的顶点、路径的初始点和终点定义。该图称为可见性图，因为节点之间的链接由一条直线给出，该直线不与任何障碍物相交。换句话说，两个连接的节点可以“看到”对方。因此，无碰撞路径是通过该图从智能体的初始位置到最终位置的最短路径。从理论角度来看，如果考虑的智能体是一个点，这种方法表现良好。对于维度不可忽略的一般情况，通过考虑障碍物的“生长”（通常通过Minkowski和）扩展上述方法。上述方法的主要缺点是计算负担，特别是在复杂环境中（具有大量复杂形状的障碍物），顶点数量是穷尽的。

第二类方法试图通过避免显式表示环境来缓解这个问题。这种类型的方法在文献中也被称为基于采样的算法（参见LaValle (2006)关于该主题的完整文献综述）。因此，显式表示被消除，而重点放在碰撞检测模块上。该模块检查连接无障碍空间中随机采样点的轨迹的可行性。图被构建为两个节点之间的链接表示（多障碍）环境中的可行路径。

注释18. 对于基于采样的算法，一个有趣的概念是概率完备性（Barraquand等人，1997年；Ladd和Kavraki，2004年）。也就是说，如果采样点的数量足够大（ $\rightarrow \infty$ ），算法返回可行解的概率趋于1。这在Hsu, Latombe和Kurniawati (2007年)中得到了经验证明. ♦

在文献中，有两种重要的基于采样的算法：PRM（概率路图）(Hsu 等人，2007) 和 RRT（快速探索随机树）(Weiss, Danielson, Berntorp, Kolmanovsky, & Cairano, 2017)。它们之间的区别在于构建图的方法。前者（PRM）是一种多查询方法，在构建路图（一组丰富的可行路径）后，通过计算图中的最优路径来回答查询。因此，如果环境具有意识地图（Karaman & Frazzoli, 2011），PRM 就是一种有用的方法。PRM 有许多变体，每个变体都代表了一种宝贵的改进，例如 Ladd 和 Kavraki (2004)、Karaman 和 Frazzoli (2011)。

宝贵的改进 Ladd 和 Kavraki (2004)，Karaman 和 Frazzoli (2011)。

当环境事先生成未知时，RRT方法更适用。在这种方法中，图的构建是增量式的，当获得足够多的无碰撞路径时算法停止。因此，无碰撞样本会作为节点添加到图中，并与周围节点连接。得到的图实际上是一棵树。与PRM类似，RRT也有多种版本。有些版本考虑运动方程并生成可达路径（LaValle和Kuffner，2001），另一些版本仅生成几何路径，这些路径成为低级控制器的参考轨迹。此外，一些版本针对复杂和/或不稳定的动力学（Leonard等人，2008）或不确定动力学（Weiss等人，2017）进行了定制。

除了经典的基于图的方法外，文献中还存在多种结合标准图算法与先进控制策略的方法。例如，Berntorp、Danielson、Weiss和Di Cairano (2018) 提出了一种实时集成运动规划方法，该方法使用反馈控制、正不变集和闭环系统的平衡轨迹。为了生成无碰撞路径，该方法以离线方式在参考路径上进行图搜索，每个参考路径都与一个允许的正不变集相关联。接下来，他们使用预设计的无约束线性二次控制器来跟踪参考路径。

同样地，Althé和Fortelle (2017) 使用一种将可行（“无碰撞”）区域划分的方法来处理轨迹规划问题，该方法允许将NP难问题分解为在一个精心设计的图中进行路径查找问题，然后对二次凸成本函数进行（简单）优化阶段（“以MPC方式”）进行优化。此外，Franzè和Lucia (2015) 处理了在未知环境中进行障碍物规避的问题（考虑的代理是UGV - 自主地面车辆）。所提出的方法包括两个部分：一个离线部分，用于计算所有可能场景下的一步可控集的椭圆近似（这些近似保证了在多障碍物环境中存在一条可行路径），以及一个在线部分，涉及开发基于MPC的策略，以使代理保持在椭圆集序列中。

势场公式化

虽然基于MIP的方法明确考虑了约束，并导致约束优化问题，但基于势场的公式 Chen, Luo, Mei, Yu, 和 Su (2016b) 通过在成本中添加惩罚项来放宽约束。本质上，势场方法依赖于构建一个标量函数（所谓的，势）。当智能体停留在禁区时，该函数取高值。在无碰撞工作空间中，该函数向目标配置减少（即，与目标点相关联的势是最小的）。因此，智能体可以在势的负梯度方向移动以到达最终点。Rimon 和 Koditschek (1992) 提供了关于势场公式的历史（以及更详细的）回顾，以及这种方法如何涉及在运动规划中。一个有趣的特点是，势场公式经常用于分散式或分布式控制策略 Filotheou, Nikou, 和 Dimarogonas (2018)。

7.2. 优化问题松弛

如前所述，MIP 最重要的能力之一是处理非凸最优控制问题中的非凸约束。解决这类问题的自然方法是通过扩展用于凸最优控制的方法和技术。通常，MI 公式会采用启发式方法求解（例如，Quaritsch 等人 (2010) 应用了遗传算法）或通过迭代求解进行松弛（优化问题被分解为“合理”的子问题，这些子问题

solved iteratively). As an example, Xu et al. (2018) employs binary variables to model the link between an UAV and a ground sensor but relaxes the formulation through time allocation tactics and channel communication pre-scheduling. Thus, in the literature, a considerable effort was put in finding a technique which allows the translation/relaxation from non-convex to convex formulations without any major gap. This is referred as convex relaxation or convexification of the non-convex optimization problem. In the literature there are various works providing methods for convexification under several labels: e.g., convex relaxation, successive convexification or time-varying constraints.

As its name suggests, the basic idea of successive convexification is to solve a non-convex optimal control problem via a sequence of convex sub-problems. The non-convexity comes either from having non-linear dynamics (Mao, Szmuk, & Acikmese, 2016) and/or from non-convex state (and/or control) constraints (Mao, Dueri, Szmuk, & Açikmeşe, 2017). In both cases the same technique is applied: linearization, commonly, using a first order Taylor approximation (in a successive manner). Therefore, a preliminary condition on the functions which generate the non-convexity is necessary: they have to be differentiable. In an iterative manner, the linearization is done about the solution obtained at the previous step.

Although the linearization procedure leads to convex formulations, it also introduces two new issues, namely *artificial infeasibility*²⁵ and *approximation error*. These two drawbacks are addressed in the literature, a variety of algorithms was developed and, recently, a convergence analysis was elaborated Liu and Lu (2014).

For instance, Dueri, Mao, Mian, Ding, and Açikmeşe (2017) deals with the problem of the trajectory optimization for autonomous vehicles in an environment containing cylindrical and ellipsoidal obstacles. The approach employs the *Successive Convexification* technique which is used to solve the non-convex optimal control problem via a convergent sequence of convex optimization problems. It considers a discrete-time, finite horizon constrained optimization problem with a number of non-convex state constraints. Several assumptions are necessary in order to employ the technique at hand. The first assumption can be readily satisfied and involves that the obstacles boundaries should not be in contact with the ones of any other constraint. A second problematical assumption builds on a dynamic with a finite number of stationary points. Having the general formulation for the non-convex problem, the successive convexification technique is applied in order to obtain the sequence of convex sub-problems, each being linearized in an iterative procedure. This linearization results in a convex problem but, at the same time, introduces two drawbacks: approximation error and artificial infeasibility. To mitigate these two issues of the convexification, the authors introduce trust regions and penalty functions, respectively. The drawbacks of the convexification procedure and the ways to alleviate them are treated in-depth in the references such as Harris and Açikmeşe (2014).

An interesting alternative for handling the non-convexity of the collision avoidance problem is based on time-varying constraints. The idea is mentioned in Frasch et al. (2013) and is used in, e.g., Janeček et al. (2017a) and Yu et al. (2014). In few words, the non-convex domain is decomposed in a sequence of convex regions and switching instants are introduced. At each moment the agent should stay in one of the convex regions. For instance, in Janeček et al. (2017a) the switching instants are steps of the prediction horizon in the MPC controller. Moreover, this approach is coupled with a heuristic black box which establishes the get-around direction and the sequence of convex sub-domains. This method can be seen as a particularization of the successive convexification. The difference comes from the way of obtaining the sequence of convex sub-problems/sub-domains.

Time-varying constraints approaches consider an a priori known number of sub-domains, since the successive convexification is an iterative method, the sequence is growing until a feasible solution is obtained.

In Rey and Hijazi (2017) a different approach for convex relaxation is presented. For instance, therein the authors treat the aircraft conflict problem²⁶. Basically, the provided formulation is based on complex numbers representation and it results in a tight convex relaxation for the inherently non-convex optimization problem. It is worth mentioning that the above reference includes a comprehensive literature review on the formulation of the air conflict problem as an optimal control problem using mixed-integer techniques (as MILP or as MINLP)²⁷. Coming to the application, the aircraft separation condition is stated: the relative position of two aircraft should be greater than a certain threshold. As it was expected, this condition leads to a non-convex feasible domain which is modeled using a binary variable (actually, the feasible region is composed of two convex ones and the solution should be in one of these two convex subsets). The control actions (speed variation rate and heading deviation angle) admit a natural representation in the complex number form. Even so, the non-convexity is not eliminated (the disjunction constraint is maintained) but the considered formulation is useful towards the convex relaxation approach. This approach is extensively treated in Coffrin, Hijazi, and Hentenryck (2017), where the non-convex constraint is tackled by deriving the corresponding convex hull, the problem being transformed in a MIQCP (mixed-integer quadratically constrained program). A further relaxation is possible by entirely omitting the non-convexity. An algorithm which included the relaxation is presented and tested (with excellent results) on two classical benchmark problems for conflict resolution.

As well, Huang and Peng (2017) solve the MI optimization problem using sequential convex optimization method (i.e., search the local optimum by forming convex sub-problems), avoiding the eventual curse of dimensionality. Likewise, Papen et al. (2017) solve a MILP by using a relaxation of the constraints and fine-tuning the complexity in order to limit the computation time w.r.t. a classical MILP resolution.

8. Conclusions and future challenges

In the preceding material we provided our evaluation on the state-of-the-art for MIP-based motion planning and also we aim to identify active topics and open problems in this field. It is important to mention that the valuable insights in the description of a non-convex feasible region represent an useful modeling tool not only for the motion planning but also for broader control fields.

As mentioned before, although the history of MIP starts almost 60 years ago, the interest of the control and robotic community on this topic is relatively recent and the research in this area is quite active. It is obvious that there exists a substantial progress in all of the topics mentioned in Section 3 and Section 4. However, there are many points that can be further enhanced since the developing of new and performing methods to provide exact solution for MIP is exponentially growing.

As open and active problems regarding mainly obstacle and collision avoidance topic we can mention the trade off between conservativeness and complexity in non-convex region modeling and representation. Even though there exist several valuable improvements on classical MIP formulations, the complexity remains an arduous issue imposing restrictions: only a small size problem can be solved in a real-time manner. The improvements may be accomplished by exploiting the underlying combinatorial structure of the MIP formulation. Nevertheless, whenever

会迭代求解)。例如, Xu 等人 (2018) 采用二进制变量来模拟无人机与地面传感器之间的联系, 但通过时间分配策略和信道通信预调度来松弛公式。因此, 在文献中, 人们投入了大量精力寻找一种技术, 能够将非凸公式转换为凸公式, 且没有任何主要差距。这被称为非凸优化问题的凸松弛或凸化。在文献中, 有各种工作在多个标签下提供凸化的方法: 例如, 凸松弛、逐次凸化或时变约束。

如其名称所示, 连续凸化 (successive convexification) 的基本思想是通过一系列凸子问题来解决非凸最优控制问题。非凸性要么来源于非线性动力学 (Mao, Szmuk, & Acikmese, 2016), 要么来源于非凸状态 (和/或控制) 约束 (Mao, Dueri, Szmuk, & Açikmes.e, 2017)。在这两种情况下, 都应用相同的技术: 线性化, 通常使用一阶泰勒展开式 (以连续方式)。因此, 对产生非凸性的函数必须有一个先决条件: 它们必须是可微分的。以迭代方式, 在先前步骤获得的解处进行线性化。

尽管线性化过程产生了凸形式, 但也引入了两个新问题, 即人工不可行性²⁵和近似误差。这两个缺点在文献中得到了解决, 开发了许多算法, 最近还进行了收敛性分析 (Liu and Lu, 2014)。

例如, Dueri, Mao, Mian, Ding 和 Açikmes.e (2017) 研究了在包含圆柱形和椭圆形障碍物的环境中, 自动驾驶车辆的轨迹优化问题。该方法采用连续凸化技术, 通过一系列收敛的凸优化问题来解决非凸最优控制问题。该方法考虑了一个具有多个非凸状态约束的离散时间、有限时间窗口的约束优化问题。为了应用该技术, 需要一些必要的假设。第一个假设可以很容易地满足, 它涉及障碍物边界不应与其他任何约束的边界接触。第二个有问题的假设基于一个具有有限个静态点的动态系统。在非凸问题的通用公式下, 连续凸化技术被应用于获得一系列凸子问题, 每个子问题都在迭代过程中线性化。这种线性化会得到一个凸问题, 但同时也引入了两个缺点: 近似误差和人工不可行性。为了缓解连续凸化的这两个问题, 作者分别引入了信任域和惩罚函数。连续凸化过程的缺点及其缓解方法在 Harris 和 Açikmes.e (2014) 等参考文献中有深入探讨。

处理碰撞避免问题非凸性的一个有趣替代方案是基于时变约束。该想法在 Frasch 等人 (2013) 中提及, 并在例如 Janeček 等人 (2017a) 和 Yu 等人 (2014) 中使用。简而言之, 非凸域被分解为一系列凸域, 并引入切换时刻。在每个时刻, 智能体应停留在其中一个凸域内。例如, 在 Janeček 等人 (2017a) 中, 切换时刻是 MPC 控制器预测周期的步长。此外, 这种方法与一个启发式黑盒耦合, 该黑盒建立绕行方向和凸子域序列。这种方法可以看作是连续凸化的特例。区别在于获取凸子问题/子域序列的方式。

时变约束方法考虑一个预先已知的子域数量, 因为连续凸化是一种迭代方法, 序列会增长直到获得可行解。

在 Rey 和 Hijazi (2017) 中, 提出了一种不同的凸松弛方法。例如, 在该文中, 作者们处理了飞机冲突问题²⁶。基本上, 所提供的公式基于复数表示, 并针对固有的非凸优化问题产生了一个紧密的凸松弛。值得一提的是, 上述参考文献包含了一份关于将空冲突问题作为最优控制问题 (使用混合整数技术, 如 MILP 或 MINLP) 的全面文献综述²⁷。在应用方面, 飞机分离条件被陈述为: 两架飞机的相对位置应大于某个阈值。正如预期的那样, 该条件导致一个非凸的可行域, 该域使用二进制变量进行建模 (实际上, 可行区域由两个凸区域组成, 解应在其中之一)。控制动作 (速度变化率和航向偏差角) 在复数形式中具有自然的表示。尽管如此, 非凸性并未消除 (析取约束被保留), 但所考虑的公式对凸松弛方法是有用的。这种方法在 Coffrin, Hijazi 和 Hentenryck (2017) 中得到了广泛处理, 其中通过推导相应的凸包来处理非凸约束, 问题被转化为一个 MIQCP (混合整数二次约束规划)。通过完全省略非凸性, 可以进行进一步的松弛。一个包含松弛的算法被提出并在两个经典的冲突解决基准问题上进行了测试 (并取得了优异的结果)。

此外, 黄和彭 (2017) 使用序列凸优化方法 (即通过构建凸子问题搜索局部最优解) 解决了混合整数规划 (MIP) 优化问题, 避免了最终的维度灾难。同样地, Papen 等人 (2017) 通过放宽约束并微调复杂度来解决混合整数线性规划 (MILP) 问题, 以限制相对于经典 MILP 求解的计算时间。

8. 结论与未来挑战

在前面的内容中, 我们提供了基于 MIP 的运动规划的当前研究进展的评估, 并且我们旨在确定该领域的活跃主题和开放问题。重要的是要提到, 在非凸可行区域的描述中的宝贵见解不仅代表了一种有用的建模工具, 而且也适用于更广泛的控制领域。

如前所述, 虽然 MIP 的历史几乎有 60 年, 但控制和机器人社区对这个主题的兴趣相对较新, 该领域的研究非常活跃。很明显, 在 3 节和 4 节中提到的所有主题中都取得了实质性进展。然而, 由于开发新的、性能更好的方法以提供 MIP 的精确解正在呈指数增长, 因此还有许多方面可以进一步改进。

在主要涉及障碍物和碰撞避免主题的开放和活跃问题中, 我们可以提及非凸区域建模和表示中保守性与复杂性的权衡。尽管在经典 MIP 公式上存在一些有价值的改进, 但复杂性仍然是一个艰巨的问题, 施加了限制: 只有规模较小的问题才能以实时方式解决。这些改进可以通过利用 MIP 公式的底层组合结构来实现。然而, 每当

²⁵ A solution of the original non-convex problem can become infeasible for the sequence of the convex sub-problems.

²⁶ According to the air traffic rules, the aircraft have to be separated by at least 5 nautical miles horizontally and 1000 ft vertically, otherwise they are in conflict.

²⁷ A more detailed review on the air conflict detection and resolution can be found in Kuchar and Yang (2000).

²⁵ 原非凸问题的解对于凸子问题的序列可能变得不可行。

²⁶ 根据空中交通规则, 飞机必须在水平方向上至少相隔 5 海里, 垂直方向上至少相隔 1000 英尺, 否则就会发生冲突。²⁷ 关于空中冲突检测和解决更详细的综述, 可以参考 Kuchar 和 Yang (2000) 的研究。

the problems are inherently non-convex and/or they involve alternative choices, then mixed-integer representations provide an useful and powerful tool, but we have to proceed with caution evaluating the structural properties that may lead to compact formulations. As well, important progress may be expected in the direction of the MIP resolution algorithms, which can be adapted and improved in order to attenuate the inherent complexity of the mixed-integer approaches by capitalizing on the different particularities of the problems to be solved. Additionally, another aspect that can be viewed as an open issue and, in some sense, is generated by the previous ones is that there are not so many works in the field of MIP-based motion planning which validate the theoretical results over, at least, experimental platforms, even though there are many applications able to benefit from their use, e.g., search and rescue, environmental measurements tasks, area coverage and the like.

This article has reviewed some recent research and development in MIP-based motion planning. In addition to the results reviewed above, there are many other publications treating MIP formulations in feedback decision making that could not find a direct link with the scope of the motion planning in spite of our best effort. Covering all the variety of application and often inhomogenous problem formulations involving MIP is a task that can be challenging even for a review.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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问题时本质上是非凸的并且/或者它们涉及替代选择, 那么混合整数表示提供了一个有用且强大的工具, 但我们必须谨慎评估可能导致紧凑公式产生的结构属性。同样, 在MIP求解算法方向上可能取得重要进展, 这些算法可以适应和改进, 以利用待解决问题的不同特性来减轻混合整数方法的固有复杂性。此外, 另一个可以被视为开放问题, 并且在某种程度上是由前一个问题产生的方面是, 在基于MIP的运动规划领域, 验证理论结果至少在实验平台上有效的工作并不多, 尽管有许多应用能够从其使用中受益, 例如搜索和救援、环境测量任务、区域覆盖等。

本文回顾了基于MIP的运动规划的一些近期研究与发展。除了上述回顾的结果外, 尽管我们已尽力, 但仍有许多其他处理反馈决策中MIP公式的出版物未能与运动规划的范围建立直接联系。涵盖MIP的各种应用以及通常非同质化的问题公式是一项挑战, 即使是对于一篇综述来说也是如此。

利益冲突声明

作者声明他们没有已知的利益冲突或个人关系可能影响本文报告的工作。

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